P20890_Angola — Fire & Gas PLC Ladder Program

Target: GE Fanuc VersaMax PLC (IC200CPU001) | HMI: QuickPanel (Target 2) | Saved with PME Build 2425

Hardware Configuration (from Hardware.hwcfg):

Rack 0 (IC200RAC001): PWR IC200PWR002 | CPU IC200CPU001 | Slot 3-5: 3x IC200ALG262 (8AI %AI0001-%AI0024) | Slot 6: IC200MDL650 (32DI %I0001-%I0032)

Rack 1 (IC200RAC001): PWR IC200PWR002 | Exp Receiver IC200ERM002 | Slot 2-3: 2x IC200MDL940 (16 Relay DO %Q0001-%Q0032)

Program Execution Flow: _MAIN -> Init (1st scan), Input (DI_1, DI_2, AI_1, AI_2, AI_3, Vote_1), Logic (CE_LOG, Sound), Output (Indicat, Shutdow, Protect, Loc_A

Ladder Diagram Blocks (24 Blocks):

Block	Rungs	Subroutine Calls	Description
_MAIN	9	Init, Input, Output, Logic	F&G Angola
Init	49		This block reset alarms and clears timers the first scan.
Input	13	DI_1, DI_2, AI_1, AI_2, AI_3, Vot…	This block calls the routines for input handling.
DI_1	42		This routine handles the inputs connected to the IC200MDL650 digital input…
DI_2	1		This block gets all inputs from 1 - 32 and do all functions on the inputs.
AI_1	76	CalcVa1, Ai_AL_1	This routine handles the inputs connected to the IC200ALG262 analogue inpu…
AI_2	75	CalcVa1, Ai_AL_1, CalcVa3, Ai_AL_3	This routine handles the inputs connected to the IC200ALG262 analogue inpu…
AI_3	62	Ai_AL_1, CalcVa2, CalcVa1, Ai_AL_2	This routine handles the inputs connected to the IC200ALG262 analogue inpu…
CalcVa1	9		This routine converts input value from 0 to 32000 to -25 to 100.
CalcVa2	9		This routine converts input value from 0 to 32000 to -37 to 150 (0 - 150 P…
CalcVa3	9		This routine converts input value from 0 to 32000 to -1,25 to 5,00 (0 - 5,…
Ai_AL_1	5		This routine calculates the Lo Alarm, Hi Alarm, Hi Fault(Short circuit) an…
Ai_AL_2	5		This routine calculates the Lo Alarm, Hi Alarm, Hi Fault(Short circuit) an…
Ai_AL_3	8		FOR REAL VALUES: This routine calculates the Lo Alarm, Hi Alarm, Hi Fault…
Vote_1	10		Voting of Low Alarm on Gas detectors pump house (ALLI).
Logic	16	CE_LOG, Sound	This routine calls the voting and C&E routines for the various fireareas o…
CE_LOG	27		This block do all Cause & Effect logic for this Fire & Gas system.
Sound	50		This block generates sound and mute of sound.
Output	11	Indicat, Shutdow, Protect, Loc_Al…	This block calls the output routines where the physical outputs are set.
Indicat	4		Indication Outputs for local panel set in this block.
Shutdow	6		Shutdown Outputs set in this block.
Protect	13		Fire Protection Outputs set in this block.
Loc_Al	4		Local sound alarm Outputs set in this block.
FLT_1	7		Line Fault on Outputs done in this block.

LD BLOCK: _MAIN

Calls: Init, Input, Output, Logic

Rung 1

```
F&G Angola
Programmed on Machine Edition 3.00 ServicePack 6.
Program History:
- Programming start date: 07. march 2003.

Revision:                       Date:           Programmed by:          Comment:
P208090_Angola-A        13.mars 2003    Vidar Pedersen - Origo-e
P208090_Angola-B        11.april 2003   Vidar Pedersen - Origo-e        As on FAT
P208090_Angola-C        23.april 2003   Vidar Pedersen - Origo-e        As when panel sent to Angola.
P208090_Angola-D        12.juni 2003    Vidar Pedersen - Origo-e
P208090_Angola-E        28.juni 2003    Vidar Pedersen - Origo-e        Changed according to input from Erik Dahl in
Angola!
P208090_Angola-F        29.juni 2003    Vidar Pedersen - Origo-e        Correction of AI_14 for 7404-DG011R_VAL.

History on changes after 23.april-03:
- 12.juni-03:  Timer for start of pump B (second and third time) changed from 30 to 60 sec. Done on 3 timers. In Block
CE_LOG.
- 12.juni-03:  - 12.juni-03:  New input (6401-ESDV-001) to also trip PumpShutdown (%Q08), Oil Booster Pumps (%Q03) and
Water Booster Pumps (%Q04). In Block CE_LOG.
- 12.juni-03:  Output %Q0008 (PumpShutdown) inverted. In Block Shutdown.
- 28.juni-03:  %AI_17 (7002-PST-001), changed LoAlarm setpoint from 60 to 50.   Vidar Pedersen
- 28.juni-03:  Quickpanel alarm endring. "6401-ESDV-001  ESD Fuel Gas Button - Activated" endret fra å gi alarm på Off til
å gi alarm på On.   Vidar Pedersen
- 28.juni-03:  INHIBIT_ALL_OUTPUTS changed to NOT trig sound in Pump house.     Vidar Pedersen
- 28.juni-03:  %AI_19 (PST_7503_001), Short circuit alarm limit changed from 98 to 102.   Vidar Pedersen
- 28.juni-03:  7404-DG011R (LineGasDetector), changed from %AI_11 til %AI_14.   Vidar Pedersen
- 29.juni-03:  7404-DG011R (for AI_14) endret fra Real til INT (som AI_11). MOVE_REAL blokk byttet ut med INT_to_REAL i
blokka AI_2.
             7404-DG011R_VAL endret fra Scientific notation til Floating point.   Vidar Pedersen
```

Rung 2

```
Call initiate
```

Rung 3



Rung 4

```
Call Input routine
```

Rung 5



Rung 6

```
Call LogicCE (Cause & Effect) routine
```

Rung 7



Rung 8

```
Call Output routine
```

Rung 9

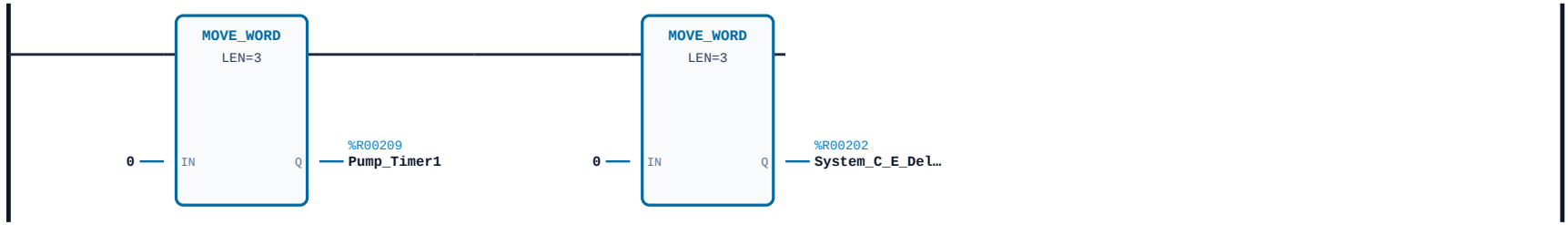


LD BLOCK: Init

Rung 1

This block reset alarms and clears timers the first scan.

Rung 2



Rung 3

Reset sound in first scan

Rung 4



Rung 5

Reset timers in first scan.

Rung 6



Rung 7 COIL DG_7404_001_ALL (%M00401)

Rung 8 COIL DG_7404_002_ALH (%M00354)

Rung 9 COIL DG_7404_002_ALL (%M00402)

Rung 10 COIL DG_7404_003_ALH (%M00355)

Rung 11 COIL DG_7404_003_ALL (%M00403)

```
Rung 12   COIL DG_7404_004_ALH (%M00356)
Rung 13   COIL DG_7404_004_ALL (%M00404)
Rung 14   COIL DG_7404_005_ALH (%M00357)
Rung 15   COIL DG_7404_005_ALL (%M00405)
Rung 16   COIL DG_7404_006_ALH (%M00358)
Rung 17   COIL DG_7404_006_ALL (%M00406)
Rung 18   COIL DG_7404_007_ALH (%M00359)
Rung 19   COIL DG_7404_007_ALL (%M00407)   |   NOCON #ALW_OFF (%S00008)
Rung 20   COIL DG_7404_008_ALH (%M00360)   |   NOCON #ALW_OFF (%S00008)
Rung 21   COIL DG_7404_008_ALL (%M00408)   |   NOCON #ALW_OFF (%S00008)
Rung 22   COIL DG_7404_009_ALH (%M00361)   |   NOCON #ALW_OFF (%S00008)
Rung 23   COIL DG_7404_009_ALL (%M00409)   |   NOCON #ALW_OFF (%S00008)
Rung 24   COIL DG_7404_010_ALH (%M00362)   |   NOCON #ALW_OFF (%S00008)
Rung 25   NOCON #ALW_OFF (%S00008)
Rung 26   NOCON #ALW_OFF (%S00008)
Rung 27   NOCON #ALW_OFF (%S00008)
Rung 28   NOCON #ALW_OFF (%S00008)
Rung 29   NOCON #ALW_OFF (%S00008)
Rung 30   NOCON #ALW_OFF (%S00008)
Rung 31   NOCON #ALW_OFF (%S00008)
Rung 32   NOCON #ALW_OFF (%S00008)
Rung 33   NOCON #ALW_OFF (%S00008)
Rung 34   NOCON #ALW_OFF (%S00008)
```

```
Rung 35   NOCON #ALW_OFF (%S00008)
Rung 36   NOCON #ALW_OFF (%S00008)
Rung 37   NOCON #ALW_OFF (%S00008)
Rung 38   NOCON #ALW_OFF (%S00008)
Rung 39   NOCON #ALW_OFF (%S00008)
Rung 40   NOCON #ALW_OFF (%S00008)
Rung 41   NOCON #ALW_OFF (%S00008)
Rung 42   NOCON #ALW_OFF (%S00008)
Rung 43   NOCON #ALW_OFF (%S00008)
Rung 44   NOCON #ALW_OFF (%S00008)
Rung 45   NOCON #ALW_OFF (%S00008)
Rung 46   NOCON #ALW_OFF (%S00008)
Rung 47   NOCON #ALW_OFF (%S00008)
Rung 48   NOCON #ALW_OFF (%S00008)
Rung 49   NOCON #ALW_OFF (%S00008)
```

LD BLOCK: Input

Calls: DI_1, DI_2, AI_1, AI_2, AI_3, Vote_1

Rung 1

```
This block calls the routines for input handling.
Each block call handles 16 digital inputs and 8 analog inputs.
```

Rung 2

```
Inputs %I0001 to %I0016 PLC Input module.
```

Rung 3



Rung 4

```
Inputs %I0017 to %I0032 PLC Input module.
```

Rung 5



Rung 6

```
Inputs %AI0001 to %AI008 PLC Input module.
```

Rung 7



Rung 8

```
Inputs %AI0009 to %AI016 PLC Input module.
```

Rung 9



Rung 10

```
Inputs %AI0017 to %AI024 PLC Input module.
```

Rung 11



Rung 12

```
Voting of input signals.
```

Rung 13



LD BLOCK: DI_1

Rung 1

```
This routine handles the inputs connected to the IC200MDL650 digital input module in PLC slot 4.
Range: %I0001 - %I0016
```

Rung 2

```
Psaudo code for inputs to be latched:

If input high then latch _AL bit until system reset
_ALI = (_AL and not _INH)


Psaudo code for other inputs:

_AL = input
_ALI = (_AL and not _INH)
```

Rung 3

```
%I0001.   Generates no latched alarms.
```

Rung 4



Rung 5



Rung 6

```
%I0002.   Generates no latched alarms.
```

Rung 7



Rung 8



Rung 9

```
%I0003.   Generates no latched alarms.
```

Rung 10



Rung 11



Rung 12

```
%I0004.   Generates latched alarms.
```

Rung 13



Rung 14



Rung 15

```
%I0005.   Generates latched alarms.
```

Rung 16



Rung 17



Rung 18

```
%I0006.   Generates nolatched alarms.
```

Rung 19



Rung 20



Rung 21

```
%I0007.   Generates nolatched alarms.
```

Rung 22



Rung 23



Rung 24

```
%I0008.   Generates nolatched alarms.
```

Rung 25



Rung 26



Rung 27

```
%I0009.   Generates nolatched alarms.
```

Rung 28



Rung 29



Rung 30

```
%I0010.   Generates nolatched alarms.
```

Rung 31



Rung 32



Rung 33

%I0011. Generates latched alarms.

Rung 34



Rung 35



Rung 36

%I0012. Generates latched alarms.

Rung 37



Rung 38



Rung 39

%I0013. Generates latched alarms.

Rung 40



Rung 41



Rung 42

```
%I0014.   Generates latched alarms.
```

LD BLOCK: DI_2

Rung 1

This block gets all inputs from 1 - 32 and do all functions on the inputs.

LD BLOCK: AI_1

Calls: CalcVa1, Ai_AL_1

Rung 1

```
This routine handles the inputs connected to the IC200ALG262 analogue input module in PLC slot 3. Channel %AI0001 to
%AI0008.
```

Rung 2

```
Standard psaudo code:

Move Analogue input value to VALUE_IN
Call CALCVAL                          //Converts input value.
_VAL = VALUE_IN
Call AI_ALARM                         //Calculates low and high alarm and fault

if LOALARM then latch _ALL bit until system reset
_ALLI = (_ALL and not _INH)
if HIALARM then latch _ALH bit until system reset
_ALHI = (_ALH and not _INH)
If (Any_actions and _ALHI) then latch _FRUP until system reset
_SC  = HIFAULT
_OC = LOFAULT


Psaudo code for digital detectors on analogue inputs:
(No scaling of input value, and only high alarm.)

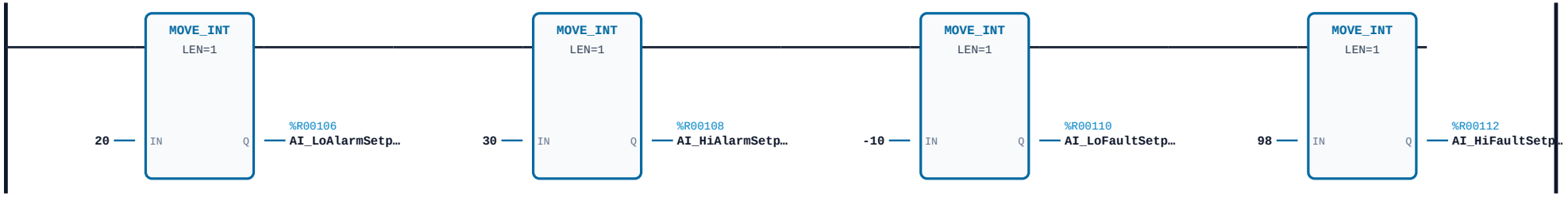
Set special alarm limits
Move Analogue input value to VALUE_IN
Call AI_ALARM                         //Calculates low and high alarm and fault

_ALL = 0
_ALLI = 0
if HIALARM then latch _ALH bit until system reset
_ALHI = (_ALH and not _INH)
If (Any_actions and _ALHI) then latch _FRUP until system reset
_SC  = HIFAULT
_OC = LOFAULT
```

Rung 3

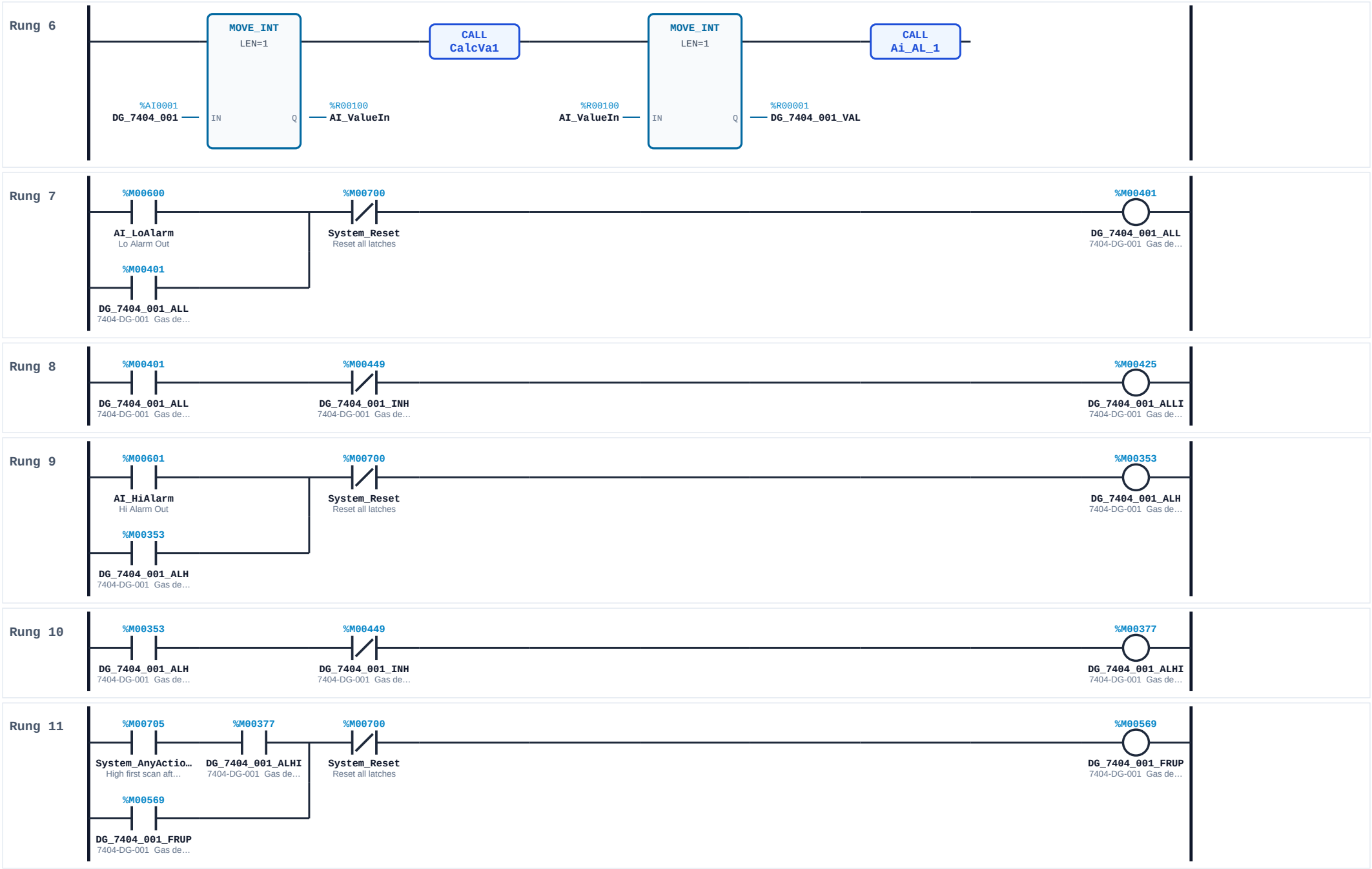
```
First set alarm limits:
```

Rung 4



Rung 5

```
%AI0001.  Generate alarms and fults.
```


Rung 6
MOVE_INT
LEN=1
%AI0001
DG_7404_001
IN
Q
%R00100
AI_ValueIn
CALL
CalcVa1
MOVE_INT
LEN=1
%R00100
AI_ValueIn
IN
Q
%R00001
DG_7404_001_VAL
CALL
Ai_AL_1
Rung 7
%M00600
AI_LoAlarm
Lo Alarm Out
%M00700
System_Reset
Reset all latches
%M00401
DG_7404_001_ALL
7404-DG-001 Gas de…
%M00401
DG_7404_001_ALL
7404-DG-001 Gas de…
Rung 8
%M00401
DG_7404_001_ALL
7404-DG-001 Gas de…
%M00449
DG_7404_001_INH
7404-DG-001 Gas de…
%M00425
DG_7404_001_ALLI
7404-DG-001 Gas de…
Rung 9
%M00601
AI_HiAlarm
Hi Alarm Out
%M00700
System_Reset
Reset all latches
%M00353
DG_7404_001_ALH
7404-DG-001 Gas de…
%M00353
DG_7404_001_ALH
7404-DG-001 Gas de…
Rung 10
%M00353
DG_7404_001_ALH
7404-DG-001 Gas de…
%M00449
DG_7404_001_INH
7404-DG-001 Gas de…
%M00377
DG_7404_001_ALHI
7404-DG-001 Gas de…
Rung 11
%M00705
System_AnyActio…
High first scan aft…
%M00377
DG_7404_001_ALHI
7404-DG-001 Gas de…
%M00700
System_Reset
Reset all latches
%M00569
DG_7404_001_FRUP
7404-DG-001 Gas de…
%M00569
DG_7404_001_FRUP
7404-DG-001 Gas de…

Rung 12



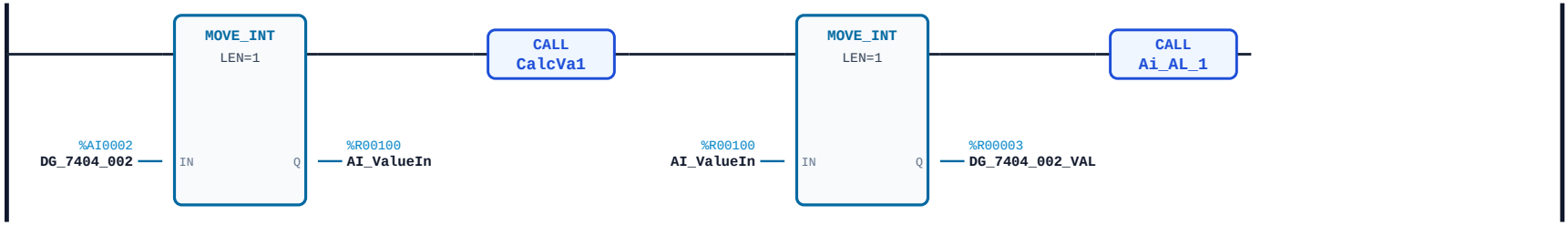
Rung 13



Rung 14

```
%AI0002.  Generate alarms and fults.
```

Rung 15



Rung 16



Rung 17



Rung 18



Rung 19



Rung 20



Rung 21



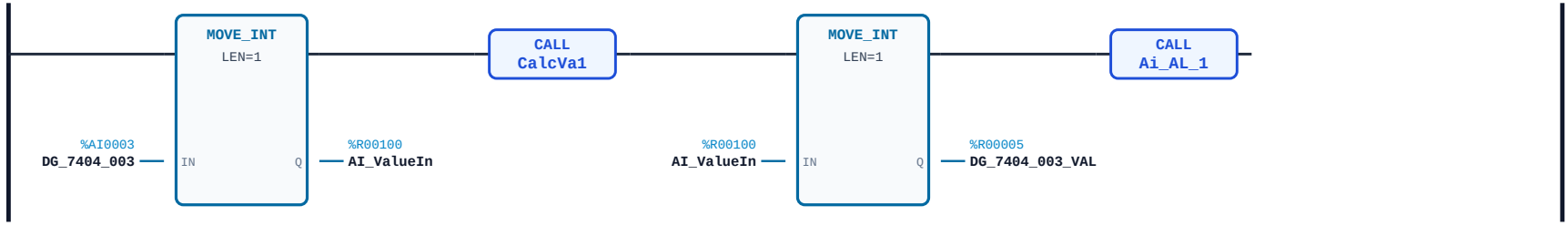
Rung 22



Rung 23

```
%AI0003.  Generate alarms and fults.
```

Rung 24



Rung 25



Rung 26



Rung 27



Rung 28



Rung 29



Rung 30

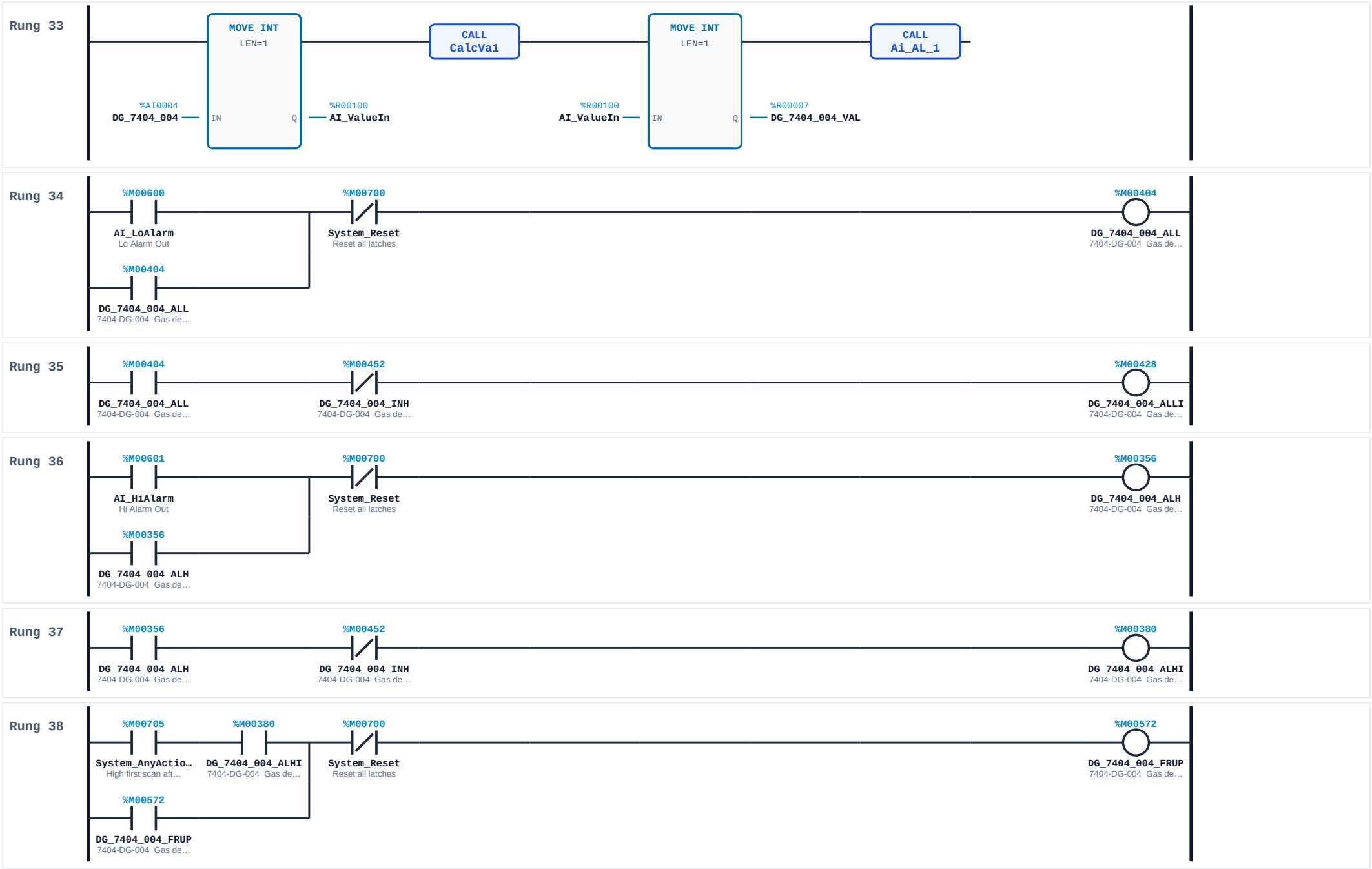


Rung 31



Rung 32

```
%AI0004.  Generate alarms and fults.
```


Rung 33
MOVE_INT
LEN=1
%AI0004
DG_7404_004
IN
Q
%R00100
AI_ValueIn
CALL
CalcVa1
MOVE_INT
LEN=1
%R00100
AI_ValueIn
IN
Q
%R00007
DG_7404_004_VAL
CALL
Ai_AL_1
Rung 34
%M00600
AI_LoAlarm
Lo Alarm Out
%M00404
DG_7404_004_ALL
7404-DG-004 Gas de…
%M00700
System_Reset
Reset all latches
%M00404
DG_7404_004_ALL
7404-DG-004 Gas de…
Rung 35
%M00404
DG_7404_004_ALL
7404-DG-004 Gas de…
%M00452
DG_7404_004_INH
7404-DG-004 Gas de…
%M00428
DG_7404_004_ALLI
7404-DG-004 Gas de…
Rung 36
%M00601
AI_HiAlarm
Hi Alarm Out
%M00356
DG_7404_004_ALH
7404-DG-004 Gas de…
%M00700
System_Reset
Reset all latches
%M00356
DG_7404_004_ALH
7404-DG-004 Gas de…
Rung 37
%M00356
DG_7404_004_ALH
7404-DG-004 Gas de…
%M00452
DG_7404_004_INH
7404-DG-004 Gas de…
%M00380
DG_7404_004_ALHI
7404-DG-004 Gas de…
Rung 38
%M00705
System_AnyActio…
High first scan aft…
%M00380
DG_7404_004_ALHI
7404-DG-004 Gas de…
%M00700
System_Reset
Reset all latches
%M00572
DG_7404_004_FRUP
7404-DG-004 Gas de…
%M00572
DG_7404_004_FRUP
7404-DG-004 Gas de…

Rung 39



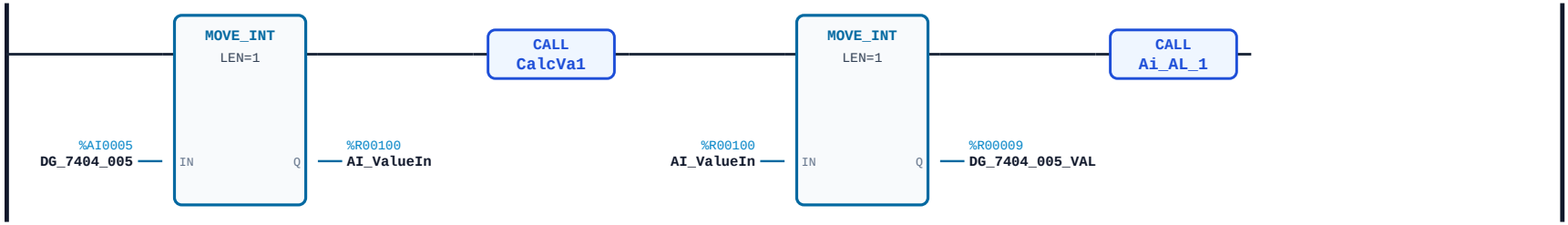
Rung 40



Rung 41

%AI0005. Generate alarms and fults.

Rung 42



Rung 43



Rung 44



Rung 45



Rung 46



Rung 47



Rung 48



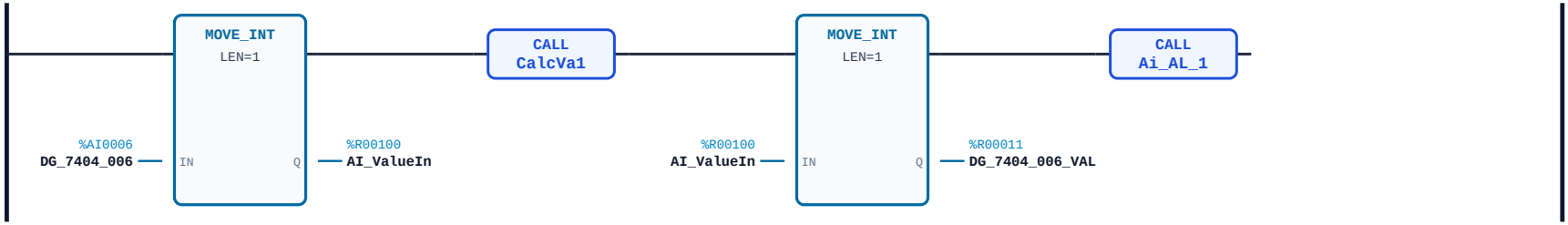
Rung 49



Rung 50

```
%AI0006.  Generate alarms and fults.
```

Rung 51



Rung 52



Rung 53



Rung 54



Rung 55



Rung 56



Rung 57

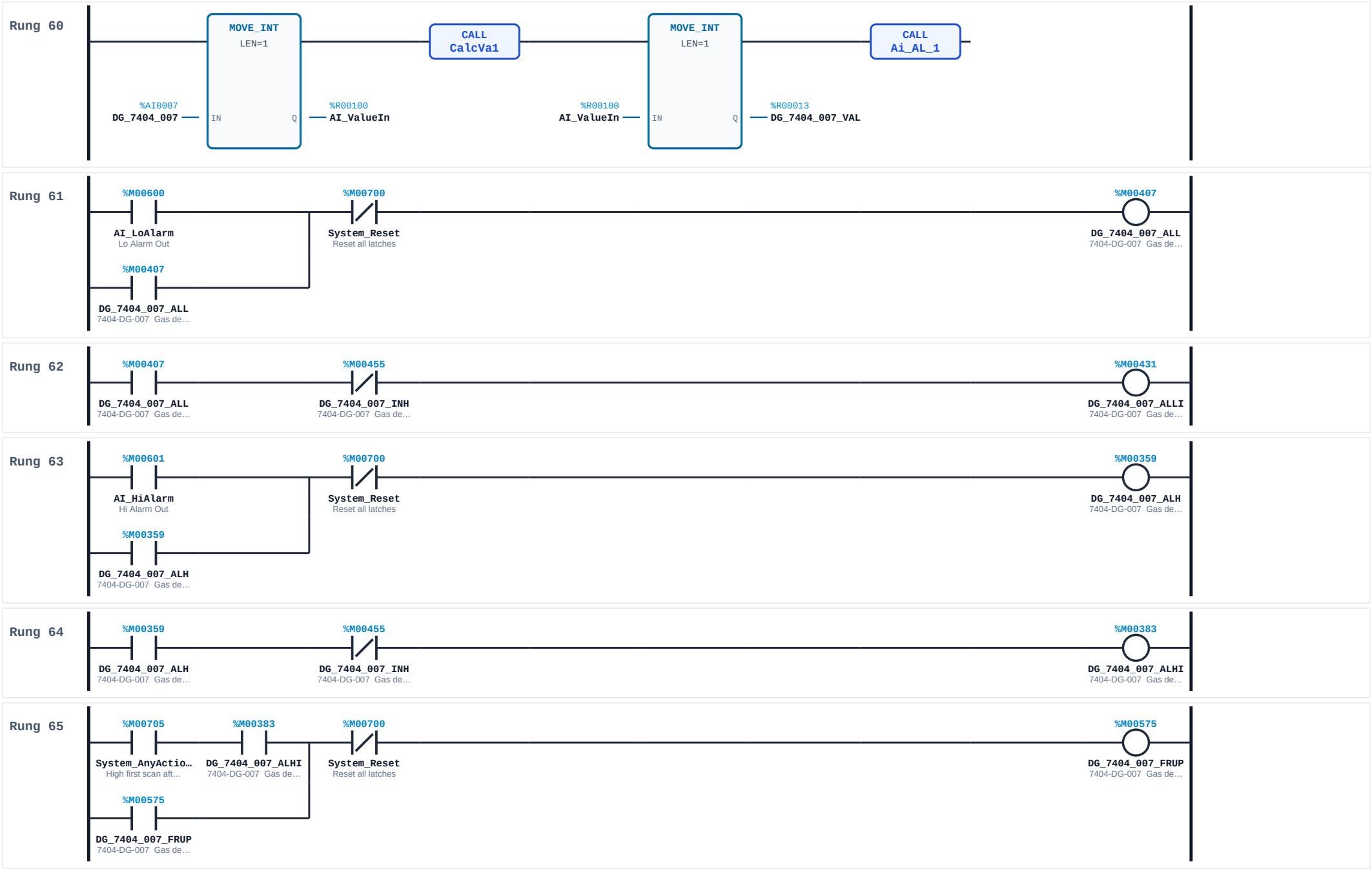


Rung 58



Rung 59

```
%AI0007.  Generate alarms and fults.
```


Rung 60
MOVE_INT
LEN=1
%AI0007
DG_7404_007
IN
Q
%R00100
AI_ValueIn
CALL
CalcVa1
MOVE_INT
LEN=1
%R00100
AI_ValueIn
IN
Q
%R00013
DG_7404_007_VAL
CALL
Ai_AL_1
Rung 61
%M00600
AI_LoAlarm
Lo Alarm Out
%M00700
System_Reset
Reset all latches
%M00407
DG_7404_007_ALL
7404-DG-007 Gas de…
%M00407
DG_7404_007_ALL
7404-DG-007 Gas de…
Rung 62
%M00407
DG_7404_007_ALL
7404-DG-007 Gas de…
%M00455
DG_7404_007_INH
7404-DG-007 Gas de…
%M00431
DG_7404_007_ALLI
7404-DG-007 Gas de…
Rung 63
%M00601
AI_HiAlarm
Hi Alarm Out
%M00700
System_Reset
Reset all latches
%M00359
DG_7404_007_ALH
7404-DG-007 Gas de…
%M00359
DG_7404_007_ALH
7404-DG-007 Gas de…
Rung 64
%M00359
DG_7404_007_ALH
7404-DG-007 Gas de…
%M00455
DG_7404_007_INH
7404-DG-007 Gas de…
%M00383
DG_7404_007_ALHI
7404-DG-007 Gas de…
Rung 65
%M00705
System_AnyActio…
High first scan aft…
%M00383
DG_7404_007_ALHI
7404-DG-007 Gas de…
%M00700
System_Reset
Reset all latches
%M00575
DG_7404_007_FRUP
7404-DG-007 Gas de…
%M00575
DG_7404_007_FRUP
7404-DG-007 Gas de…

Rung 66



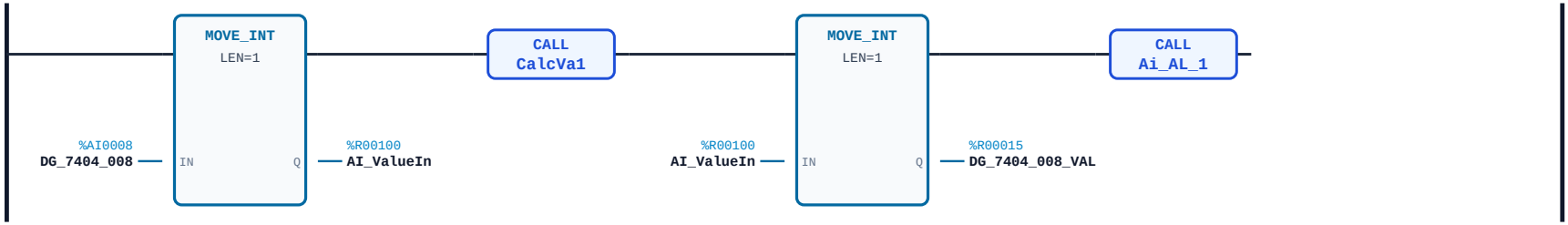
Rung 67



Rung 68

```
%AI0008.  Generate alarms and fults.
```

Rung 69



Rung 70



Rung 71



Rung 72



Rung 73



Rung 74



Rung 75



Rung 76



LD BLOCK: AI_2

Calls: CalcVa1, Ai_AL_1, CalcVa3, Ai_AL_3

Rung 1

```
This routine handles the inputs connected to the IC200ALG262 analogue input module in PLC slot 4.  Channel %AI0009 to
%AI0016.
```

Rung 2

```
Standard psaudo code:

Move Analogue input value to VALUE_IN
Call CALCVAL                          //Converts input value.
_VAL = VALUE_IN
Call AI_ALARM                         //Calculates low and high alarm and fault

if LOALARM then latch _ALL bit until system reset
_ALLI = (_ALL and not _INH)
if HIALARM then latch _ALH bit until system reset
_ALHI = (_ALH and not _INH)
If (Any_actions and _ALHI) then latch _FRUP until system reset
_SC  = HIFAULT
_OC = LOFAULT


Psaudo code for digital detectors on analogue inputs:
(No scaling of input value, and only high alarm.)

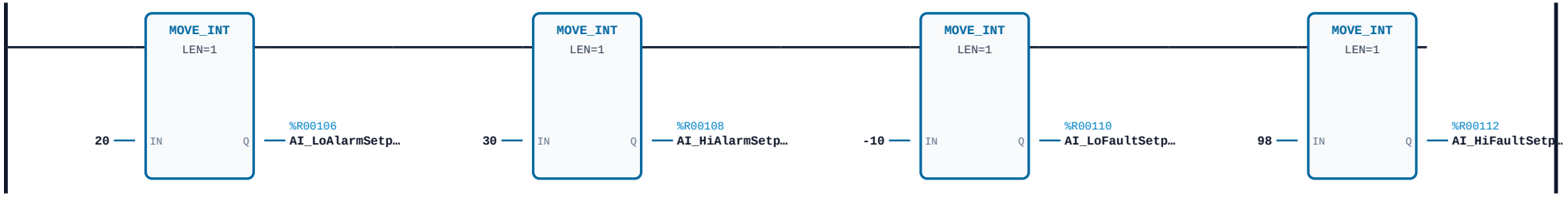
Set special alarm limits
Move Analogue input value to VALUE_IN
Call AI_ALARM                         //Calculates low and high alarm and fault

_ALL = 0
_ALLI = 0
if HIALARM then latch _ALH bit until system reset
_ALHI = (_ALH and not _INH)
If (Any_actions and _ALHI) then latch _FRUP until system reset
_SC  = HIFAULT
_OC = LOFAULT
```

Rung 3

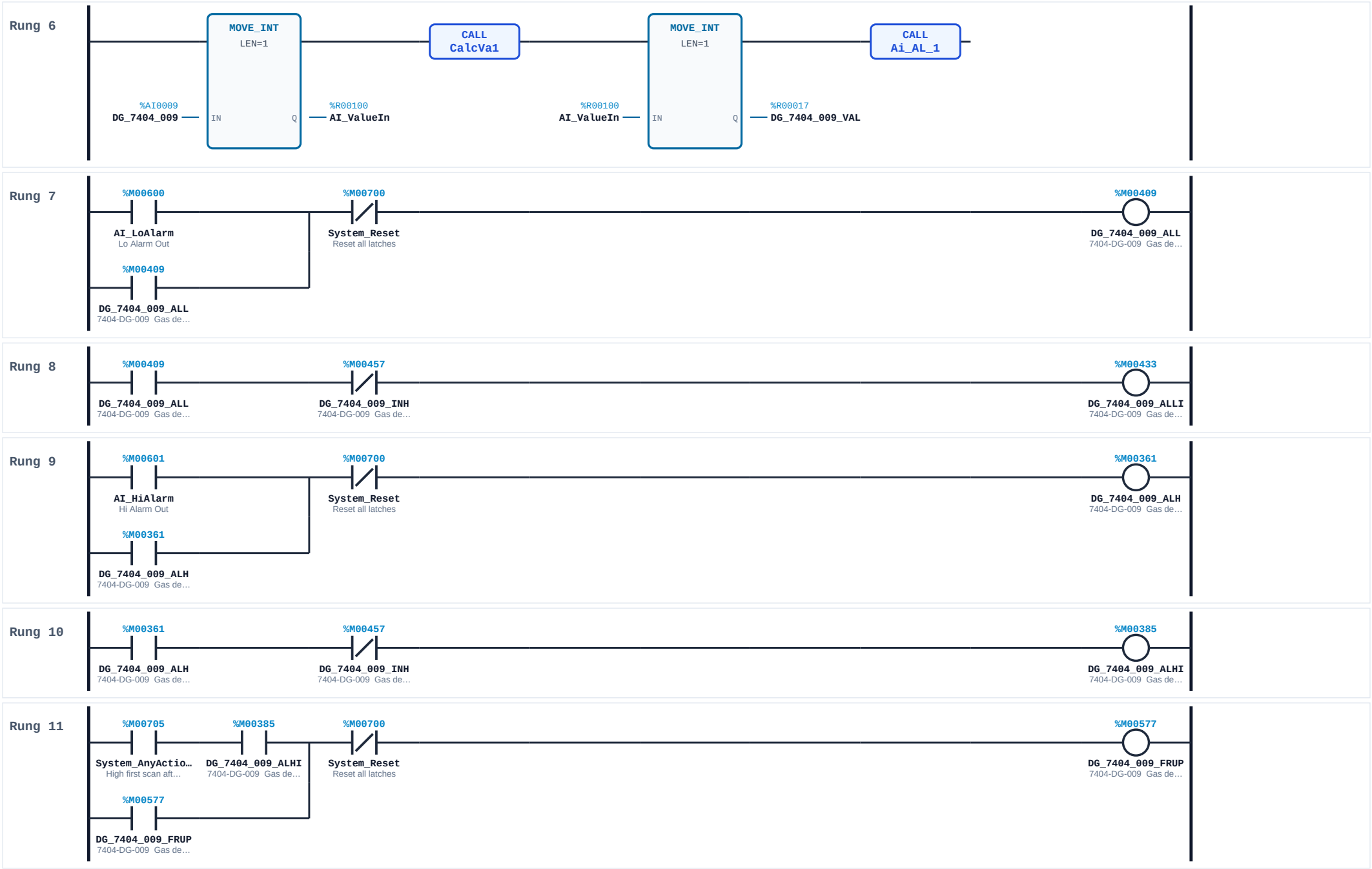
```
First set alarm limits:
```

Rung 4



Rung 5

```
%AI0009.  Generate alarms and fults.
```


Rung 6
MOVE_INT
LEN=1
%AI0009
DG_7404_009
IN
Q
%R00100
AI_ValueIn
CALL
CalcVa1
MOVE_INT
LEN=1
%R00100
AI_ValueIn
IN
Q
%R00017
DG_7404_009_VAL
CALL
Ai_AL_1
Rung 7
%M00600
AI_LoAlarm
Lo Alarm Out
%M00700
System_Reset
Reset all latches
%M00409
DG_7404_009_ALL
7404-DG-009 Gas de…
%M00409
DG_7404_009_ALL
7404-DG-009 Gas de…
Rung 8
%M00409
DG_7404_009_ALL
7404-DG-009 Gas de…
%M00457
DG_7404_009_INH
7404-DG-009 Gas de…
%M00433
DG_7404_009_ALLI
7404-DG-009 Gas de…
Rung 9
%M00601
AI_HiAlarm
Hi Alarm Out
%M00700
System_Reset
Reset all latches
%M00361
DG_7404_009_ALH
7404-DG-009 Gas de…
%M00361
DG_7404_009_ALH
7404-DG-009 Gas de…
Rung 10
%M00361
DG_7404_009_ALH
7404-DG-009 Gas de…
%M00457
DG_7404_009_INH
7404-DG-009 Gas de…
%M00385
DG_7404_009_ALHI
7404-DG-009 Gas de…
Rung 11
%M00705
System_AnyActio…
High first scan aft…
%M00385
DG_7404_009_ALHI
7404-DG-009 Gas de…
%M00700
System_Reset
Reset all latches
%M00577
DG_7404_009_FRUP
7404-DG-009 Gas de…
%M00577
DG_7404_009_FRUP
7404-DG-009 Gas de…

Rung 12



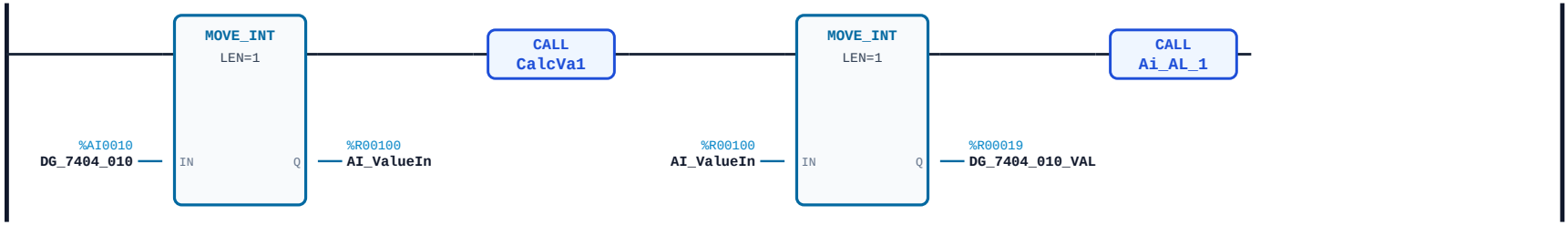
Rung 13



Rung 14

```
%AI0010.  Generate alarms and fults.
```

Rung 15



Rung 16



Rung 17



Rung 18



Rung 19



Rung 20



Rung 21



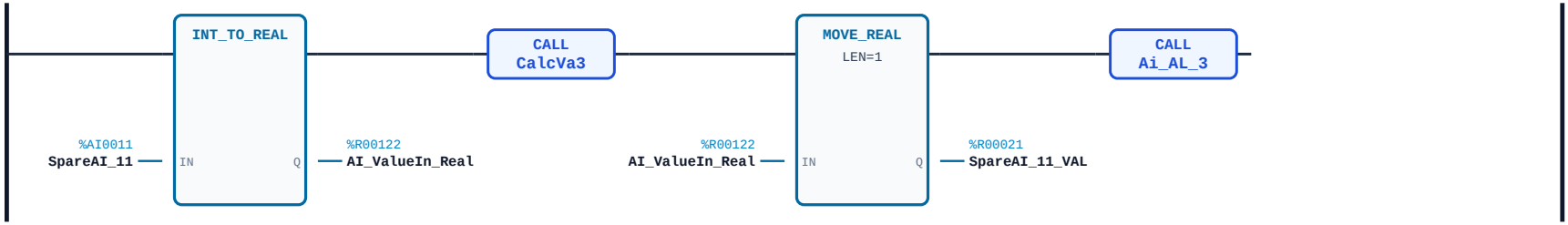
Rung 22



Rung 23

```
%AI0011.  Generate alarms and fults.
```

Rung 24



Rung 25



Rung 26



Rung 27



Rung 28



Rung 29



Rung 30



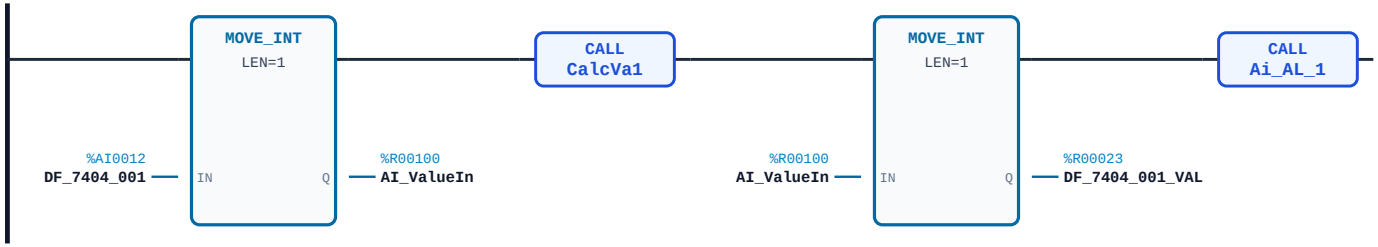
Rung 31



Rung 32

```
%AI0012.  Generate alarms and fults.
Alarm will reset if
- Pumps have been running for 60 minutes
- "Reset Fire&Gas" pushed.
- "Stop Foam" pushed.
```

Rung 33



Rung 34



Rung 35



Rung 36



Rung 37



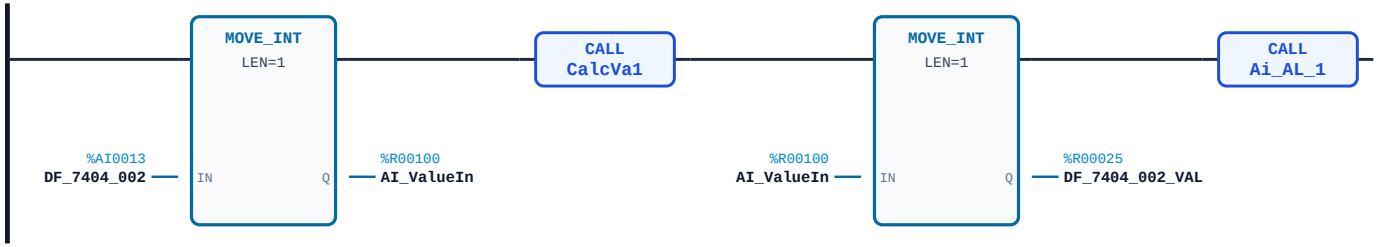
Rung 38



Rung 39

```
%AI0013.  Generate alarms and fults.
Alarm will reset if
- Pumps have been running for 60 minutes
- "Reset Fire&Gas" pushed.
- "Stop Foam" pushed.
```

Rung 40



Rung 41



Rung 42



Rung 43



Rung 44



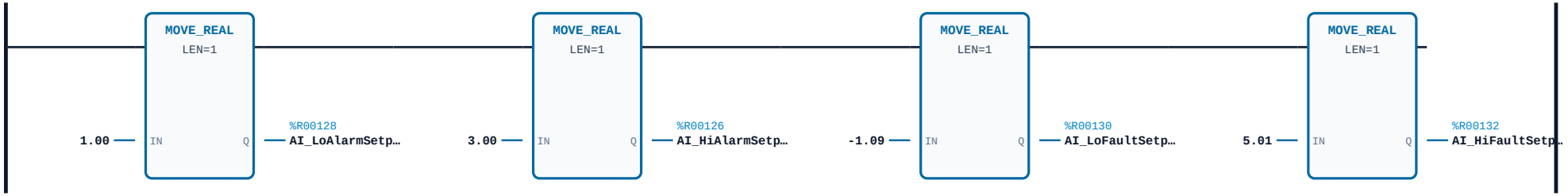
Rung 45



Rung 46

```
%AI0014.  Range -1,25 - 5,00 (0-5,00 LELm = 4-20 mA).
First set alarm limits for this spesific analog input.
```

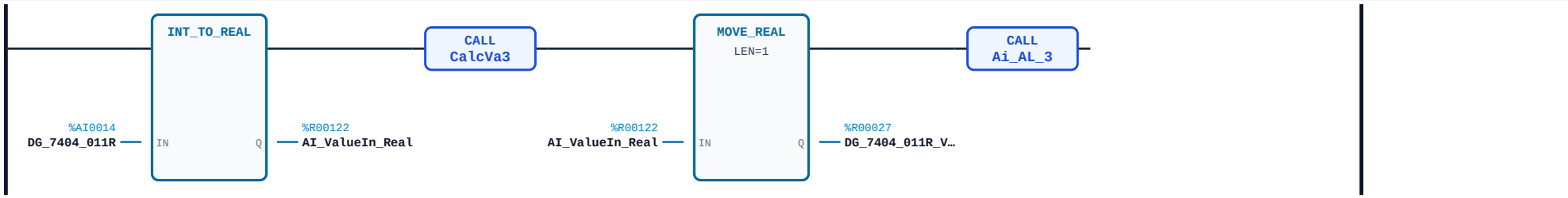
Rung 47



Rung 48

```
%AI0014.  Generate alarms and fults.
```

Rung 49



Rung 50



Rung 51

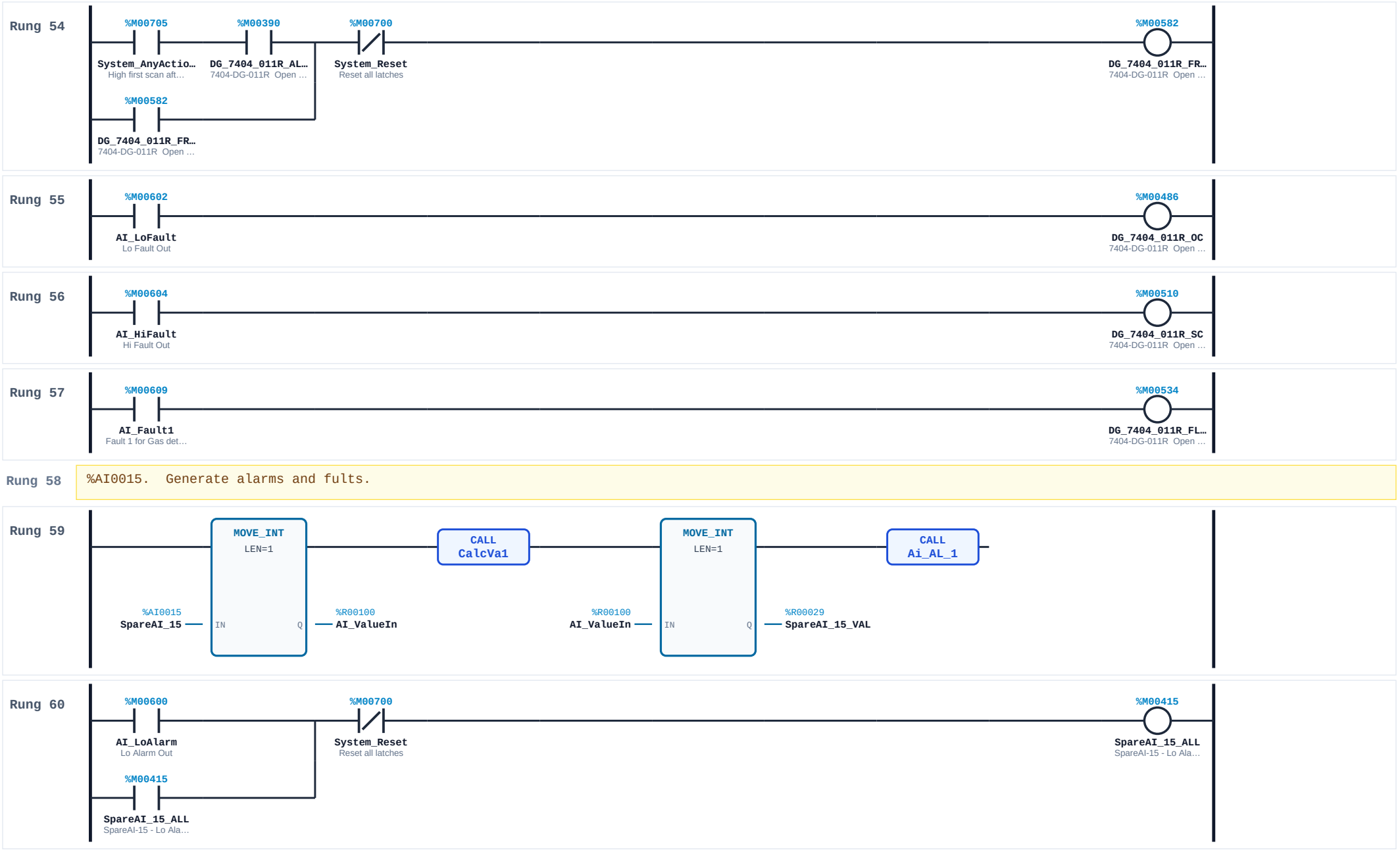


Rung 52



Rung 53




Rung 54
%M00705
System_AnyActio…
High first scan aft…
%M00390
DG_7404_011R_AL…
7404-DG-011R Open …
%M00700
System_Reset
Reset all latches
%M00582
DG_7404_011R_FR…
7404-DG-011R Open …
%M00582
DG_7404_011R_FR…
7404-DG-011R Open …
Rung 55
%M00602
AI_LoFault
Lo Fault Out
%M00486
DG_7404_011R_OC
7404-DG-011R Open …
Rung 56
%M00604
AI_HiFault
Hi Fault Out
%M00510
DG_7404_011R_SC
7404-DG-011R Open …
Rung 57
%M00609
AI_Fault1
Fault 1 for Gas det…
%M00534
DG_7404_011R_FL…
7404-DG-011R Open …
Rung 58
%AI0015. Generate alarms and fults.
Rung 59
MOVE_INT
LEN=1
%AI0015
SpareAI_15
IN
Q
%R00100
AI_ValueIn
CALL
CalcVa1
MOVE_INT
LEN=1
%R00100
AI_ValueIn
IN
Q
%R00029
SpareAI_15_VAL
CALL
Ai_AL_1
Rung 60
%M00600
AI_LoAlarm
Lo Alarm Out
%M00700
System_Reset
Reset all latches
%M00415
SpareAI_15_ALL
SpareAI-15 - Lo Ala…
%M00415
SpareAI_15_ALL
SpareAI-15 - Lo Ala…

Rung 61



Rung 62



Rung 63



Rung 64



Rung 65

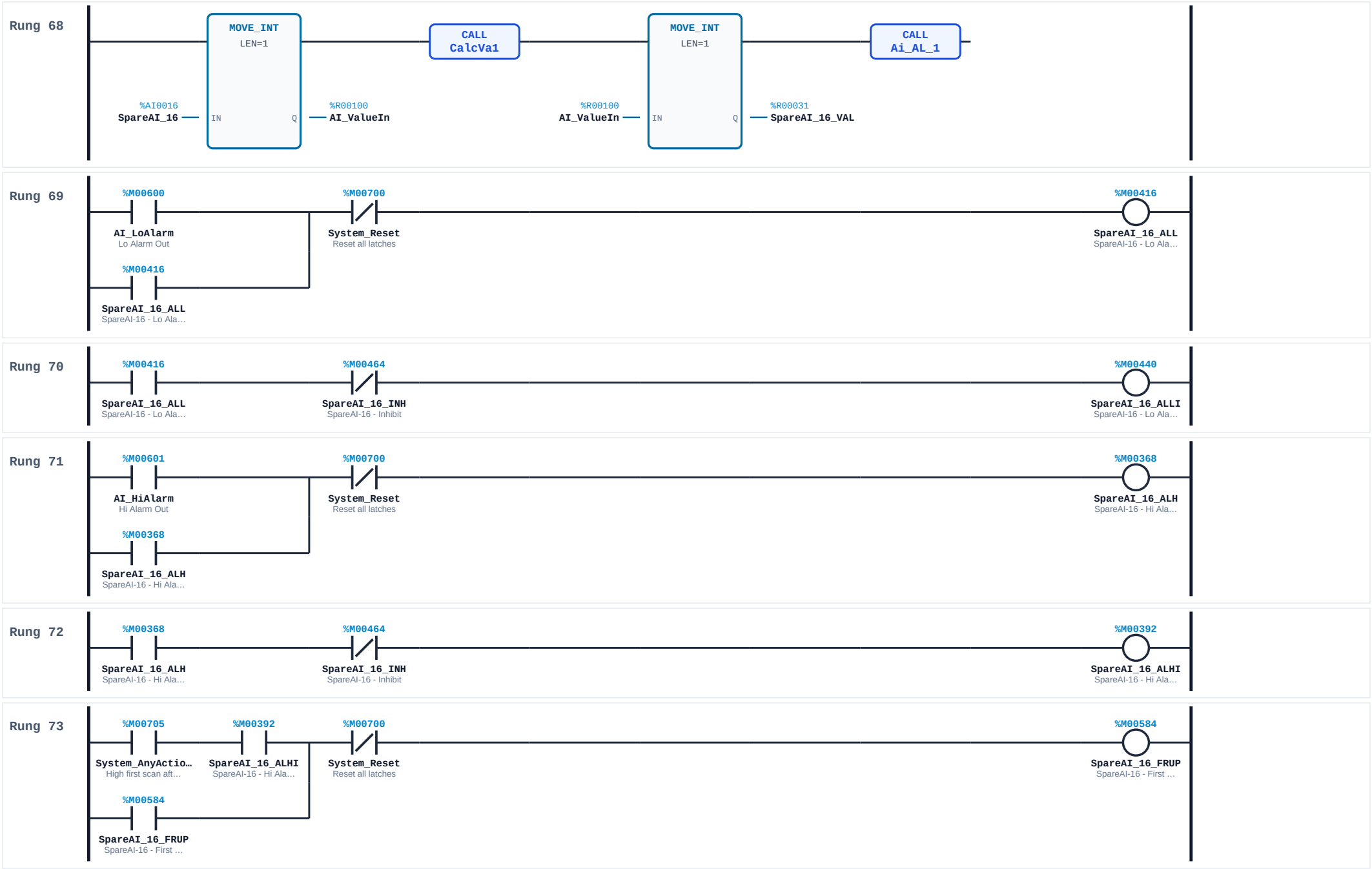


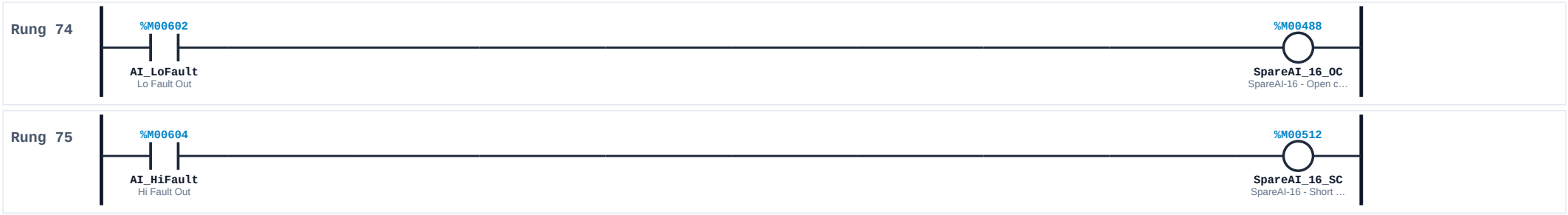
Rung 66



Rung 67

```
%AI0016.  Generate alarms and fults.
```


Rung 68
MOVE_INT
LEN=1
%AI0016
SpareAI_16
IN
Q
%R00100
AI_ValueIn
CALL
CalcVa1
MOVE_INT
LEN=1
%R00100
AI_ValueIn
IN
Q
%R00031
SpareAI_16_VAL
CALL
Ai_AL_1
Rung 69
%M00600
AI_LoAlarm
Lo Alarm Out
%M00700
System_Reset
Reset all latches
%M00416
SpareAI_16_ALL
SpareAI-16 - Lo Ala…
%M00416
SpareAI_16_ALL
SpareAI-16 - Lo Ala…
%M00416
SpareAI_16_ALL
SpareAI-16 - Lo Ala…
%M00464
SpareAI_16_INH
SpareAI-16 - Inhibit
%M00440
SpareAI_16_ALLI
SpareAI-16 - Lo Ala…
Rung 71
%M00601
AI_HiAlarm
Hi Alarm Out
%M00700
System_Reset
Reset all latches
%M00368
SpareAI_16_ALH
SpareAI-16 - Hi Ala…
%M00368
SpareAI_16_ALH
SpareAI-16 - Hi Ala…
Rung 72
%M00368
SpareAI_16_ALH
SpareAI-16 - Hi Ala…
%M00464
SpareAI_16_INH
SpareAI-16 - Inhibit
%M00392
SpareAI_16_ALHI
SpareAI-16 - Hi Ala…
Rung 73
%M00705
System_AnyActio…
High first scan aft…
%M00392
SpareAI_16_ALHI
SpareAI-16 - Hi Ala…
%M00700
System_Reset
Reset all latches
%M00584
SpareAI_16_FRUP
SpareAI-16 - First …
%M00584
SpareAI_16_FRUP
SpareAI-16 - First …


Rung 74
%M00602
AI_LoFault
Lo Fault Out
%M00488
SpareAI_16_OC
SpareAI-16 - Open c...
Rung 75
%M00604
AI_HiFault
Hi Fault Out
%M00512
SpareAI_16_SC
SpareAI-16 - Short ...

LD BLOCK: AI_3

Calls: Ai_AL_1, CalcVa2, CalcVa1, Ai_AL_2

Rung 1

```
This routine handles the inputs connected to the IC200ALG262 analogue input module in PLC slot 5.  Channel %AI0017 to
%AI0024.
```

Rung 2

```
History on changes after 23.april-03:
- 28.juni-03:  %AI_17, changed LoAlarm setpoint from 60 to 50.   Vidar Pedersen
- 28.juni-03:  %AI_19 (PST_7503_001), Short circuit alarm limit changed from 98 to 102.   Vidar Pedersen
```

Rung 3

```
Standard psaudo code:

Move Analogue input value to VALUE_IN
Call CALCVAL                          //Converts input value.
_VAL = VALUE_IN
Call AI_ALARM                         //Calculates low and high alarm and fault

if LOALARM then latch _ALL bit until system reset
_ALLI = (_ALL and not _INH)
if HIALARM then latch _ALH bit until system reset
_ALHI = (_ALH and not _INH)
If (Any_actions and _ALHI) then latch _FRUP until system reset
_SC  = HIFAULT
_OC = LOFAULT


Psaudo code for digital detectors on analogue inputs:
(No scaling of input value, and only high alarm.)

Set special alarm limits
Move Analogue input value to VALUE_IN
Call AI_ALARM                         //Calculates low and high alarm and fault

_ALL = 0
_ALLI = 0
if HIALARM then latch _ALH bit until system reset
_ALHI = (_ALH and not _INH)
If (Any_actions and _ALHI) then latch _FRUP until system reset
_SC  = HIFAULT
_OC = LOFAULT
```

Rung 4

```
%AI0017.  Range -25 - 100 (0-100 Psi = 4-20 mA).
First set alarm limits for this spesific analog input.   This input will never set Hi Alarm bit because of this is set
higher than hi range.
```

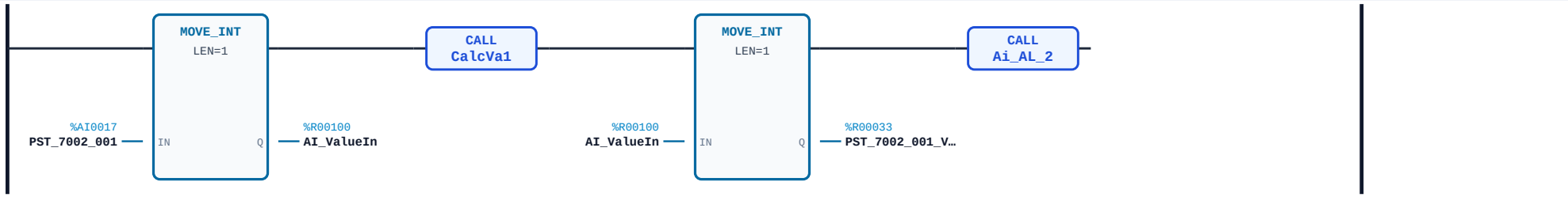
Rung 5



Rung 6

```
%AI0017.  Generate alarms and fults.
```

Rung 7



Rung 8



Rung 9



Rung 10



Rung 11



Rung 12



Rung 13



Rung 14



Rung 15

```
%AI0018.  Range -37 - 150 (0-150 Psi = 4-20 mA).
First set alarm limits for this spesific analog input.   This input will never set Lo Alarm bit because of this is set
higher than hi range.
```

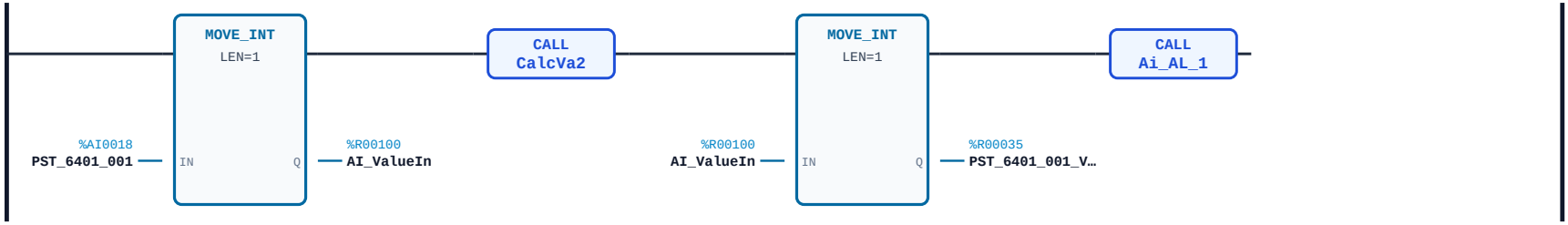
Rung 16

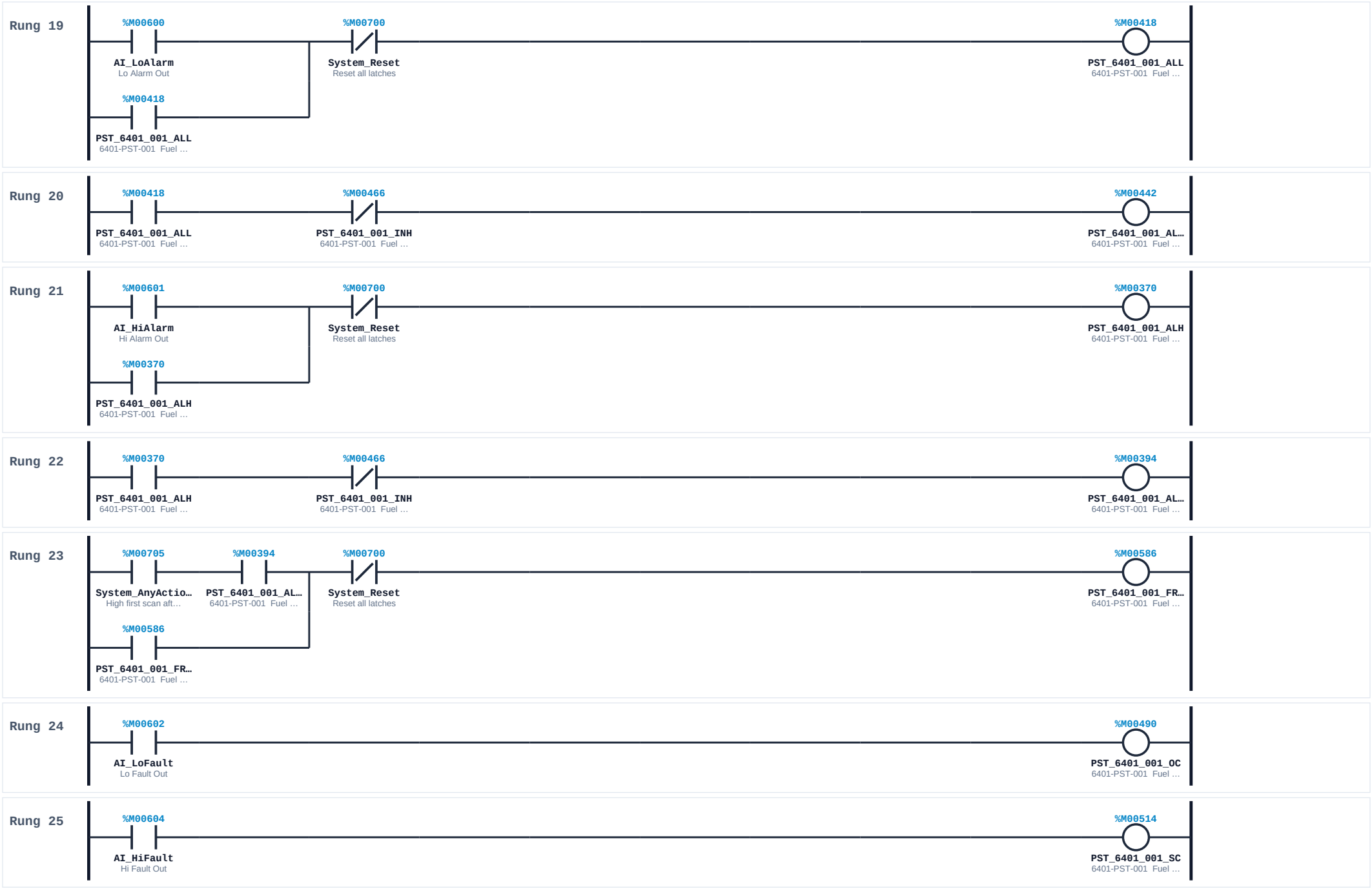


Rung 17

```
%AI0018.  Generate alarms and fults.
```

Rung 18




Rung 19
%M00600
AI_LoAlarm
Lo Alarm Out
%M00700
System_Reset
Reset all latches
%M00418
PST_6401_001_ALL
6401-PST-001 Fuel ...
%M00418
PST_6401_001_ALL
6401-PST-001 Fuel ...
Rung 20
%M00418
PST_6401_001_ALL
6401-PST-001 Fuel ...
%M00466
PST_6401_001_INH
6401-PST-001 Fuel ...
%M00442
PST_6401_001_AL...
6401-PST-001 Fuel ...
Rung 21
%M00601
AI_HiAlarm
Hi Alarm Out
%M00700
System_Reset
Reset all latches
%M00370
PST_6401_001_ALH
6401-PST-001 Fuel ...
%M00370
PST_6401_001_ALH
6401-PST-001 Fuel ...
Rung 22
%M00370
PST_6401_001_ALH
6401-PST-001 Fuel ...
%M00466
PST_6401_001_INH
6401-PST-001 Fuel ...
%M00394
PST_6401_001_AL...
6401-PST-001 Fuel ...
Rung 23
%M00705
System_AnyActio...
High first scan aft...
%M00394
PST_6401_001_AL...
6401-PST-001 Fuel ...
%M00700
System_Reset
Reset all latches
%M00586
PST_6401_001_FR...
6401-PST-001 Fuel ...
%M00586
PST_6401_001_FR...
6401-PST-001 Fuel ...
Rung 24
%M00602
AI_LoFault
Lo Fault Out
%M00490
PST_6401_001_OC
6401-PST-001 Fuel ...
Rung 25
%M00604
AI_HiFault
Hi Fault Out
%M00514
PST_6401_001_SC
6401-PST-001 Fuel ...

Rung 26

%AI0019. Range -25 - 100 (0-100 Psi = 4-20 mA).
First set alarm limits for this spesific analog input. This input will never set Lo Alarm bit because of this is set higher than hi range.

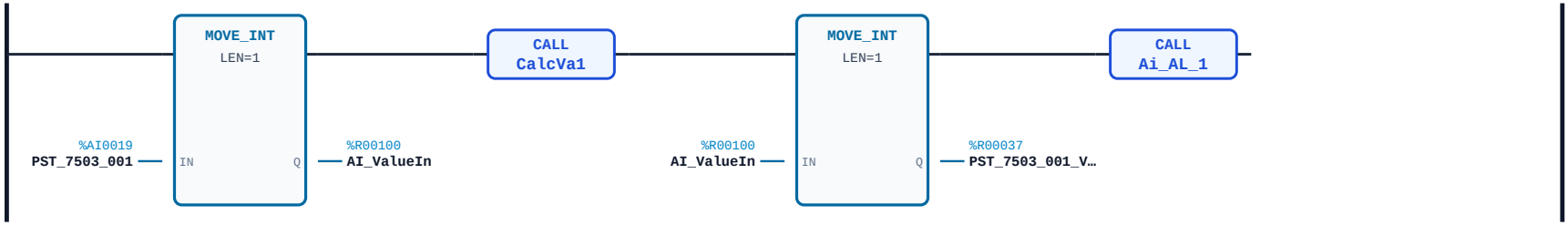
Rung 27



Rung 28

%AI0019. Generate alarms and fults.

Rung 29



Rung 30



Rung 31



Rung 32



Rung 33



Rung 34



Rung 35



Rung 36



Rung 37

```
%AI0020.  Range -25 - 100 (0-100 % = 4-20 mA).
First set alarm limits for this spesific analog input.  This input will never set Lo Alarm bit because of this is set
higher than hi range.
```

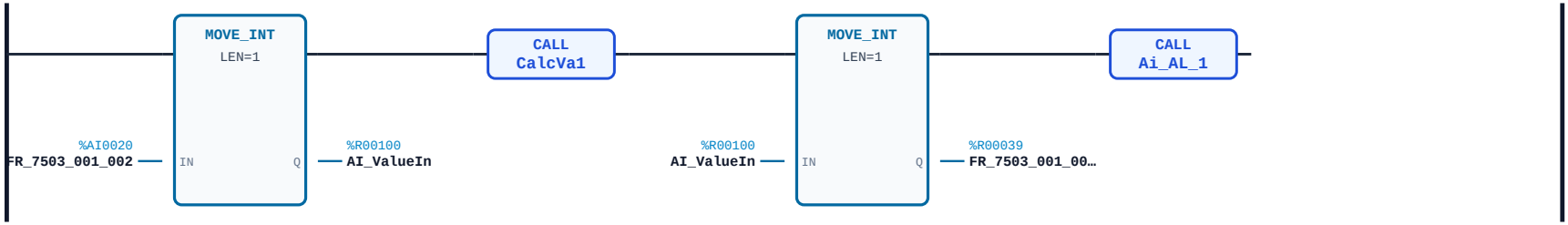
Rung 38



Rung 39

```
%AI0020.  Generate alarms and fults.
Alarm will reset if
- Pumps have been running for 60 minutes
- "Reset Fire&Gas" pushed.
- "Stop Foam" pushed.
```

Rung 40



Rung 41



Rung 42



Rung 43



Rung 44



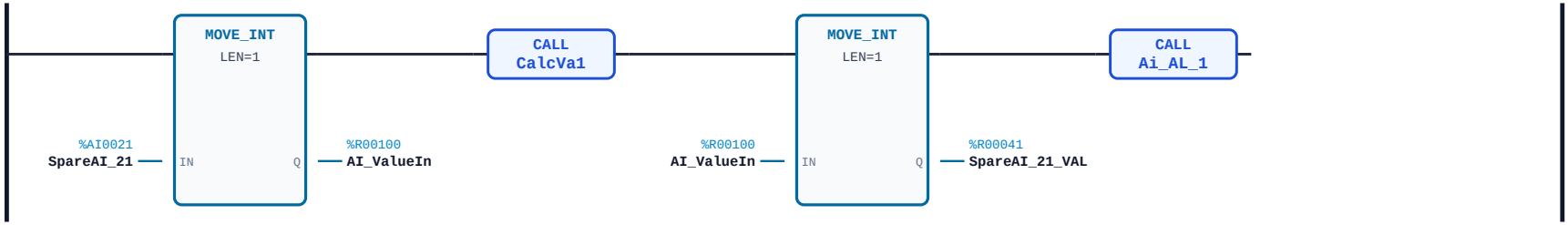
Rung 45



Rung 46

```
%AI0021.  Generate alarms and fults.
```

Rung 47



Rung 48



Rung 49



Rung 50



Rung 51



Rung 52



Rung 53



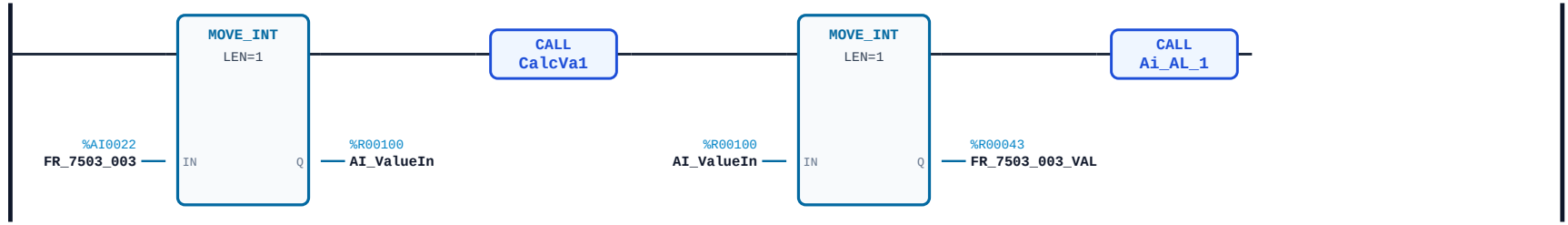
Rung 54



Rung 55

```
%AI0022.  Generate alarms and fults.
Alarm will reset if
- Pumps have been running for 60 minutes
- "Reset Fire&Gas" pushed.
- "Stop Foam" pushed.
```

Rung 56



Rung 57



Rung 58



Rung 59



Rung 60



Rung 61



LD BLOCK: CalcVa1

Rung 1

This routine converts input value from 0 to 32000 to -25 to 100.

Rung 2

Clearing internal register.

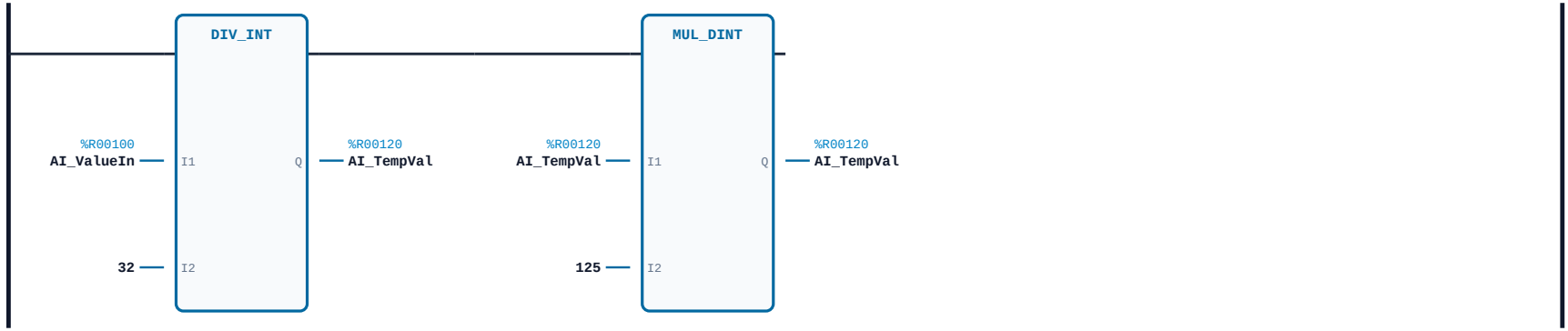
Rung 3



Rung 4

Convert analog input range 0-32000 to 0-125000 range.

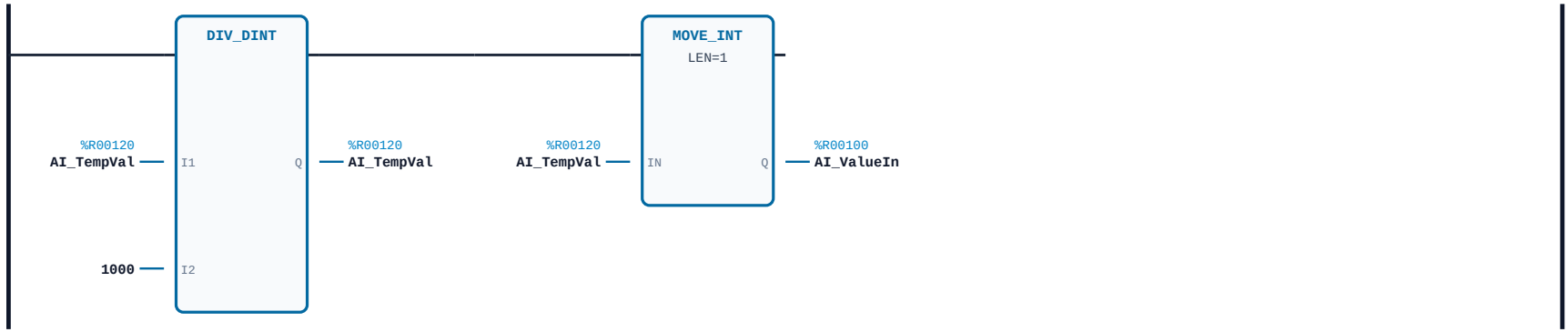
Rung 5



Rung 6

Convert from 125000 range to 125 range. This to display percent value with one desimal.

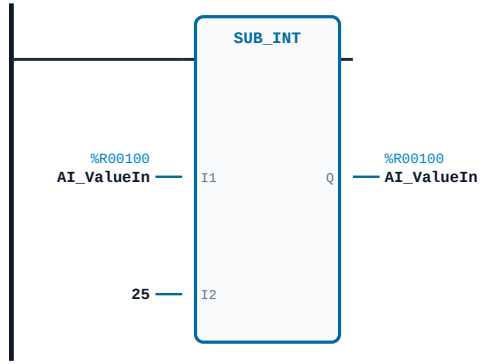
Rung 7



Rung 8

Convert from 0-125 range to -25 to 100 range. This is percent value with one decimal point.

Rung 9



LD BLOCK: CalcVa2

Rung 1

This routine converts input value from 0 to 32000 to -37 to 150 (0 - 150 Psi v/ 4-20 mA)

Rung 2

Clearing internal register.

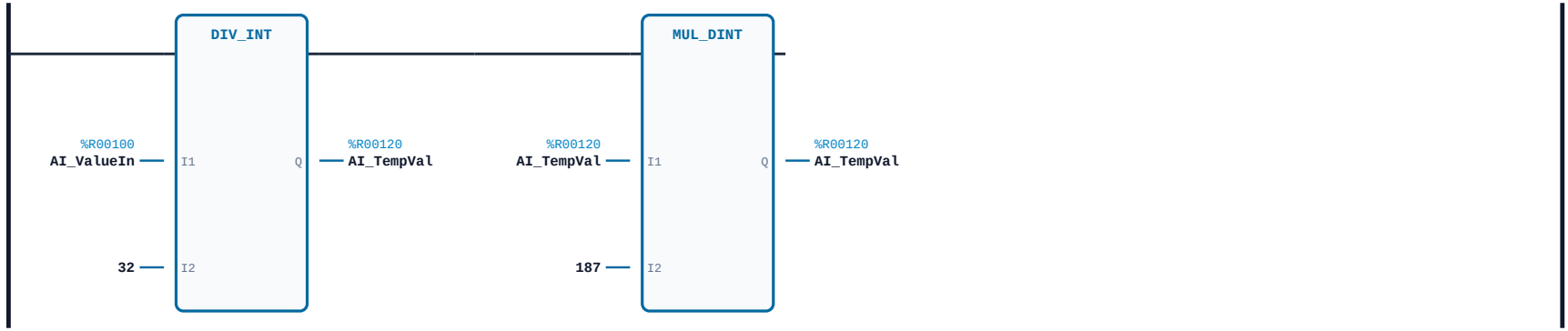
Rung 3



Rung 4

Convert analog input range 0-32000 to 0-187000 range.

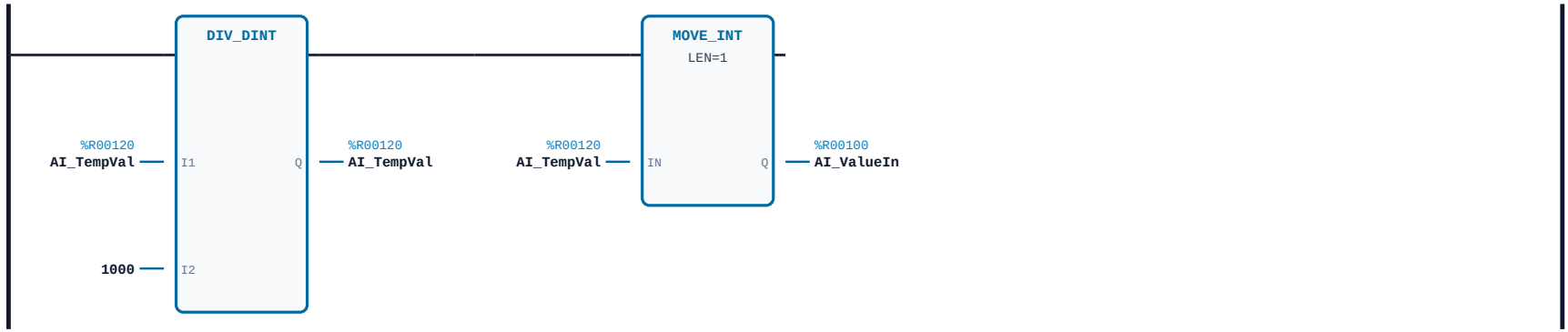
Rung 5



Rung 6

Convert from 187000 range to 187 range. This to display percent value.

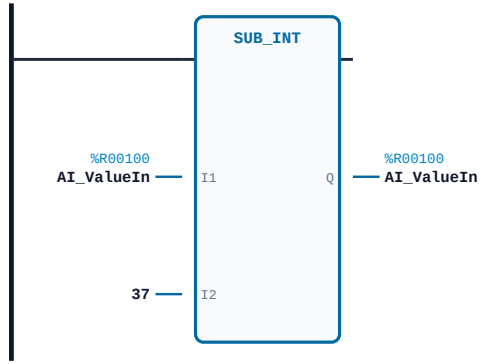
Rung 7



Rung 8

Convert from 0-187 range to -37 to 150 range. This is value (Psi, etc...).

Rung 9



LD BLOCK: CalcVa3

Rung 1

This routine converts input value from 0 to 32000 to -1,25 to 5,00 (0 - 5,00 LELm v/ 4-20 mA)

Rung 2

Clearing internal register.

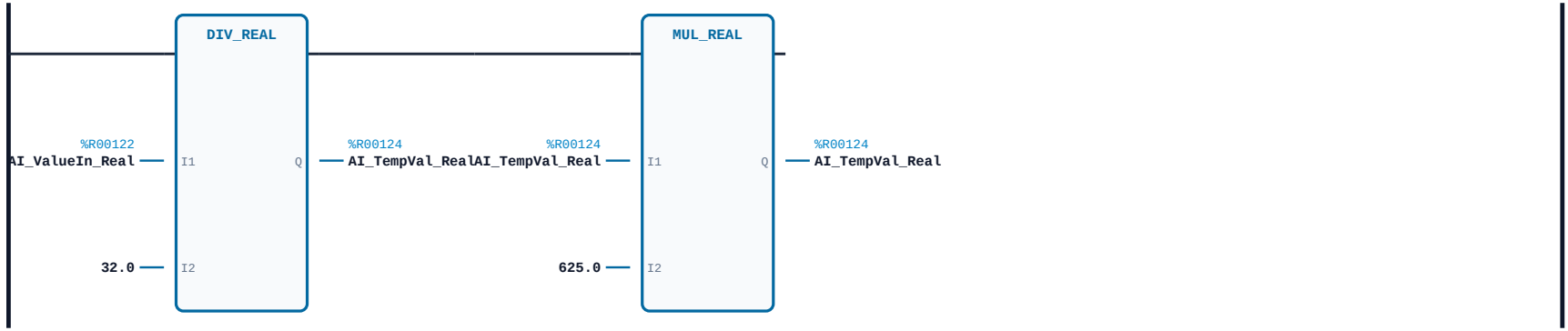
Rung 3



Rung 4

Convert analog input range 0-32000 to 0-625000 range.

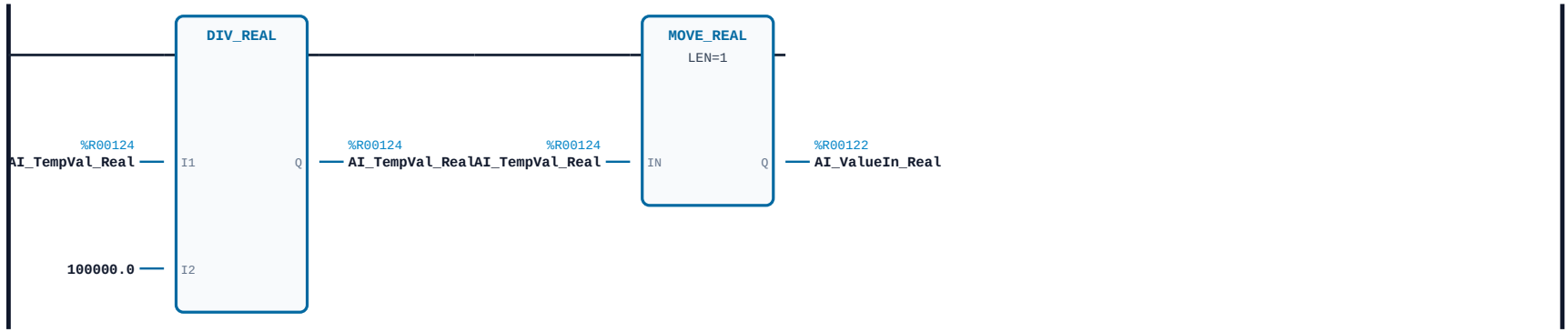
Rung 5



Rung 6

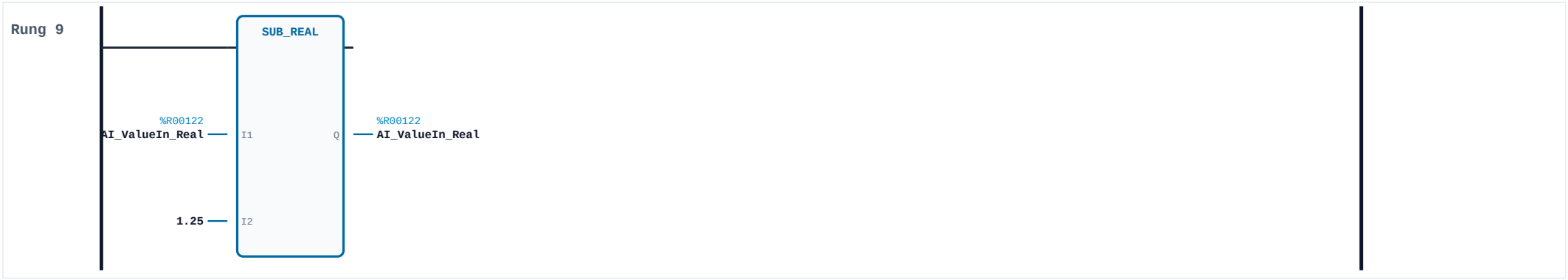
Convert from 625000 range to 6,25 range. This to display percent value with two desimals.

Rung 7



Rung 8

Convert from 0-6,25 range to -1,25 to 5,00 range. This is value (LELm, etc...) with two decimals.

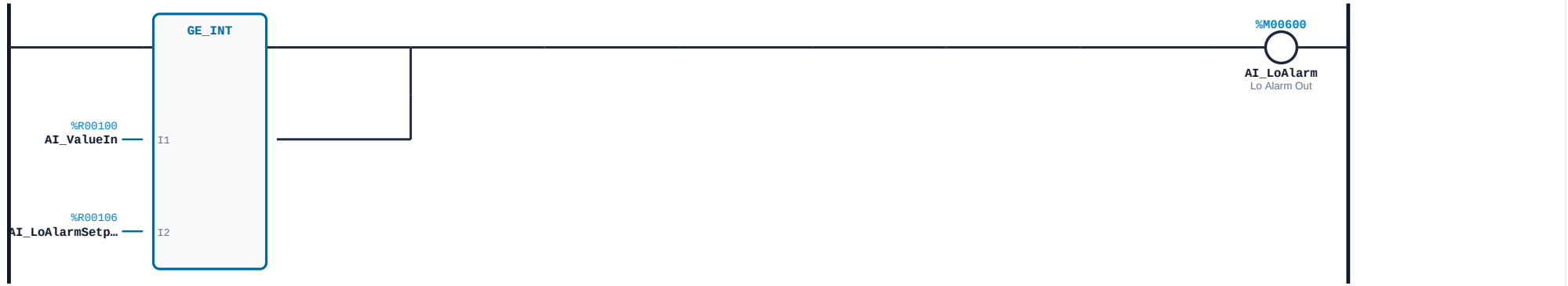

Rung 9
SUB_REAL
%R00122
AI_ValueIn_Real
I1
Q
%R00122
AI_ValueIn_Real
1.25
I2

LD BLOCK: Ai_AL_1

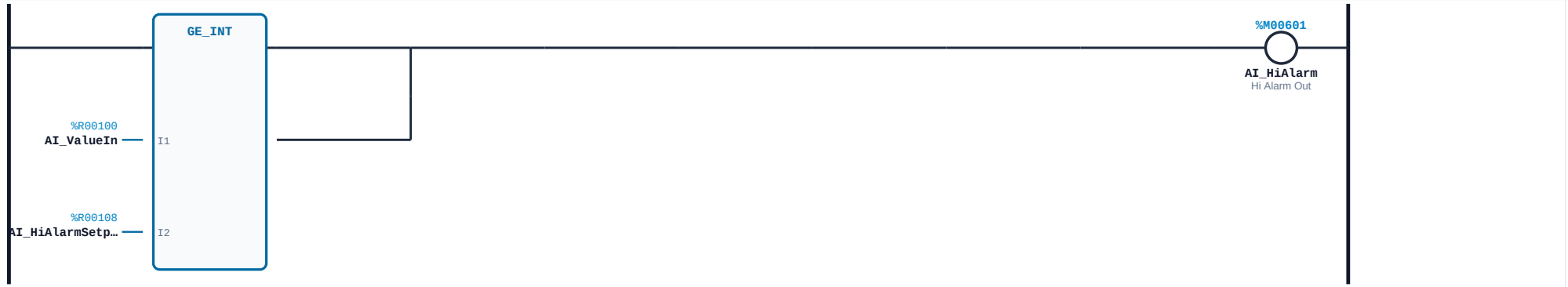
Rung 1

```
This routine calculates the Lo Alarm, Hi Alarm, Hi Fault(Short circuit) and Lo Fault(Open circuit).
- Lo Alarm when Value In is greater or equal to Lo alarm setpoint.
- Hi Alarm when Value In is greater or equal to Hi alarm setpoint.
```

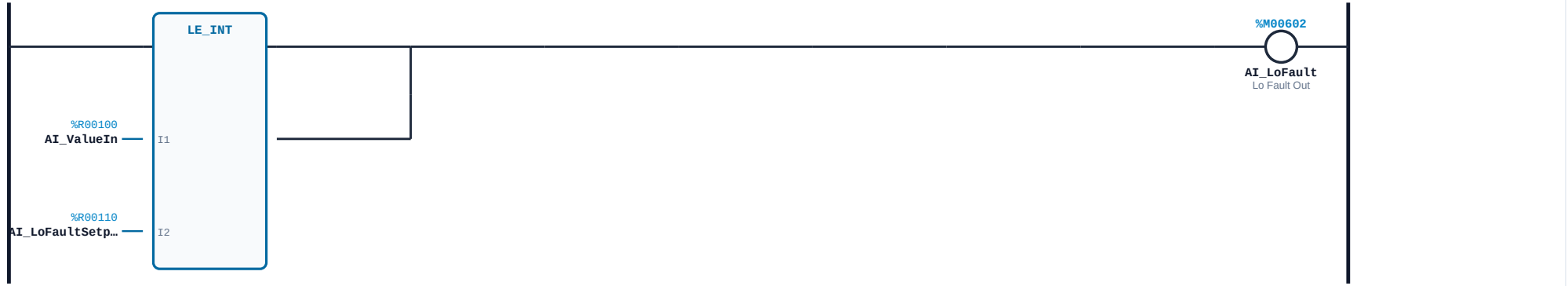
Rung 2

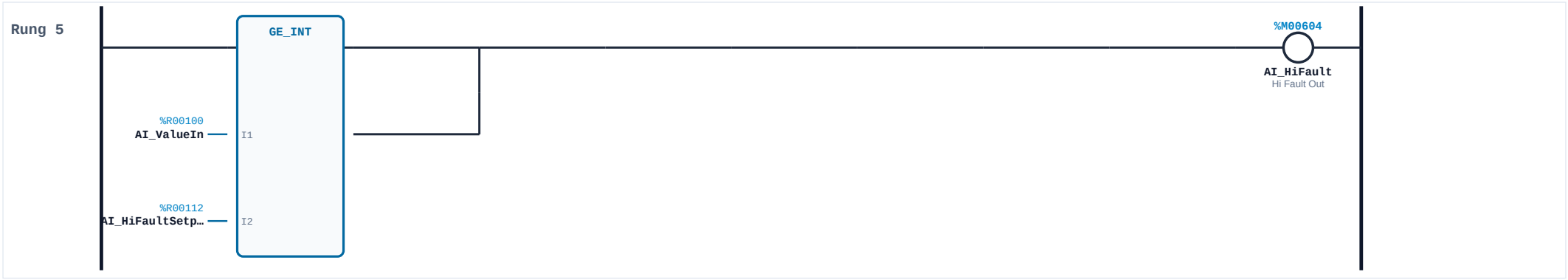


Rung 3



Rung 4



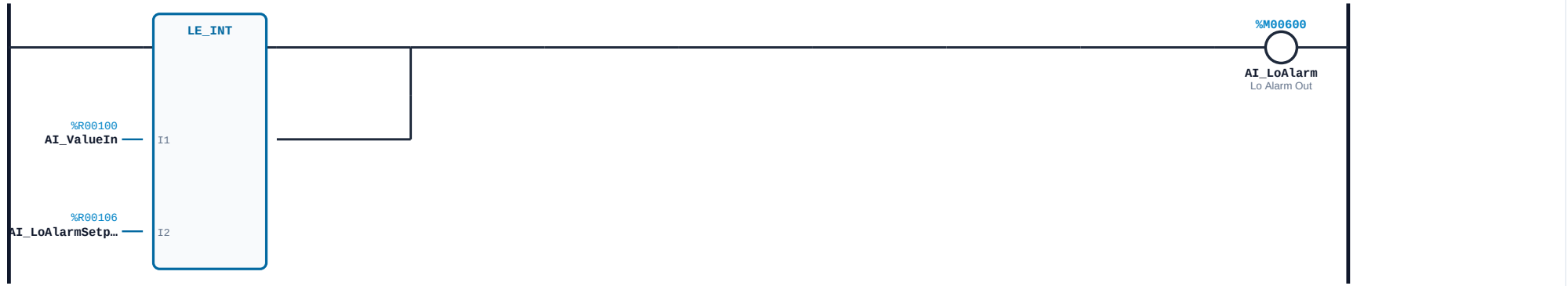

Rung 5
GE_INT
%R00100
AI_ValueIn
I1
%R00112
AI_HiFaultSetp…
I2
%M00604
AI_HiFault
Hi Fault Out

LD BLOCK: Ai_AL_2

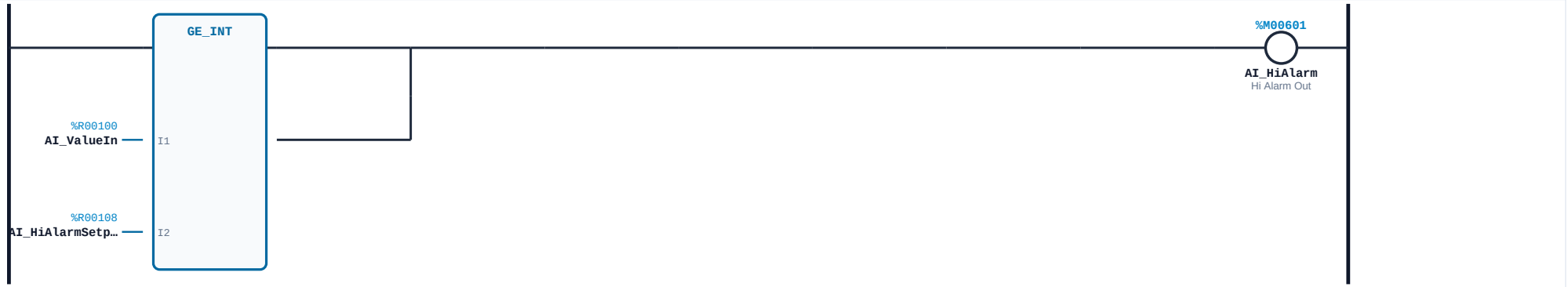
Rung 1

```
This routine calculates the Lo Alarm, Hi Alarm, Hi Fault(Short circuit) and Lo Fault(Open circuit).
- Lo Alarm when Value In is less or equal to Lo alarm setpoint.
- Hi Alarm when Value In is greater or equal to Hi alarm setpoint.
```

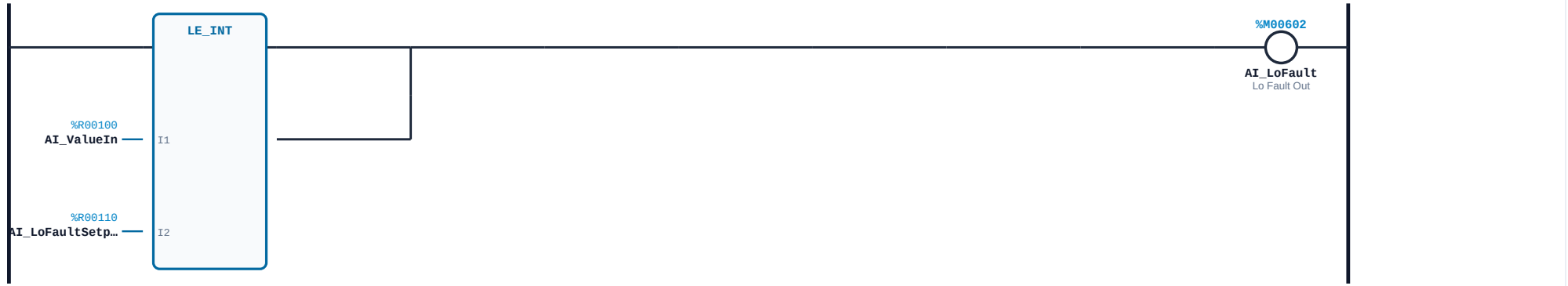
Rung 2

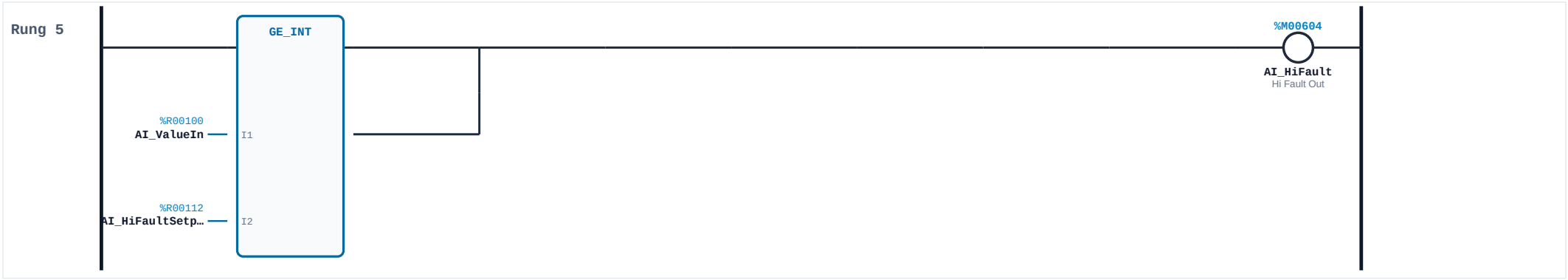


Rung 3



Rung 4



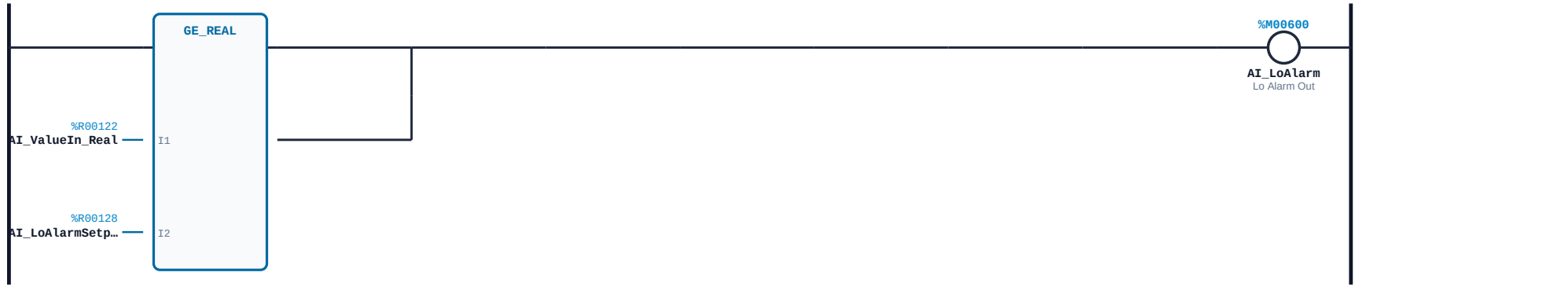

Rung 5
GE_INT
%R00100
AI_ValueIn
I1
%R00112
AI_HiFaultSetp…
I2
%M00604
AI_HiFault
Hi Fault Out

LD BLOCK: Ai_AL_3

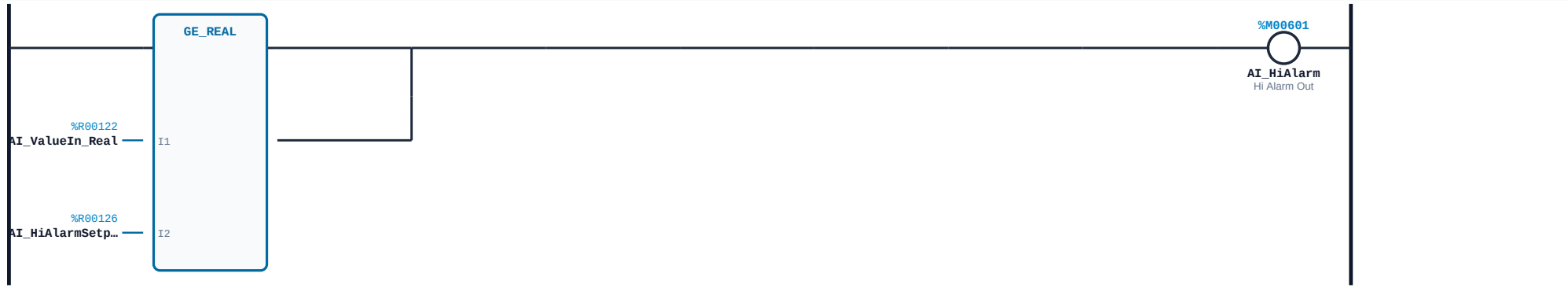
Rung 1

```
FOR REAL VALUES:  This routine calculates the Lo Alarm, Hi Alarm, Hi Fault(Short circuit) and Lo Fault(Open circuit).
- Lo Alarm when Value In is greater or equal to Lo alarm setpoint.
- Hi Alarm when Value In is greater or equal to Hi alarm setpoint.
```

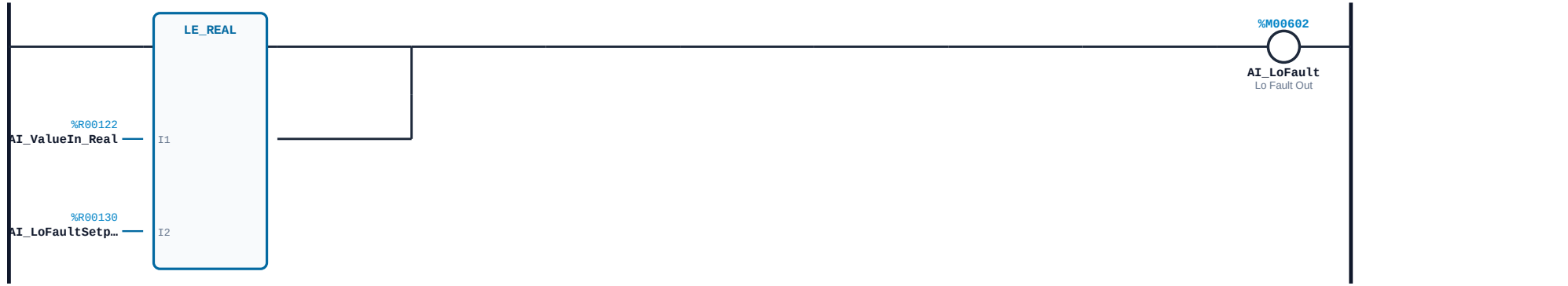
Rung 2

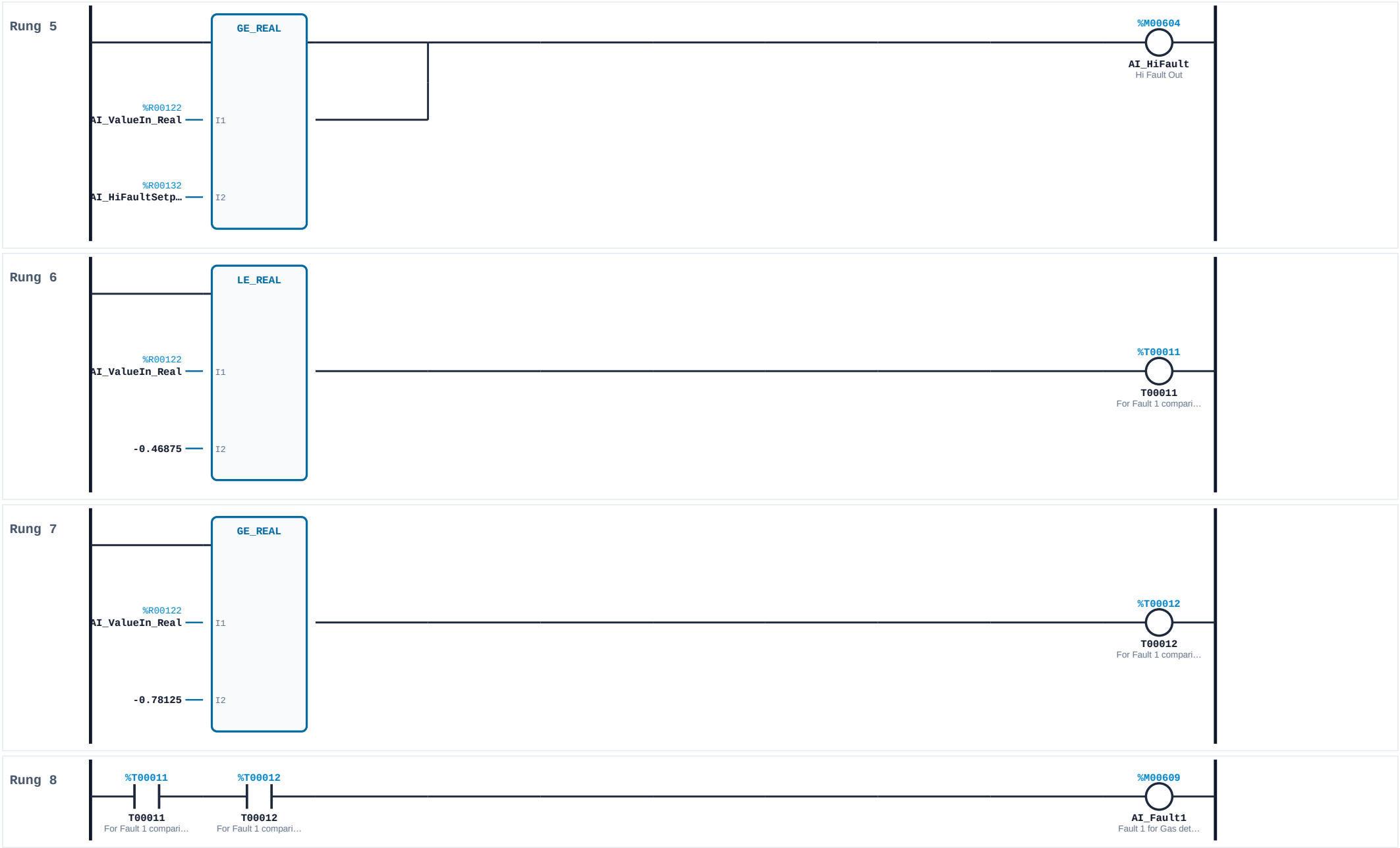


Rung 3



Rung 4



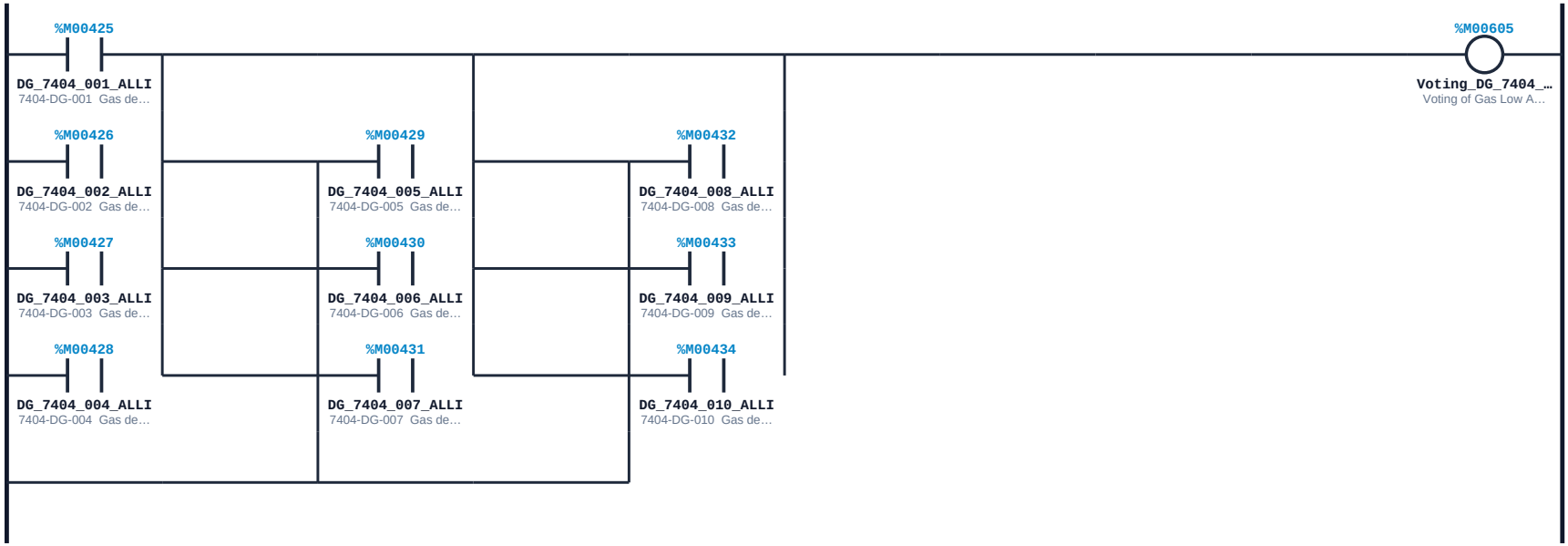

Rung 5
GE_REAL
%R00122
AI_ValueIn_Real
I1
%R00132
AI_HiFaultSetp…
I2
%M00604
AI_HiFault
Hi Fault Out
Rung 6
LE_REAL
%R00122
AI_ValueIn_Real
I1
-0.46875
I2
%T00011
T00011
For Fault 1 compari…
Rung 7
GE_REAL
%R00122
AI_ValueIn_Real
I1
-0.78125
I2
%T00012
T00012
For Fault 1 compari…
Rung 8
%T00011
T00011
For Fault 1 compari…
%T00012
T00012
For Fault 1 compari…
%M00609
AI_Fault1
Fault 1 for Gas det…

LD BLOCK: Vote_1

Rung 1

Voting of Low Alarm on Gas detectors pump house (ALLI).

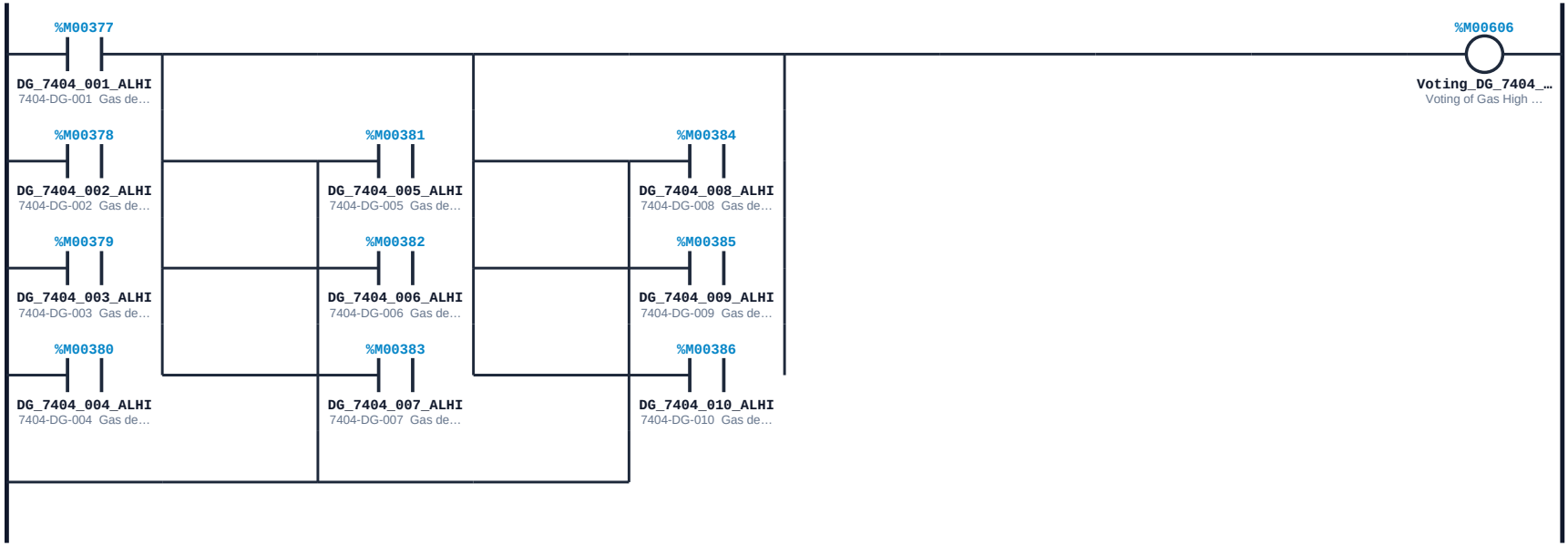
Rung 2



Rung 3

Voting of High Alarm on Gas detectors pump house (ALHI).

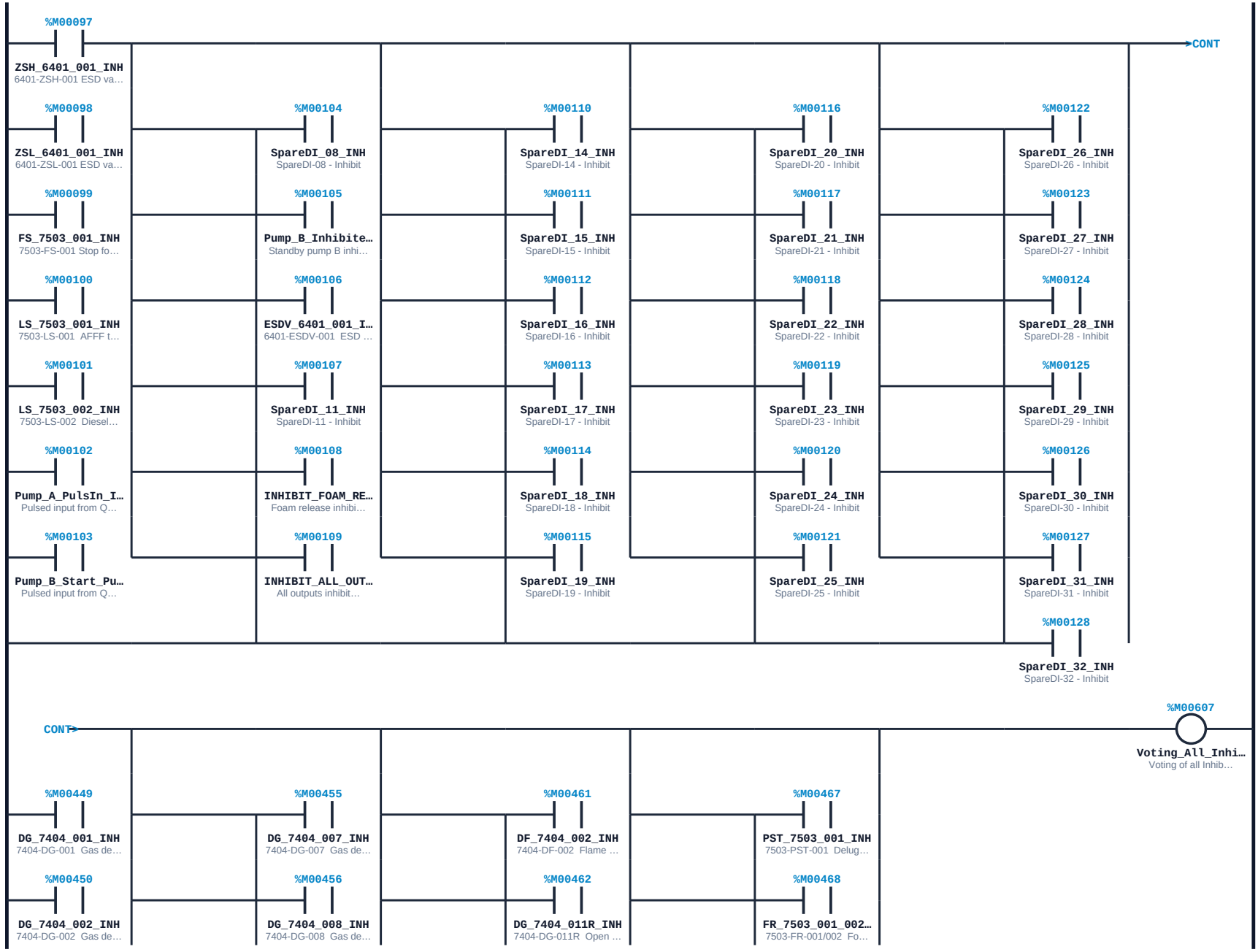
Rung 4



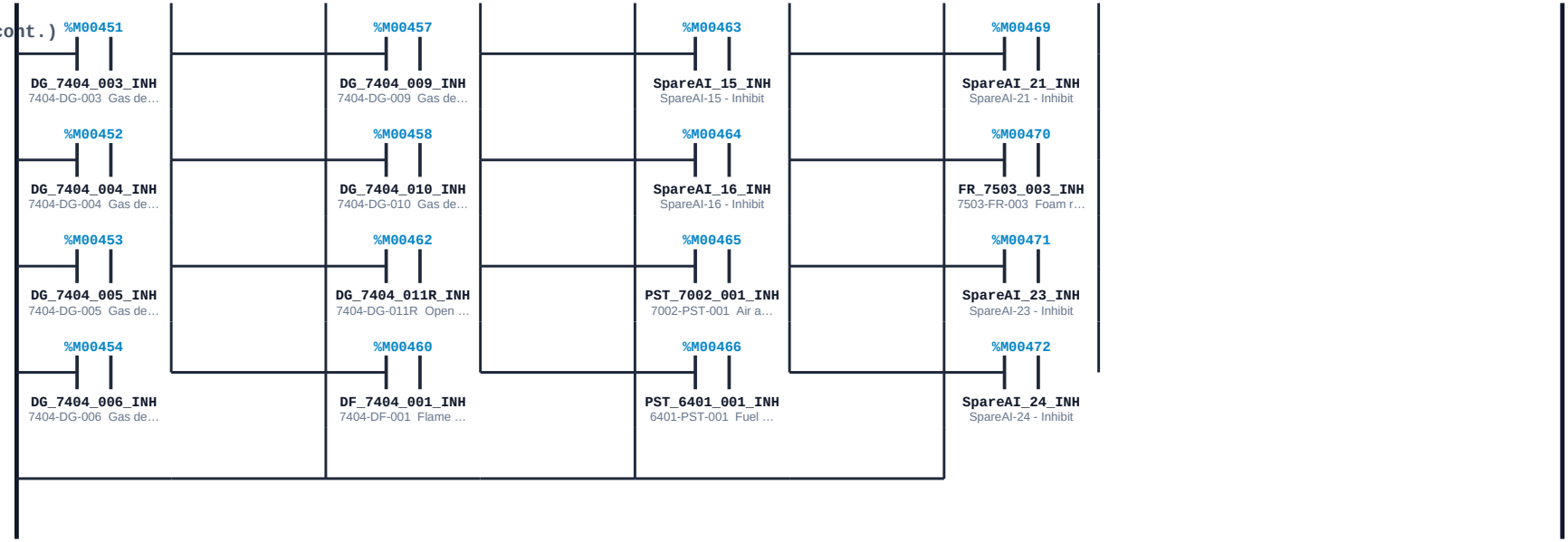
Rung 5

```
Voting of all Inhibits from all inputs.
```

Rung 6


%M00097
ZSH_6401_001_INH
6401-ZSH-001 ESD va…
%M00098
ZSL_6401_001_INH
6401-ZSL-001 ESD va…
%M00099
FS_7503_001_INH
7503-FS-001 Stop fo…
%M00100
LS_7503_001_INH
7503-LS-001 AFFF t…
%M00101
LS_7503_002_INH
7503-LS-002 Diesel…
%M00102
Pump_A_PulsIn_I…
Pulsed input from Q…
%M00103
Pump_B_Start_Pu…
Pulsed input from Q…
%M00104
SpareDI_08_INH
SpareDI-08 - Inhibit
%M00105
Pump_B_Inhibite…
Standby pump B inhi…
%M00106
ESDV_6401_001_I…
6401-ESDV-001 ESD …
%M00107
SpareDI_11_INH
SpareDI-11 - Inhibit
%M00108
INHIBIT_FOAM_RE…
Foam release inhibi…
%M00109
INHIBIT_ALL_OUT…
All outputs inhibit…
%M00110
SpareDI_14_INH
SpareDI-14 - Inhibit
%M00111
SpareDI_15_INH
SpareDI-15 - Inhibit
%M00112
SpareDI_16_INH
SpareDI-16 - Inhibit
%M00113
SpareDI_17_INH
SpareDI-17 - Inhibit
%M00114
SpareDI_18_INH
SpareDI-18 - Inhibit
%M00115
SpareDI_19_INH
SpareDI-19 - Inhibit
%M00116
SpareDI_20_INH
SpareDI-20 - Inhibit
%M00117
SpareDI_21_INH
SpareDI-21 - Inhibit
%M00118
SpareDI_22_INH
SpareDI-22 - Inhibit
%M00119
SpareDI_23_INH
SpareDI-23 - Inhibit
%M00120
SpareDI_24_INH
SpareDI-24 - Inhibit
%M00121
SpareDI_25_INH
SpareDI-25 - Inhibit
%M00122
SpareDI_26_INH
SpareDI-26 - Inhibit
%M00123
SpareDI_27_INH
SpareDI-27 - Inhibit
%M00124
SpareDI_28_INH
SpareDI-28 - Inhibit
%M00125
SpareDI_29_INH
SpareDI-29 - Inhibit
%M00126
SpareDI_30_INH
SpareDI-30 - Inhibit
%M00127
SpareDI_31_INH
SpareDI-31 - Inhibit
%M00128
SpareDI_32_INH
SpareDI-32 - Inhibit
CONT
%M00607
Voting_All_Inhi…
Voting of all Inhib…
CONT
%M00449
DG_7404_001_INH
7404-DG-001 Gas de…
%M00450
DG_7404_002_INH
7404-DG-002 Gas de…
%M00455
DG_7404_007_INH
7404-DG-007 Gas de…
%M00456
DG_7404_008_INH
7404-DG-008 Gas de…
%M00461
DF_7404_002_INH
7404-DF-002 Flame …
%M00462
DG_7404_011R_INH
7404-DG-011R Open …
%M00467
PST_7503_001_INH
7503-PST-001 Delug…
%M00468
FR_7503_001_002…
7503-FR-001/002 Fo…

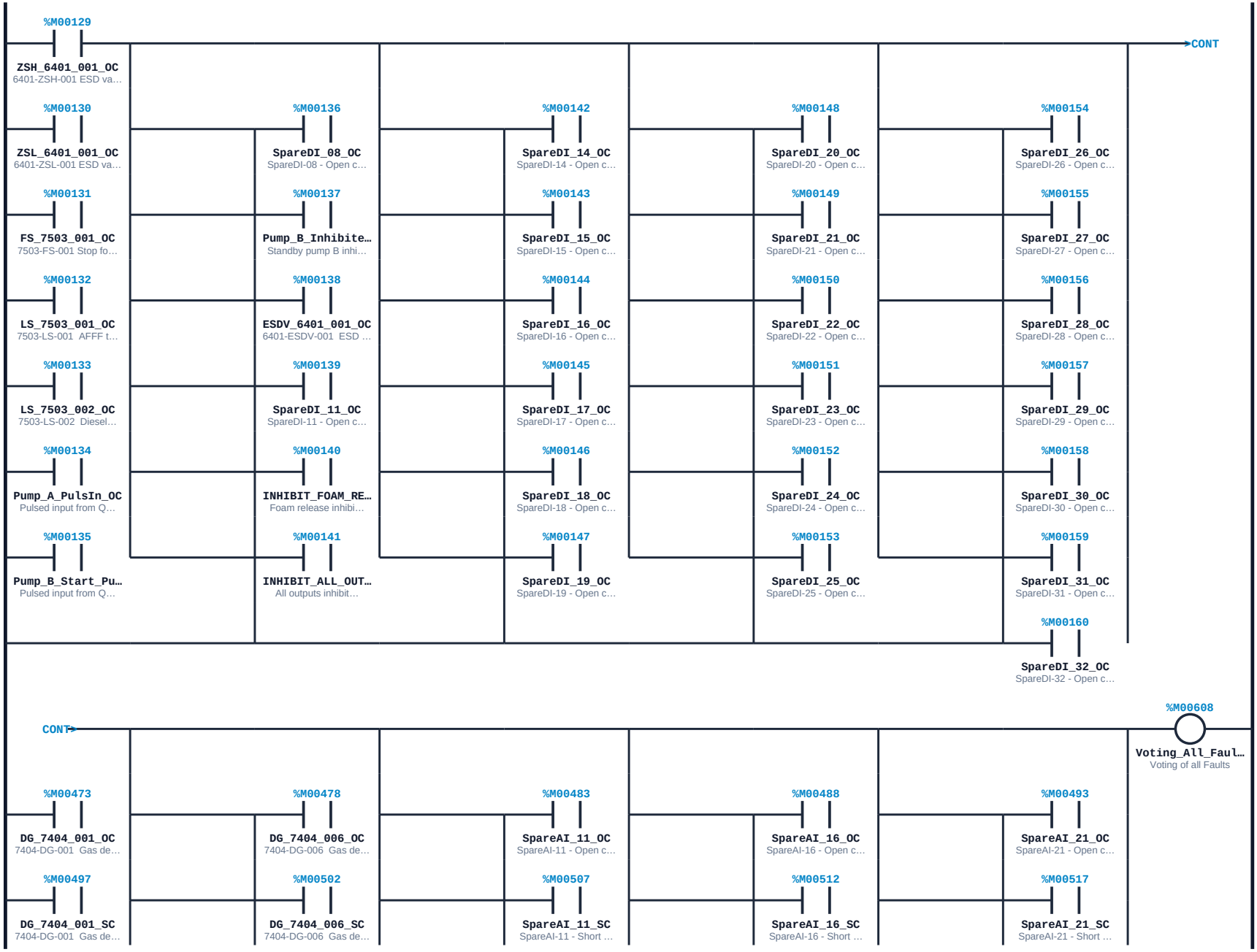
Rung 6 (cont.)



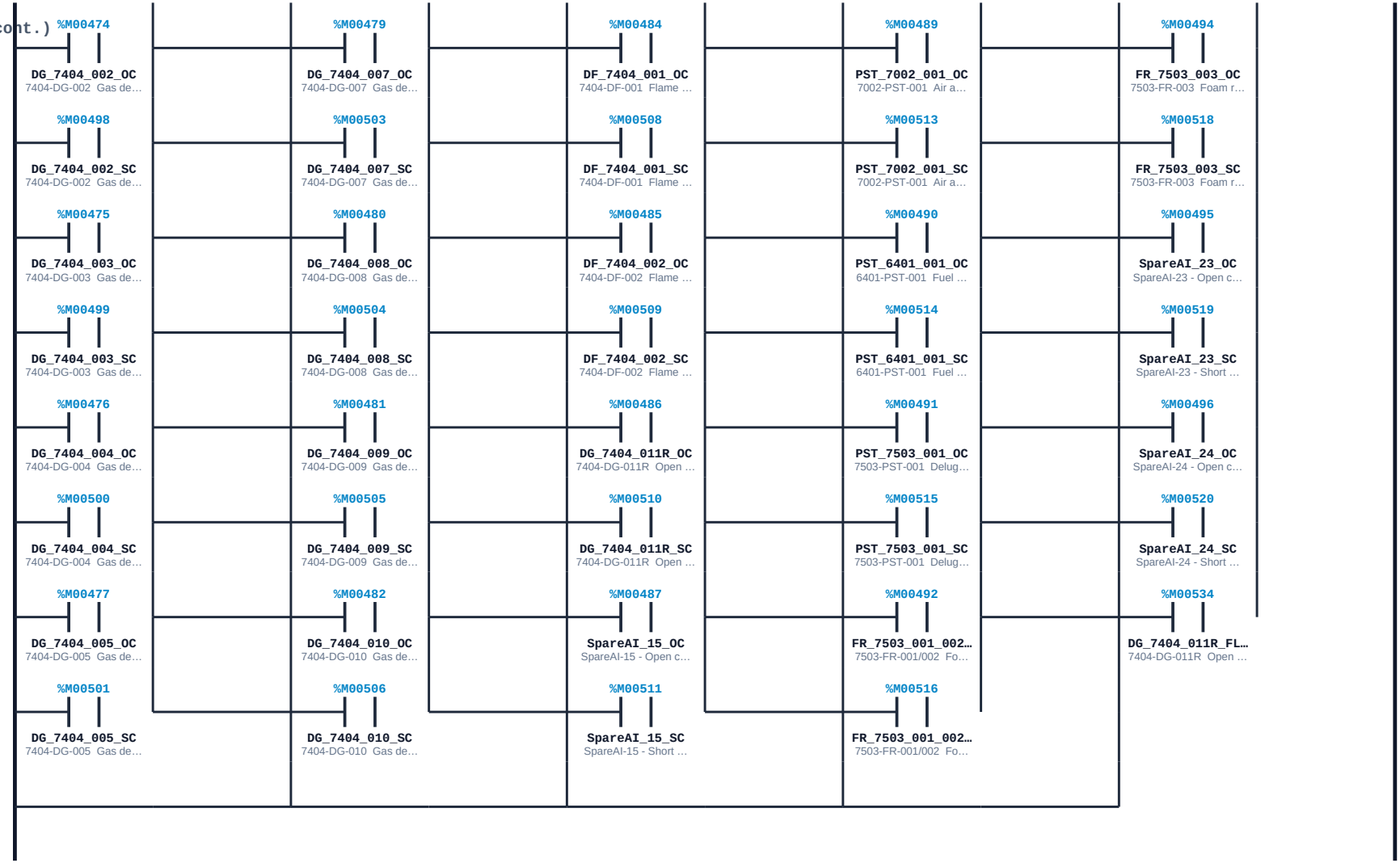
Rung 7

Voting of all Faults from all inputs.

Rung 8


%M00129
ZSH_6401_001_OC
6401-ZSH-001 ESD va…
%M00130
ZSL_6401_001_OC
6401-ZSL-001 ESD va…
%M00131
FS_7503_001_OC
7503-FS-001 Stop fo…
%M00132
LS_7503_001_OC
7503-LS-001 AFFF t…
%M00133
LS_7503_002_OC
7503-LS-002 Diesel…
%M00134
Pump_A_PulsIn_OC
Pulsed input from Q…
%M00135
Pump_B_Start_Pu…
Pulsed input from Q…
%M00136
SpareDI_08_OC
SpareDI-08 - Open c…
%M00137
Pump_B_Inhibite…
Standby pump B inhi…
%M00138
ESDV_6401_001_OC
6401-ESDV-001 ESD …
%M00139
SpareDI_11_OC
SpareDI-11 - Open c…
%M00140
INHIBIT_FOAM_RE…
Foam release inhibi…
%M00141
INHIBIT_ALL_OUT…
All outputs inhibit…
%M00142
SpareDI_14_OC
SpareDI-14 - Open c…
%M00143
SpareDI_15_OC
SpareDI-15 - Open c…
%M00144
SpareDI_16_OC
SpareDI-16 - Open c…
%M00145
SpareDI_17_OC
SpareDI-17 - Open c…
%M00146
SpareDI_18_OC
SpareDI-18 - Open c…
%M00147
SpareDI_19_OC
SpareDI-19 - Open c…
%M00148
SpareDI_20_OC
SpareDI-20 - Open c…
%M00149
SpareDI_21_OC
SpareDI-21 - Open c…
%M00150
SpareDI_22_OC
SpareDI-22 - Open c…
%M00151
SpareDI_23_OC
SpareDI-23 - Open c…
%M00152
SpareDI_24_OC
SpareDI-24 - Open c…
%M00153
SpareDI_25_OC
SpareDI-25 - Open c…
%M00154
SpareDI_26_OC
SpareDI-26 - Open c…
%M00155
SpareDI_27_OC
SpareDI-27 - Open c…
%M00156
SpareDI_28_OC
SpareDI-28 - Open c…
%M00157
SpareDI_29_OC
SpareDI-29 - Open c…
%M00158
SpareDI_30_OC
SpareDI-30 - Open c…
%M00159
SpareDI_31_OC
SpareDI-31 - Open c…
%M00160
SpareDI_32_OC
SpareDI-32 - Open c…
CONT
CONT
%M00608
Voting_All_Faul…
Voting of all Faults
%M00473
DG_7404_001_OC
7404-DG-001 Gas de…
%M00497
DG_7404_001_SC
7404-DG-001 Gas de…
%M00478
DG_7404_006_OC
7404-DG-006 Gas de…
%M00502
DG_7404_006_SC
7404-DG-006 Gas de…
%M00483
SpareAI_11_OC
SpareAI-11 - Open c…
%M00507
SpareAI_11_SC
SpareAI-11 - Short …
%M00488
SpareAI_16_OC
SpareAI-16 - Open c…
%M00512
SpareAI_16_SC
SpareAI-16 - Short …
%M00493
SpareAI_21_OC
SpareAI-21 - Open c…
%M00517
SpareAI_21_SC
SpareAI-21 - Short …

Rung 8 (cont.)


%M00474
DG_7404_002_OC
7404-DG-002 Gas de…
%M00498
DG_7404_002_SC
7404-DG-002 Gas de…
%M00475
DG_7404_003_OC
7404-DG-003 Gas de…
%M00499
DG_7404_003_SC
7404-DG-003 Gas de…
%M00476
DG_7404_004_OC
7404-DG-004 Gas de…
%M00500
DG_7404_004_SC
7404-DG-004 Gas de…
%M00477
DG_7404_005_OC
7404-DG-005 Gas de…
%M00501
DG_7404_005_SC
7404-DG-005 Gas de…
%M00479
DG_7404_007_OC
7404-DG-007 Gas de…
%M00503
DG_7404_007_SC
7404-DG-007 Gas de…
%M00480
DG_7404_008_OC
7404-DG-008 Gas de…
%M00504
DG_7404_008_SC
7404-DG-008 Gas de…
%M00481
DG_7404_009_OC
7404-DG-009 Gas de…
%M00505
DG_7404_009_SC
7404-DG-009 Gas de…
%M00482
DG_7404_010_OC
7404-DG-010 Gas de…
%M00506
DG_7404_010_SC
7404-DG-010 Gas de…
%M00484
DF_7404_001_OC
7404-DF-001 Flame …
%M00508
DF_7404_001_SC
7404-DF-001 Flame …
%M00485
DF_7404_002_OC
7404-DF-002 Flame …
%M00509
DF_7404_002_SC
7404-DF-002 Flame …
%M00486
DG_7404_011R_OC
7404-DG-011R Open …
%M00510
DG_7404_011R_SC
7404-DG-011R Open …
%M00487
SpareAI_15_OC
SpareAI-15 - Open c…
%M00511
SpareAI_15_SC
SpareAI-15 - Short …
%M00489
PST_7002_001_OC
7002-PST-001 Air a…
%M00513
PST_7002_001_SC
7002-PST-001 Air a…
%M00490
PST_6401_001_OC
6401-PST-001 Fuel …
%M00514
PST_6401_001_SC
6401-PST-001 Fuel …
%M00491
PST_7503_001_OC
7503-PST-001 Delug…
%M00515
PST_7503_001_SC
7503-PST-001 Delug…
%M00492
FR_7503_001_002…
7503-FR-001/002 Fo…
%M00516
FR_7503_001_002…
7503-FR-001/002 Fo…
%M00494
FR_7503_003_OC
7503-FR-003 Foam r…
%M00518
FR_7503_003_SC
7503-FR-003 Foam r…
%M00495
SpareAI_23_OC
SpareAI-23 - Open c…
%M00519
SpareAI_23_SC
SpareAI-23 - Short …
%M00496
SpareAI_24_OC
SpareAI-24 - Open c…
%M00520
SpareAI_24_SC
SpareAI-24 - Short …
%M00534
DG_7404_011R_FL…
7404-DG-011R Open …

LD BLOCK: Logic

Calls: CE_LOG, Sound

Rung 1

```
This routine calls the voting and C&E routines for the various fireareas on the Rig.
This routine also calculates internal variables used in the logic such as Any_Action, System_Reset, System_Reset_Delay and
Acknowledge bits.
```

Rung 2

```
Calculating system reset.
The System_Reset bit will be high when the reset pushbutton is pressed OR when reset from HMI.
HMI reset bit is reset after 5 sec.
```

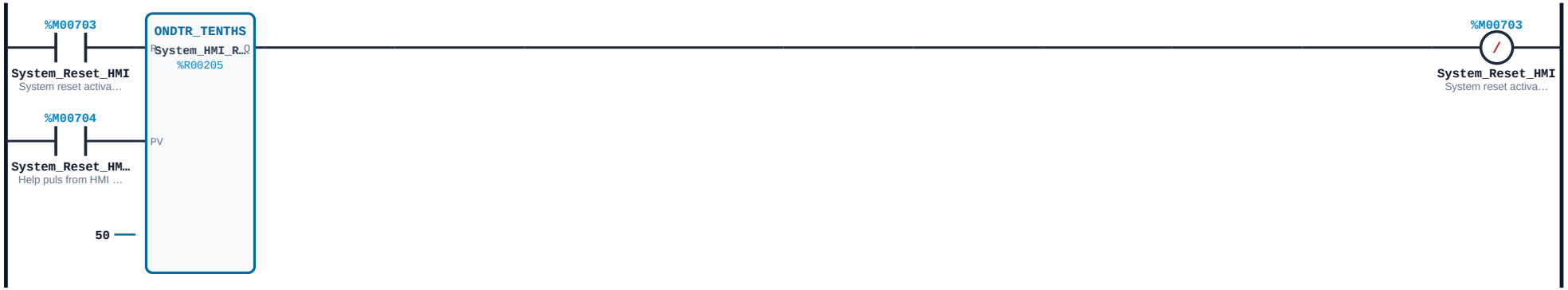
Rung 3



Rung 4



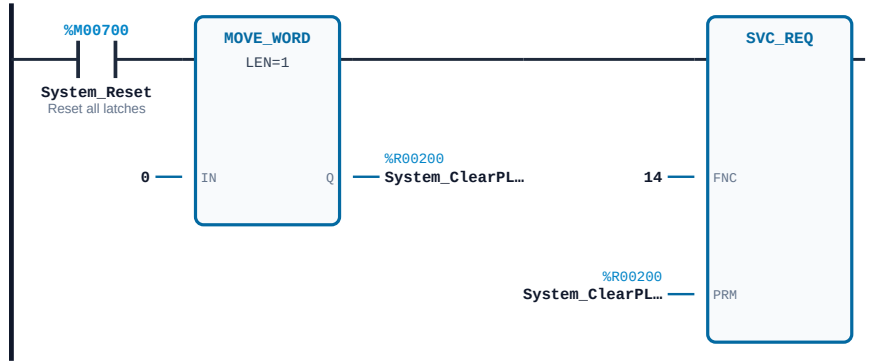
Rung 5



Rung 6

```
CLEAR  PLC FAULT  TABLE
```

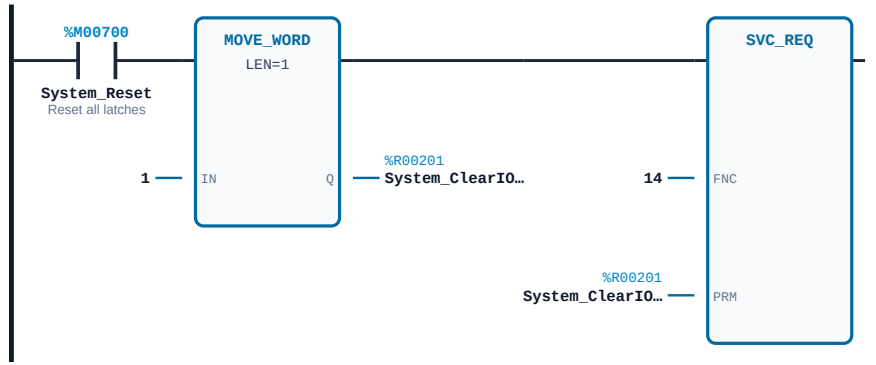
Rung 7



Rung 8

CLEAR IO FAULT TABLE

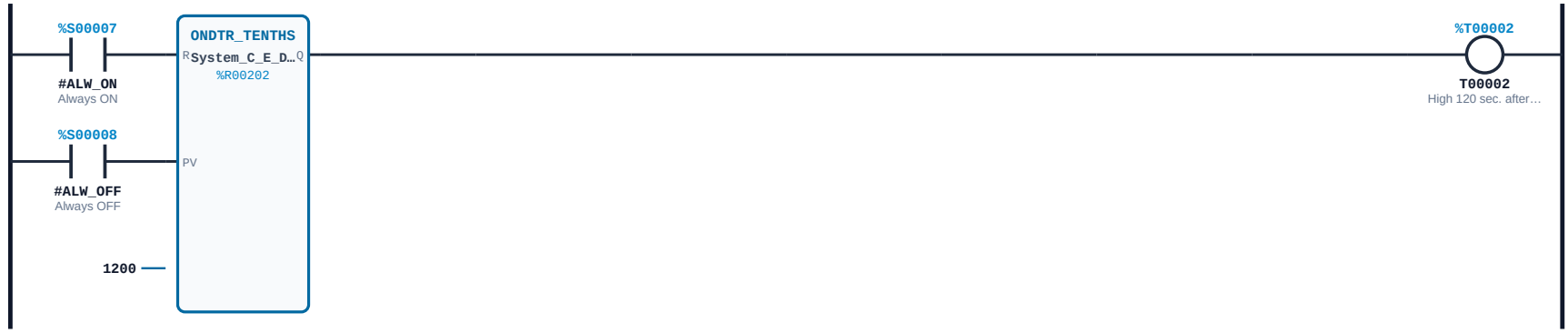
Rung 9



Rung 10

Calculating the delay before C&E is called.
This to be shure that gasdetectors and etc. have finnished the initialisation before logic is started.

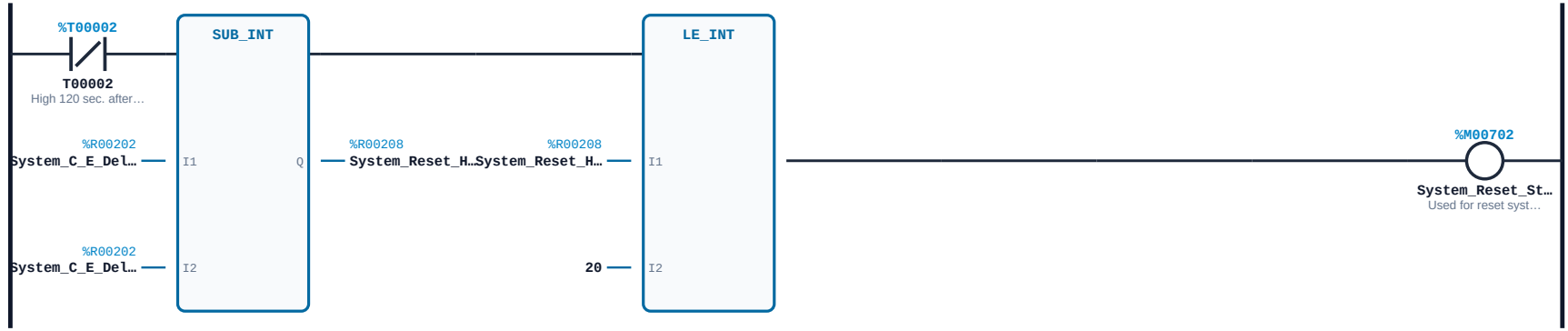
Rung 11



Rung 12

Reset the system 2 seconds before the 120 sec timer is finnish.

Rung 13



Rung 14

```
Calling Cause & Effect logic.
```

Rung 15



Rung 16



LD BLOCK: CE_LOG

Rung 1

```
This block do all Cause & Effect logic for this Fire & Gas system.
Setting OUTPUT's done in blocks called: Shutdown, Protect, Indicat, Loc_Al and FLT1.
```

Rung 2

```
History on changes after 23.april-03:
- 12.juni-03:  Timer for start of pump B (second and third time) changed from 30 to 60 sec. Done on 3 timers.
- 12.juni-03: New input (6401-ESDV-001) to also trip PumpShutdown (%Q08), Oil Booster Pumps (%Q03) and Water Booster Pumps
(%Q04).
```

Rung 3

```
Logic for Shutdown.
```

Rung 4



Rung 5



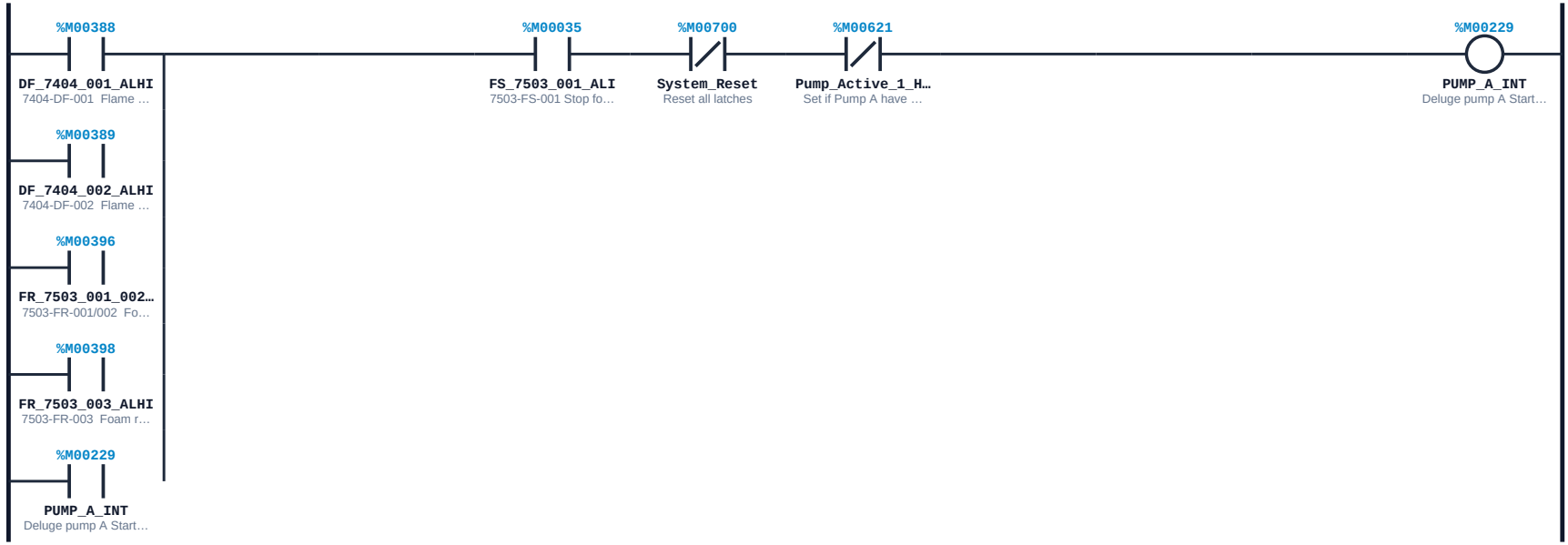
Rung 6

```
Logic for Fire Protection.
```

Rung 7

```
Start of Pump A logic.
Start signal on as long as input active. BUT:
- The pump will be active for 60 minutes and then reset fire detectors and foam release buttons.
- Will reset if system reset pushed "Reset Fire&Gas" (and inputs normal)   OR
- "Stop Foam" button pushed!
```

Rung 8

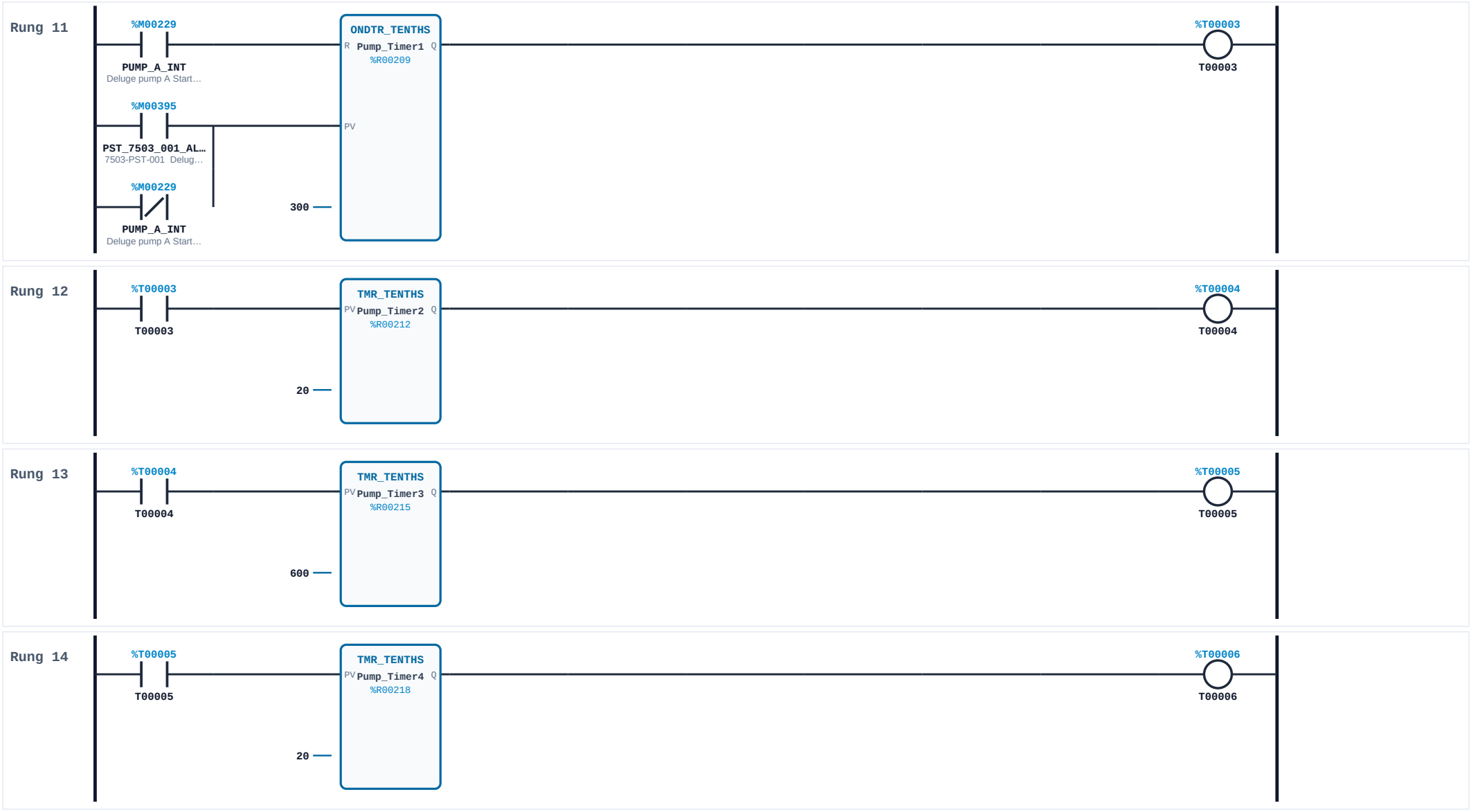


Rung 9



Rung 10

```
Start of Pump B logic.
If pump A start init. activated in 30 seconds without getting deluge released signal (pressure transmitter), then:
- puls (2 sec.) to start pump B is activated. If still no released signal within 60 sec. then:
- puls (2 sec.) to start pump B is activated. If still no released signal within 60 sec. then:
- puls (2 sec.) to start pump B is activated. If still no released signal within 60 sec. then:
- set bit to display alarm in panel: "Deluge failed to release.".
```


Rung 11
%M00229
PUMP_A_INT
Deluge pump A Start…
%M00395
PST_7503_001_AL…
7503-PST-001 Delug…
%M00229
PUMP_A_INT
Deluge pump A Start…
300
ONDTR_TENTHS
R Pump_Timer1 Q
%R00209
PV
%T00003
T00003
Rung 12
%T00003
T00003
20
TMR_TENTHS
PV Pump_Timer2 Q
%R00212
%T00004
T00004
Rung 13
%T00004
T00004
600
TMR_TENTHS
PV Pump_Timer3 Q
%R00215
%T00005
T00005
Rung 14
%T00005
T00005
20
TMR_TENTHS
PV Pump_Timer4 Q
%R00218
%T00006
T00006

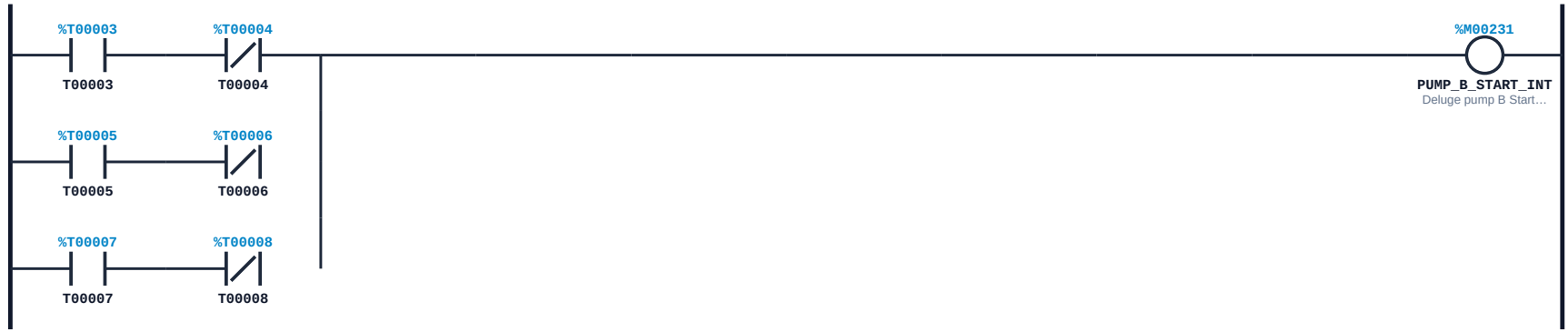
Rung 15



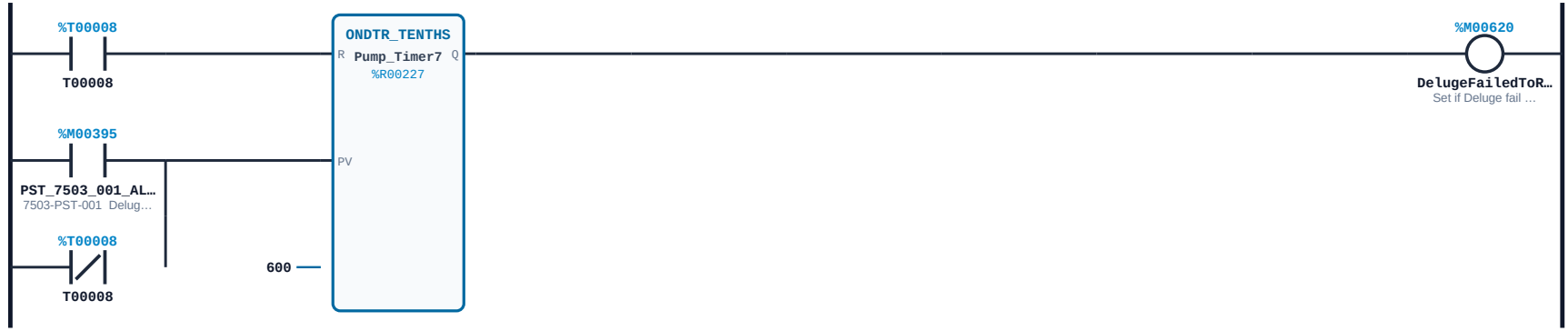
Rung 16



Rung 17



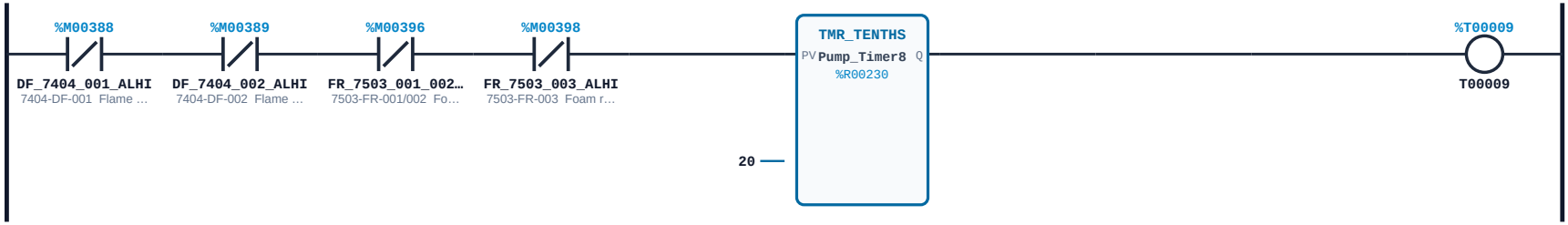
Rung 18



Rung 19

```
Stop of pump B logic.
When fire detectors return to normal OR when stop foam is pusshed, stop puls (2 sec.) is trigged.
```

Rung 20



Rung 21



Rung 22

Logic for Indication on panel (not operator panel).

Rung 23



Rung 24



Rung 25



LD BLOCK: Sound

Rung 1

```
This block generates sound and mute of sound.
In Bottom is logic for trigging of PLS System sound.
```

Rung 2

```
History on changes after 23.april-03:
- 28.juni-03:  INHIBIT_ALL_OUTPUTS changed to NOT trig sound in Pump house.     Vidar Pedersen
```

Rung 3

```
Trigging alarm sound.
```

Rung 4



Rung 5



Rung 6



Rung 7



Rung 8

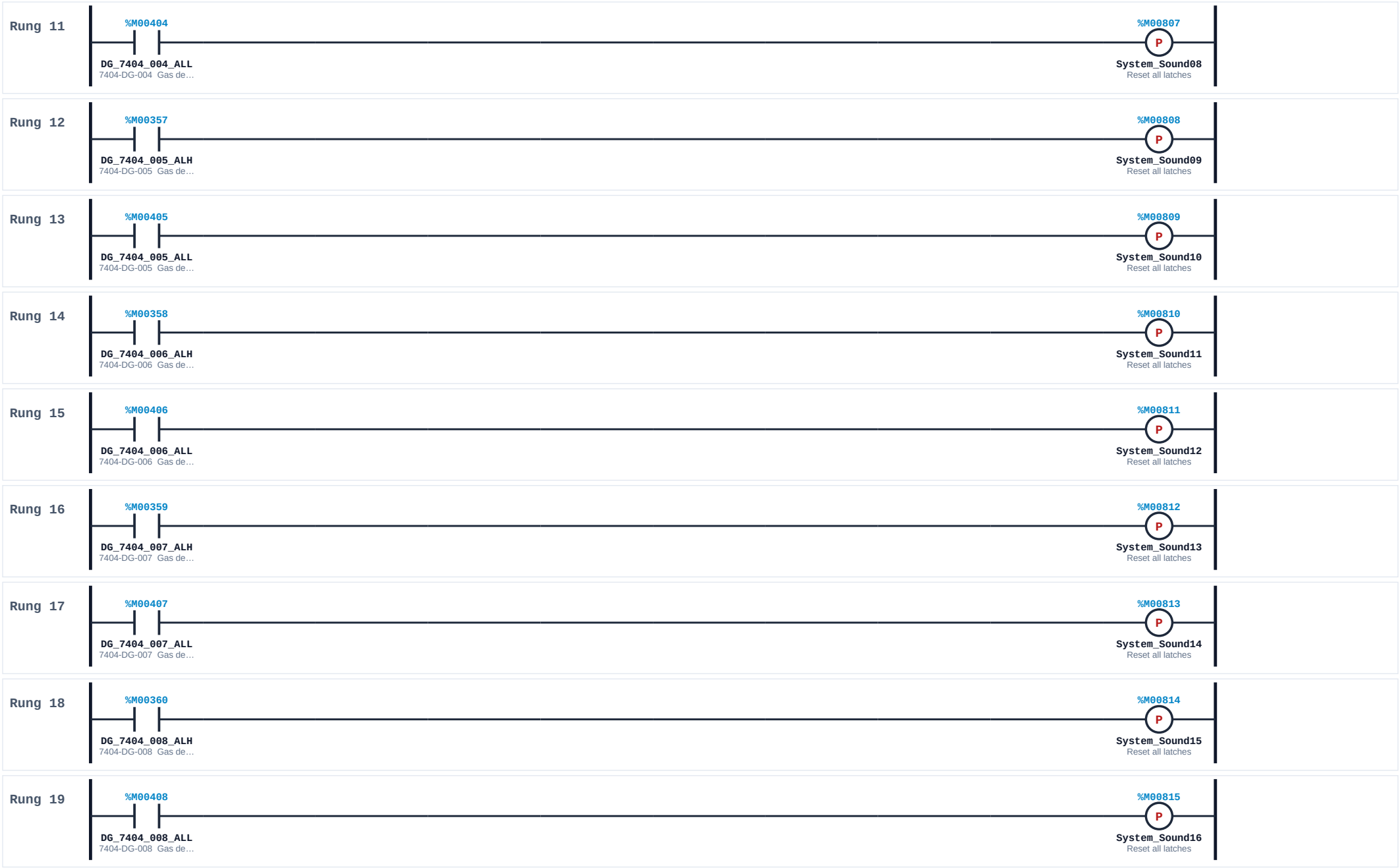


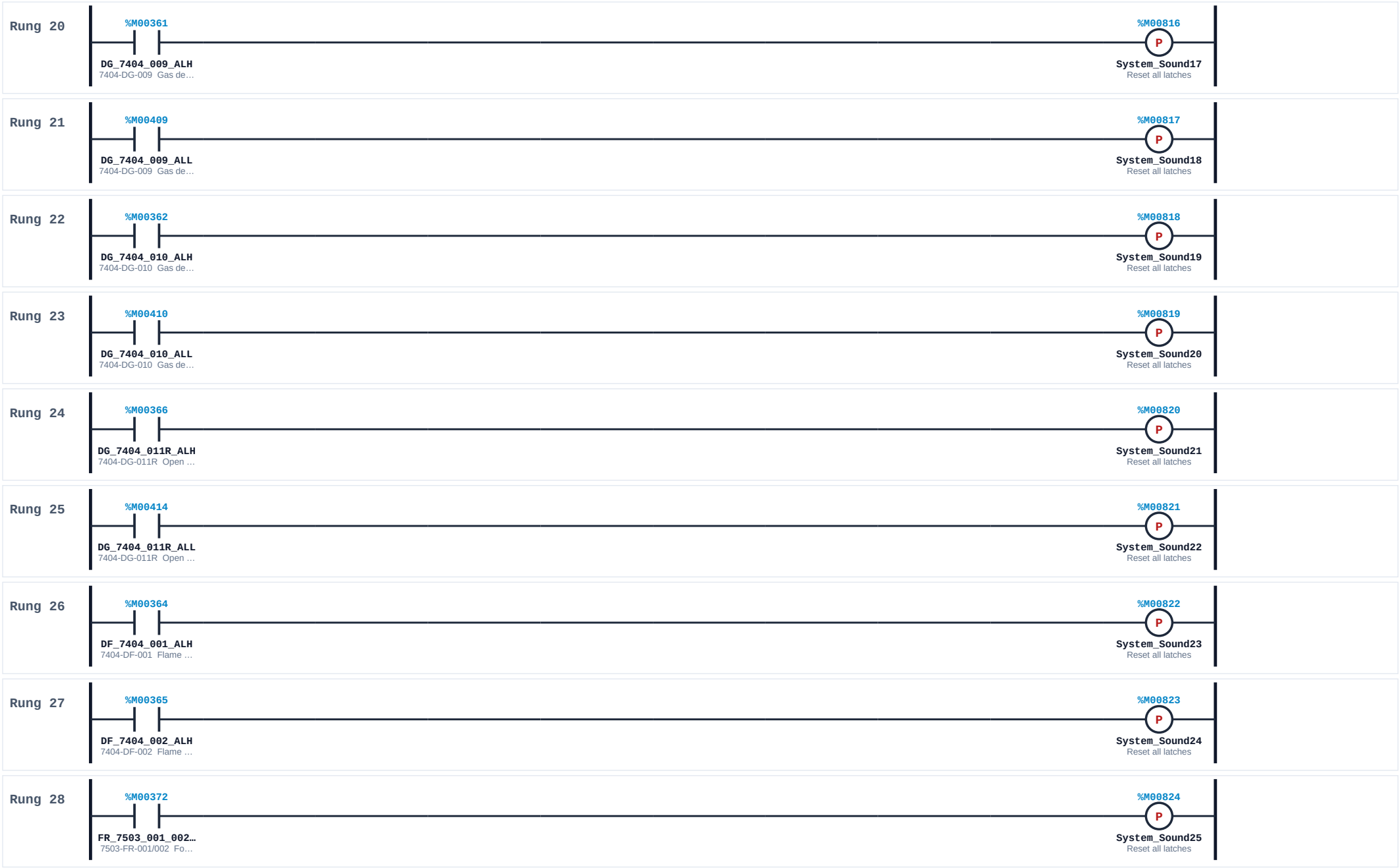
Rung 9



Rung 10




Rung 11
%M00404
DG_7404_004_ALL
7404-DG-004 Gas de…
%M00807
P
System_Sound08
Reset all latches
Rung 12
%M00357
DG_7404_005_ALH
7404-DG-005 Gas de…
%M00808
P
System_Sound09
Reset all latches
Rung 13
%M00405
DG_7404_005_ALL
7404-DG-005 Gas de…
%M00809
P
System_Sound10
Reset all latches
Rung 14
%M00358
DG_7404_006_ALH
7404-DG-006 Gas de…
%M00810
P
System_Sound11
Reset all latches
Rung 15
%M00406
DG_7404_006_ALL
7404-DG-006 Gas de…
%M00811
P
System_Sound12
Reset all latches
Rung 16
%M00359
DG_7404_007_ALH
7404-DG-007 Gas de…
%M00812
P
System_Sound13
Reset all latches
Rung 17
%M00407
DG_7404_007_ALL
7404-DG-007 Gas de…
%M00813
P
System_Sound14
Reset all latches
Rung 18
%M00360
DG_7404_008_ALH
7404-DG-008 Gas de…
%M00814
P
System_Sound15
Reset all latches
Rung 19
%M00408
DG_7404_008_ALL
7404-DG-008 Gas de…
%M00815
P
System_Sound16
Reset all latches


Rung 20
%M00361
DG_7404_009_ALH
7404-DG-009 Gas de…
%M00816
P
System_Sound17
Reset all latches
Rung 21
%M00409
DG_7404_009_ALL
7404-DG-009 Gas de…
%M00817
P
System_Sound18
Reset all latches
Rung 22
%M00362
DG_7404_010_ALH
7404-DG-010 Gas de…
%M00818
P
System_Sound19
Reset all latches
Rung 23
%M00410
DG_7404_010_ALL
7404-DG-010 Gas de…
%M00819
P
System_Sound20
Reset all latches
Rung 24
%M00366
DG_7404_011R_ALH
7404-DG-011R Open …
%M00820
P
System_Sound21
Reset all latches
Rung 25
%M00414
DG_7404_011R_ALL
7404-DG-011R Open …
%M00821
P
System_Sound22
Reset all latches
Rung 26
%M00364
DF_7404_001_ALH
7404-DF-001 Flame …
%M00822
P
System_Sound23
Reset all latches
Rung 27
%M00365
DF_7404_002_ALH
7404-DF-002 Flame …
%M00823
P
System_Sound24
Reset all latches
Rung 28
%M00372
FR_7503_001_002…
7503-FR-001/002 Fo…
%M00824
P
System_Sound25
Reset all latches

Rung 29



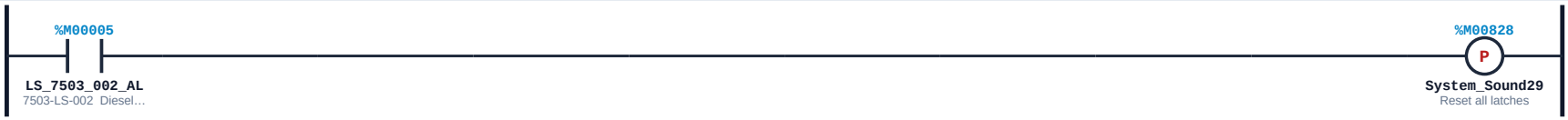
Rung 30



Rung 31



Rung 32



Rung 33

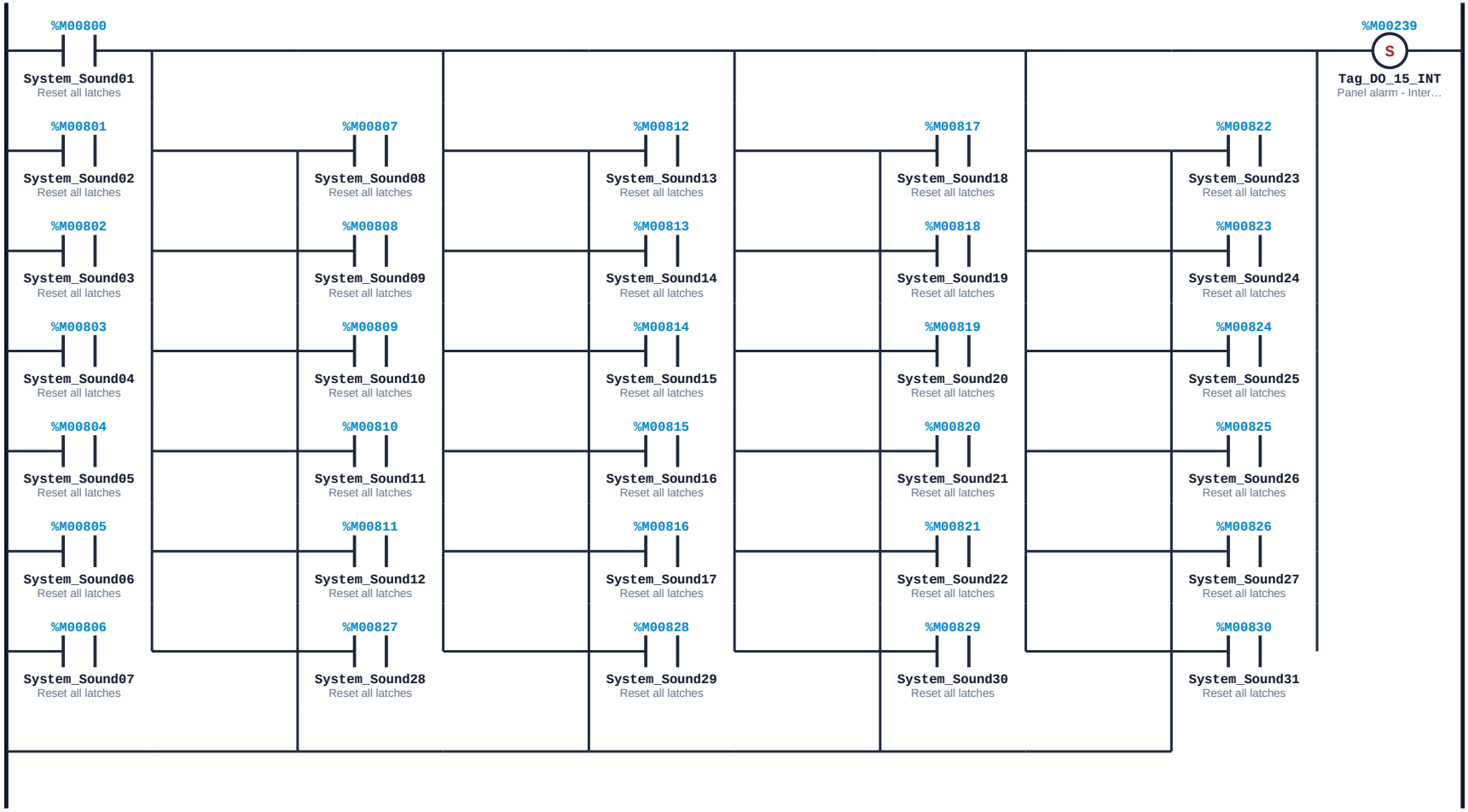


Rung 34



Rung 35 Sound on panel in Production Control Room (buzzer):

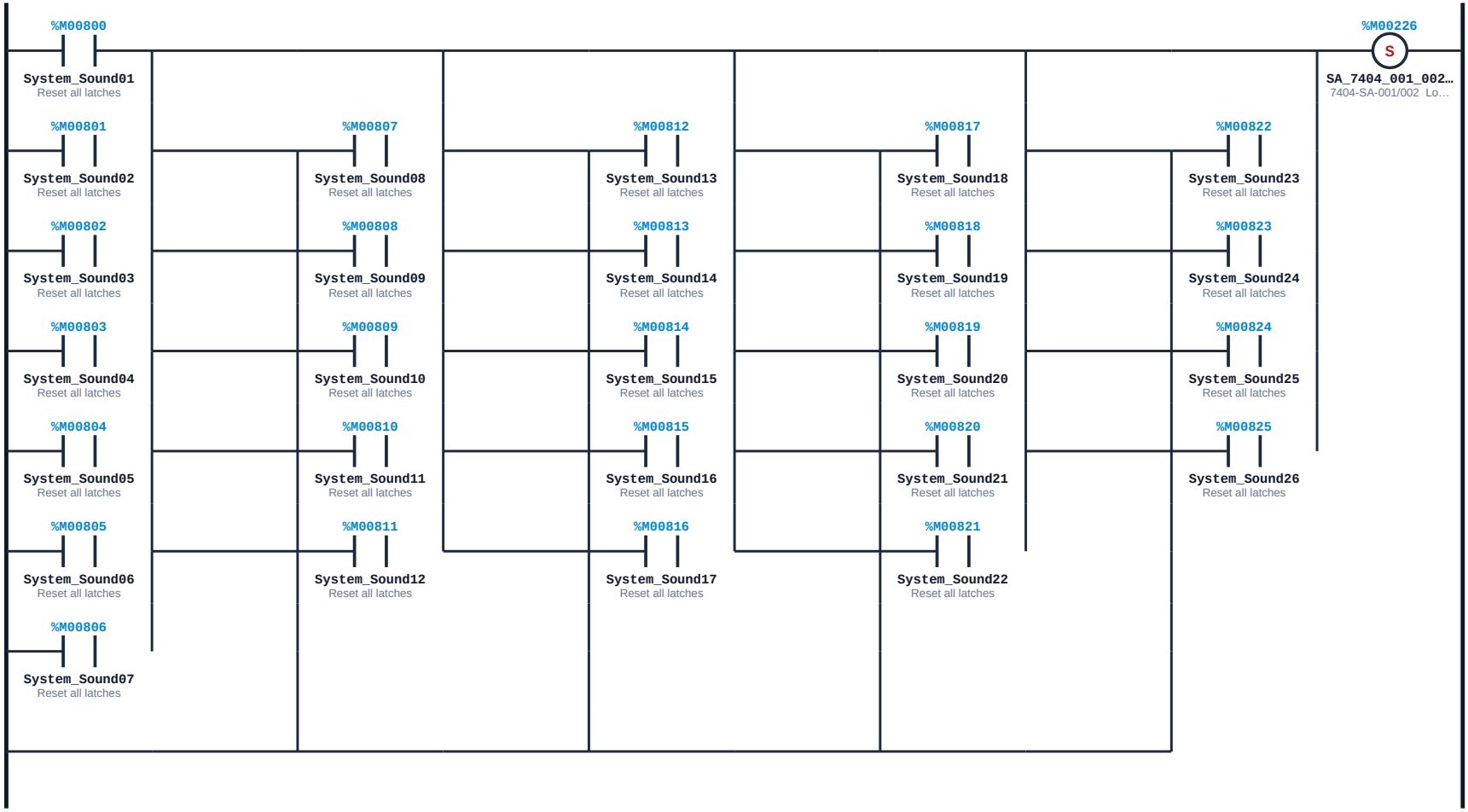
Rung 36



Rung 37

```
Alarm Bell in Pump House:
```

Rung 38



Rung 39

```
Trigging of PLC System sound.
```

Rung 40



Rung 41



Rung 42

%SC0014 — #HRD_FLT — Set when a hardware…

%M00833 (P) — System_Sound34 — Reset all latches

Rung 43

%SA0014 — #LOS_IOM — Set when an I/O mod…

%M00834 (P) — System_Sound35 — Reset all latches

Rung 44

%SA0012 — #LOS_RCK — Set when an expansi…

%M00835 (P) — System_Sound36 — Reset all latches

Rung 45

%SA0011 — #LOW_BAT — Set when a low batt…

%M00836 (P) — System_Sound37 — Reset all latches

Rung 46

%M00009 — Pump_B_Inhibite… — Standby pump B inhi…

%M00839 (P) — System_Sound40 — Reset all latches

Rung 47

%M00831 — System_Sound32 — Reset all latches

%M00832 — System_Sound33 — Reset all latches

%M00833 — System_Sound34 — Reset all latches

%M00834 — System_Sound35 — Reset all latches

%M00835 — System_Sound36 — Reset all latches

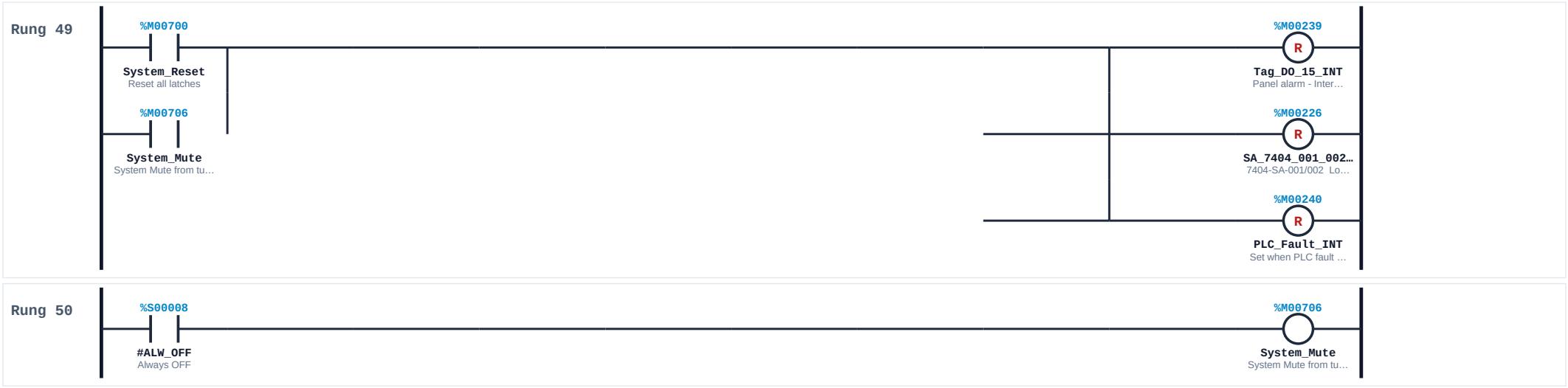
%M00836 — System_Sound37 — Reset all latches

%M00839 — System_Sound40 — Reset all latches

%M00240 (S) — PLC_Fault_INT — Set when PLC fault …

Rung 48

```
Reset of All Sounds.
```


Rung 49
%M00700
System_Reset
Reset all latches
%M00706
System_Mute
System Mute from tu…
%M00239
R
Tag_DO_15_INT
Panel alarm - Inter…
%M00226
R
SA_7404_001_002…
7404-SA-001/002 Lo…
%M00240
R
PLC_Fault_INT
Set when PLC fault …
Rung 50
%S00008
#ALW_OFF
Always OFF
%M00706
System_Mute
System Mute from tu…

LD BLOCK: Output

Calls: Indicat, Shutdow, Protect, Loc_Al, FLT_1

Rung 1

This block calls the output routines where the physical outputs are set.

Rung 2

Call Output block for Shutdown.

Rung 3



Rung 4

Call Output block for Fire Protection.

Rung 5



Rung 6

Call Output block for indication.

Rung 7



Rung 8

Call Output block for Local alarms.

Rung 9



Rung 10

Call Output block FLT_1 for Line fault on Outputs.

Rung 11



LD BLOCK: Indicat

Rung 1

Indication Outputs for local panel set in this block.

Rung 2



Rung 3



Rung 4



LD BLOCK: Shutdow

Rung 1

```
Shutdown Outputs set in this block.
```

Rung 2

```
History on changes after 23.april-03:
- 12.juni-03:  Output %Q0008 (PumpShutdown) inverted.
```

Rung 3



Rung 4



Rung 5



Rung 6



LD BLOCK: Protect

Rung 1

Fire Protection Outputs set in this block.

Rung 2



Rung 3



Rung 4



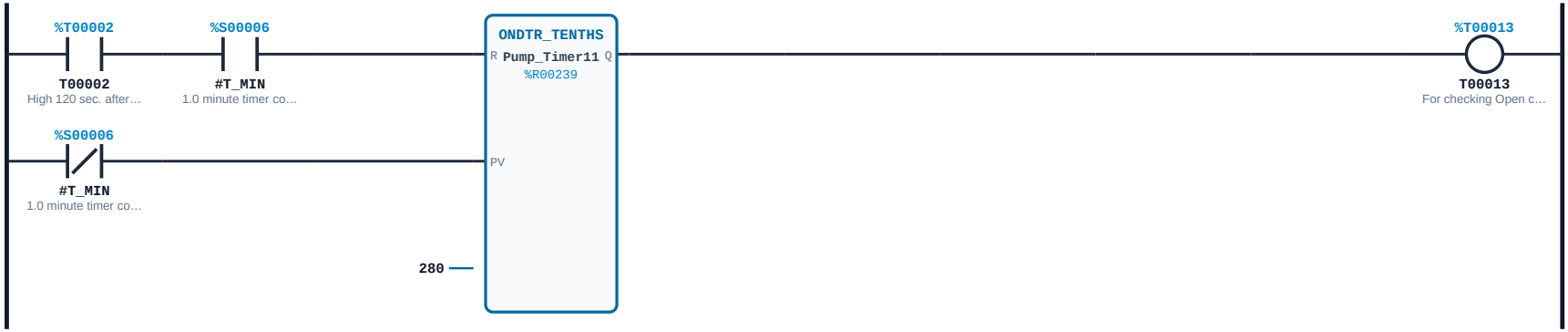
Rung 5

Pulsed outputs for Open circuit handling on %Q0005 & %Q0007.

Rung 6

OPEN CIRCUIT handling for %Q0006 - Start/Stop signal on Pump A.
- First puls output.
- Read pulsed output on input.
If puls on input not registered in 70 seconds (1min + 10 sec) -> set Open circuit bit for %Q0005.

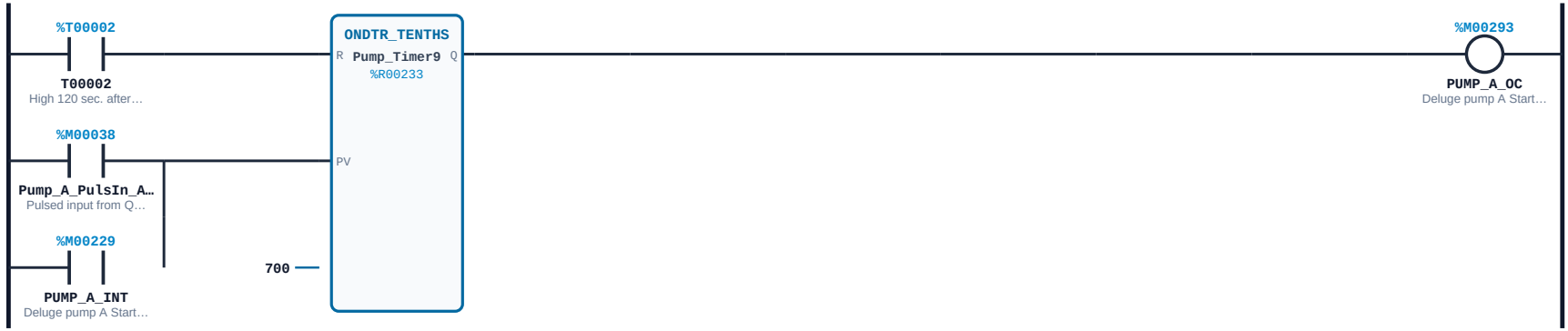
Rung 7



Rung 8



Rung 9



Rung 10

```
OPEN CIRCUIT handling for %Q0007 - Start signal on Pump B.
- First puls output.
- Read pulsed output on input.
If puls on input not registered in 70 seconds (1min + 10 sec)  -> set Open circuit bit for %Q0007.
```

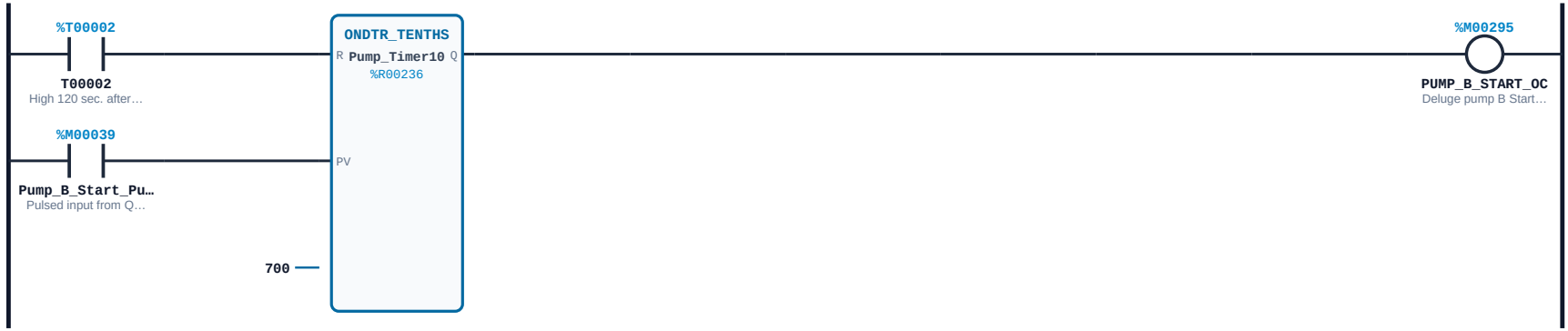
Rung 11



Rung 12



Rung 13



LD BLOCK: Loc_Al

Rung 1

Local sound alarm Outputs set in this block.

Rung 2



Rung 3



Rung 4



LD BLOCK: FLT_1

Rung 1

```
Line Fault on Outputs done in this block.
Created 9. april 2003 by Vidar Pedersen.
```

Rung 2

```
Set OPEN CIRCUIT for %Q0006 - Start/Stop signal on Pump A.
```

Rung 3



Rung 4

```
Set OPEN CIRCUIT for %Q0007 - Start signal on Pump B.
```

Rung 5



Rung 6

```
COIL Pump_B_Start_PulsOut (%Q00010)   |   NOCON PUMP_B_START (%Q00007)
```

Rung 7

```
COIL PUMP_B_START_OC (%M00295)   |   NOCON Pump_B_Start_PulsIn_ALI (%M00039)   |   NOCON PUMP_B_START (%Q00007)   |
ONDTR_TENTHS Pump_Timer10 (%R00236)
```

PLC Variable / Tag Database (760 tags)

Tag Name	Address	Description
DG_7404_001	%AI0001	7404-DG-001 Gas detector pump house
DG_7404_002	%AI0002	7404-DG-002 Gas detector pump house
DG_7404_003	%AI0003	7404-DG-003 Gas detector pump house
DG_7404_004	%AI0004	7404-DG-004 Gas detector pump house
DG_7404_005	%AI0005	7404-DG-005 Gas detector pump house
DG_7404_006	%AI0006	7404-DG-006 Gas detector pump house
DG_7404_007	%AI0007	7404-DG-007 Gas detector pump house
DG_7404_008	%AI0008	7404-DG-008 Gas detector pump house
DG_7404_009	%AI0009	7404-DG-009 Gas detector pump house
DG_7404_010	%AI0010	7404-DG-010 Gas detector pump house
SpareAI_11	%AI0011	Defekt!!!!
DF_7404_001	%AI0012	7404-DF-001 Flame detector pump house
DF_7404_002	%AI0013	7404-DF-002 Flame detector pump house
DG_7404_011R	%AI0014	7404-DG-011R Open path detector
SpareAI_15	%AI0015	SpareAI-15
SpareAI_16	%AI0016	SpareAI-16
PST_7002_001	%AI0017	7002-PST-001 Air acc. pressure
PST_6401_001	%AI0018	6401-PST-001 Fuel gas pressure
PST_7503_001	%AI0019	7503-PST-001 Deluge line pressure
FR_7503_001_002	%AI0020	7503-FR-001/002 Foam release from field
SpareAI_21	%AI0021	SpareAI-21
FR_7503_003	%AI0022	7503-FR-003 Foam release from panel
SpareAI_23	%AI0023	SpareAI-23
SpareAI_24	%AI0024	SpareAI-24
ZSH_6401_001	%I00001	6401-ZSH-001 ESD valve open
ZSL_6401_001	%I00002	6401-ZSL-001 ESD valve closed
FS_7503_001	%I00003	7503-FS-001 Stop foam button
LS_7503_001	%I00004	7503-LS-001 AFFF tank low level
LS_7503_002	%I00005	7503-LS-002 Diesel tank low level
Pump_A_PulsIn	%I00006	Pulsed input from Q05 for Pump A Out signal
Pump_B_Start_PulsIn	%I00007	Pulsed input from Q07 for Pump B Start signal
SpareDI_08	%I00008	SpareDI-08
Pump_B_Inhibited	%I00009	Standby pump B inhibited
ESDV_6401_001	%I00010	6401-ESDV-001 ESD Fuel Gas Button
SpareDI_11	%I00011	SpareDI-11
INHIBIT_FOAM_REL	%I00012	Foam release inhibit
INHIBIT_ALL_OUTPUTS	%I00013	All outputs inhibit
SpareDI_14	%I00014	SpareDI-14
SpareDI_15	%I00015	SpareDI-15
SpareDI_16	%I00016	SpareDI-16
SpareDI_17	%I00017	SpareDI-17
SpareDI_18	%I00018	SpareDI-18
SpareDI_19	%I00019	SpareDI-19
SpareDI_20	%I00020	SpareDI-20
SpareDI_21	%I00021	SpareDI-21
SpareDI_22	%I00022	SpareDI-22
SpareDI_23	%I00023	SpareDI-23

Tag Name	Address	Description
SpareDI_24	%I00024	SpareDI-24
SpareDI_25	%I00025	SpareDI-25
SpareDI_26	%I00026	SpareDI-26
SpareDI_27	%I00027	SpareDI-27
SpareDI_28	%I00028	SpareDI-28
SpareDI_29	%I00029	SpareDI-29
SpareDI_30	%I00030	SpareDI-30
SpareDI_31	%I00031	SpareDI-31
SpareDI_32	%I00032	SpareDI-32
ZSH_6401_001_AL	%M00001	6401-ZSH-001 ESD valve open - Alarm
ZSL_6401_001_AL	%M00002	6401-ZSL-001 ESD valve closed - Alarm
FS_7503_001_AL	%M00003	7503-FS-001 Stop foam button - Alarm
LS_7503_001_AL	%M00004	7503-LS-001 AFFF tank low level - Alarm
LS_7503_002_AL	%M00005	7503-LS-002 Diesel tank low level - Alarm
Pump_A_PulsIn_AL	%M00006	Pulsed input from Q05 for Pump A Out signal - Alarm
Pump_B_Start_PulsIn_AL	%M00007	Pulsed input from Q07 for Pump B Start signal - Alarm
SpareDI_08_AL	%M00008	SpareDI-08 - Alarm
Pump_B_Inhibited_AL	%M00009	Standby pump B inhibited - Alarm
ESDV_6401_001_AL	%M00010	6401-ESDV-001 ESD Fuel Gas Button - Alarm
SpareDI_11_AL	%M00011	SpareDI-11 - Alarm
INHIBIT_FOAM_REL_AL	%M00012	Foam release inhibit - Alarm
INHIBIT_ALL_OUTPUTS_AL	%M00013	All outputs inhibit - Alarm
SpareDI_14_AL	%M00014	SpareDI-14 - Alarm
SpareDI_15_AL	%M00015	SpareDI-15 - Alarm
SpareDI_16_AL	%M00016	SpareDI-16 - Alarm
SpareDI_17_AL	%M00017	SpareDI-17 - Alarm
SpareDI_18_AL	%M00018	SpareDI-18 - Alarm
SpareDI_19_AL	%M00019	SpareDI-19 - Alarm
SpareDI_20_AL	%M00020	SpareDI-20 - Alarm
SpareDI_21_AL	%M00021	SpareDI-21 - Alarm
SpareDI_22_AL	%M00022	SpareDI-22 - Alarm
SpareDI_23_AL	%M00023	SpareDI-23 - Alarm
SpareDI_24_AL	%M00024	SpareDI-24 - Alarm
SpareDI_25_AL	%M00025	SpareDI-25 - Alarm
SpareDI_26_AL	%M00026	SpareDI-26 - Alarm
SpareDI_27_AL	%M00027	SpareDI-27 - Alarm
SpareDI_28_AL	%M00028	SpareDI-28 - Alarm
SpareDI_29_AL	%M00029	SpareDI-29 - Alarm
SpareDI_30_AL	%M00030	SpareDI-30 - Alarm
SpareDI_31_AL	%M00031	SpareDI-31 - Alarm
SpareDI_32_AL	%M00032	SpareDI-32 - Alarm
ZSH_6401_001_ALI	%M00033	6401-ZSH-001 ESD valve open - Alarm inhibit
ZSL_6401_001_ALI	%M00034	6401-ZSL-001 ESD valve closed - Alarm inhibit
FS_7503_001_ALI	%M00035	7503-FS-001 Stop foam button - Alarm inhibit
LS_7503_001_ALI	%M00036	7503-LS-001 AFFF tank low level - Alarm inhibit
LS_7503_002_ALI	%M00037	7503-LS-002 Diesel tank low level - Alarm inhibit
Pump_A_PulsIn_ALI	%M00038	Pulsed input from Q05 for Pump A Out signal - Alarm inhibit
Pump_B_Start_PulsIn_ALI	%M00039	Pulsed input from Q07 for Pump B Start signal - Alarm inhibit

Tag Name	Address	Description
SpareDI_08_ALI	%M00040	SpareDI-08 - Alarm inhibit
Pump_B_Inhibited_ALI	%M00041	Standby pump B inhibited - Alarm inhibit
ESDV_6401_001_ALI	%M00042	6401-ESDV-001 ESD Fuel Gas Button - Alarm inhibit
SpareDI_11_ALI	%M00043	SpareDI-11 - Alarm inhibit
INHIBIT_FOAM_REL_ALI	%M00044	Foam release inhibit - Alarm inhibit
INHIBIT_ALL_OUTPUTS_ALI	%M00045	All outputs inhibit - Alarm inhibit
SpareDI_14_ALI	%M00046	SpareDI-14 - Alarm inhibit
SpareDI_15_ALI	%M00047	SpareDI-15 - Alarm inhibit
SpareDI_16_ALI	%M00048	SpareDI-16 - Alarm inhibit
SpareDI_17_ALI	%M00049	SpareDI-17 - Alarm inhibit
SpareDI_18_ALI	%M00050	SpareDI-18 - Alarm inhibit
SpareDI_19_ALI	%M00051	SpareDI-19 - Alarm inhibit
SpareDI_20_ALI	%M00052	SpareDI-20 - Alarm inhibit
SpareDI_21_ALI	%M00053	SpareDI-21 - Alarm inhibit
SpareDI_22_ALI	%M00054	SpareDI-22 - Alarm inhibit
SpareDI_23_ALI	%M00055	SpareDI-23 - Alarm inhibit
SpareDI_24_ALI	%M00056	SpareDI-24 - Alarm inhibit
SpareDI_25_ALI	%M00057	SpareDI-25 - Alarm inhibit
SpareDI_26_ALI	%M00058	SpareDI-26 - Alarm inhibit
SpareDI_27_ALI	%M00059	SpareDI-27 - Alarm inhibit
SpareDI_28_ALI	%M00060	SpareDI-28 - Alarm inhibit
SpareDI_29_ALI	%M00061	SpareDI-29 - Alarm inhibit
SpareDI_30_ALI	%M00062	SpareDI-30 - Alarm inhibit
SpareDI_31_ALI	%M00063	SpareDI-31 - Alarm inhibit
SpareDI_32_ALI	%M00064	SpareDI-32 - Alarm inhibit
ZSH_6401_001_FRUP	%M00065	6401-ZSH-001 ESD valve open - First up
ZSL_6401_001_FRUP	%M00066	6401-ZSL-001 ESD valve closed - First up
FS_7503_001_FRUP	%M00067	7503-FS-001 Stop foam button - First up
LS_7503_001_FRUP	%M00068	7503-LS-001 AFFF tank low level - First up
LS_7503_002_FRUP	%M00069	7503-LS-002 Diesel tank low level - First up
Pump_A_PulsIn_FRUP	%M00070	Pulsed input from Q05 for Pump A Out signal - First up
Pump_B_Start_PulsIn_FRUP	%M00071	Pulsed input from Q07 for Pump B Start signal - First up
SpareDI_08_FRUP	%M00072	SpareDI-08 - First up
Pump_B_Inhibited_FRUP	%M00073	Standby pump B inhibited - First up
ESDV_6401_001_FRUP	%M00074	6401-ESDV-001 ESD Fuel Gas Button - First up
SpareDI_11_FRUP	%M00075	SpareDI-11 - First up
INHIBIT_FOAM_REL_FRUP	%M00076	Foam release inhibit - First up
INHIBIT_ALL_OUTPUTS_FRUP	%M00077	All outputs inhibit - First up
SpareDI_14_FRUP	%M00078	SpareDI-14 - First up
SpareDI_15_FRUP	%M00079	SpareDI-15 - First up
SpareDI_16_FRUP	%M00080	SpareDI-16 - First up
SpareDI_17_FRUP	%M00081	SpareDI-17 - First up
SpareDI_18_FRUP	%M00082	SpareDI-18 - First up
SpareDI_19_FRUP	%M00083	SpareDI-19 - First up
SpareDI_20_FRUP	%M00084	SpareDI-20 - First up
SpareDI_21_FRUP	%M00085	SpareDI-21 - First up
SpareDI_22_FRUP	%M00086	SpareDI-22 - First up
SpareDI_23_FRUP	%M00087	SpareDI-23 - First up

Tag Name	Address	Description
SpareDI_24_FRUP	%M00088	SpareDI-24 - First up
SpareDI_25_FRUP	%M00089	SpareDI-25 - First up
SpareDI_26_FRUP	%M00090	SpareDI-26 - First up
SpareDI_27_FRUP	%M00091	SpareDI-27 - First up
SpareDI_28_FRUP	%M00092	SpareDI-28 - First up
SpareDI_29_FRUP	%M00093	SpareDI-29 - First up
SpareDI_30_FRUP	%M00094	SpareDI-30 - First up
SpareDI_31_FRUP	%M00095	SpareDI-31 - First up
SpareDI_32_FRUP	%M00096	SpareDI-32 - First up
ZSH_6401_001_INH	%M00097	6401-ZSH-001 ESD valve open - Inhibit
ZSL_6401_001_INH	%M00098	6401-ZSL-001 ESD valve closed - Inhibit
FS_7503_001_INH	%M00099	7503-FS-001 Stop foam button - Inhibit
LS_7503_001_INH	%M00100	7503-LS-001 AFFF tank low level - Inhibit
LS_7503_002_INH	%M00101	7503-LS-002 Diesel tank low level - Inhibit
Pump_A_PulsIn_INH	%M00102	Pulsed input from Q05 for Pump A Out signal - Inhibit
Pump_B_Start_PulsIn_INH	%M00103	Pulsed input from Q07 for Pump B Start signal - Inhibit
SpareDI_08_INH	%M00104	SpareDI-08 - Inhibit
Pump_B_Inhibited_INH	%M00105	Standby pump B inhibited - Inhibit
ESDV_6401_001_INH	%M00106	6401-ESDV-001 ESD Fuel Gas Button - Inhibit
SpareDI_11_INH	%M00107	SpareDI-11 - Inhibit
INHIBIT_FOAM_REL_INH	%M00108	Foam release inhibit - Inhibit
INHIBIT_ALL_OUTPUTS_INH	%M00109	All outputs inhibit - Inhibit
SpareDI_14_INH	%M00110	SpareDI-14 - Inhibit
SpareDI_15_INH	%M00111	SpareDI-15 - Inhibit
SpareDI_16_INH	%M00112	SpareDI-16 - Inhibit
SpareDI_17_INH	%M00113	SpareDI-17 - Inhibit
SpareDI_18_INH	%M00114	SpareDI-18 - Inhibit
SpareDI_19_INH	%M00115	SpareDI-19 - Inhibit
SpareDI_20_INH	%M00116	SpareDI-20 - Inhibit
SpareDI_21_INH	%M00117	SpareDI-21 - Inhibit
SpareDI_22_INH	%M00118	SpareDI-22 - Inhibit
SpareDI_23_INH	%M00119	SpareDI-23 - Inhibit
SpareDI_24_INH	%M00120	SpareDI-24 - Inhibit
SpareDI_25_INH	%M00121	SpareDI-25 - Inhibit
SpareDI_26_INH	%M00122	SpareDI-26 - Inhibit
SpareDI_27_INH	%M00123	SpareDI-27 - Inhibit
SpareDI_28_INH	%M00124	SpareDI-28 - Inhibit
SpareDI_29_INH	%M00125	SpareDI-29 - Inhibit
SpareDI_30_INH	%M00126	SpareDI-30 - Inhibit
SpareDI_31_INH	%M00127	SpareDI-31 - Inhibit
SpareDI_32_INH	%M00128	SpareDI-32 - Inhibit
ZSH_6401_001_OC	%M00129	6401-ZSH-001 ESD valve open - Open circuit
ZSL_6401_001_OC	%M00130	6401-ZSL-001 ESD valve closed - Open circuit
FS_7503_001_OC	%M00131	7503-FS-001 Stop foam button - Open circuit
LS_7503_001_OC	%M00132	7503-LS-001 AFFF tank low level - Open circuit
LS_7503_002_OC	%M00133	7503-LS-002 Diesel tank low level - Open circuit
Pump_A_PulsIn_OC	%M00134	Pulsed input from Q05 for Pump A Out signal - Open circuit
Pump_B_Start_PulsIn_OC	%M00135	Pulsed input from Q07 for Pump B Start signal - Open circuit

Tag Name	Address	Description
SpareDI_08_OC	%M00136	SpareDI-08 - Open circuit
Pump_B_Inhibited_OC	%M00137	Standby pump B inhibited - Open circuit
ESDV_6401_001_OC	%M00138	6401-ESDV-001 ESD Fuel Gas Button - Open circuit
SpareDI_11_OC	%M00139	SpareDI-11 - Open circuit
INHIBIT_FOAM_REL_OC	%M00140	Foam release inhibit - Open circuit
INHIBIT_ALL_OUTPUTS_OC	%M00141	All outputs inhibit - Open circuit
SpareDI_14_OC	%M00142	SpareDI-14 - Open circuit
SpareDI_15_OC	%M00143	SpareDI-15 - Open circuit
SpareDI_16_OC	%M00144	SpareDI-16 - Open circuit
SpareDI_17_OC	%M00145	SpareDI-17 - Open circuit
SpareDI_18_OC	%M00146	SpareDI-18 - Open circuit
SpareDI_19_OC	%M00147	SpareDI-19 - Open circuit
SpareDI_20_OC	%M00148	SpareDI-20 - Open circuit
SpareDI_21_OC	%M00149	SpareDI-21 - Open circuit
SpareDI_22_OC	%M00150	SpareDI-22 - Open circuit
SpareDI_23_OC	%M00151	SpareDI-23 - Open circuit
SpareDI_24_OC	%M00152	SpareDI-24 - Open circuit
SpareDI_25_OC	%M00153	SpareDI-25 - Open circuit
SpareDI_26_OC	%M00154	SpareDI-26 - Open circuit
SpareDI_27_OC	%M00155	SpareDI-27 - Open circuit
SpareDI_28_OC	%M00156	SpareDI-28 - Open circuit
SpareDI_29_OC	%M00157	SpareDI-29 - Open circuit
SpareDI_30_OC	%M00158	SpareDI-30 - Open circuit
SpareDI_31_OC	%M00159	SpareDI-31 - Open circuit
SpareDI_32_OC	%M00160	SpareDI-32 - Open circuit
EY_6401_001_002_MT	%M00161	6401-EY-001/002 ESD fuel gas valve shutdown - Manual trip
SA_7404_001_002_MT	%M00162	7404-SA-001/002 Local alarm pump house - Manual trip
P2100_P2200_MT	%M00163	P2100/P2200 Oil booster pump shutdown - Manual trip
P3600_P3700_MT	%M00164	P3600/P3700 Water booster pump shutdown - Manual trip
PUMP_A_MT	%M00165	Deluge pump A Start/Stop - Manual trip
PUMP_B_STOP_MT	%M00166	Deluge pump B Stop - Manual trip
PUMP_B_START_MT	%M00167	Deluge pump B Start - Manual trip
PumpShutdown_MT	%M00168	Fire & Gas pump shutdown - Manual trip
Pump_A_PulsOut_MT	%M00169	Pulsed output for open circuit check on Pump A Out signal - Manual trip
Pump_B_Start_PulsOut_MT	%M00170	Pulsed output for open circuit check on Pump B Start signal - Manual trip
SpareDO_11_MT	%M00171	SpareDO-11 - Manual trip
Tag_DO_12_MT	%M00172	EDS valve open indication - Manual trip
Tag_DO_13_MT	%M00173	EDS valve closed indication - Manual trip
Tag_DO_14_MT	%M00174	EDS valve low air indication - Manual trip
Tag_DO_15_MT	%M00175	Panel alarm - Manual trip
PLC_Fault_MT	%M00176	Set when PLC fault exists - Manual trip
SpareDO_17_MT	%M00177	SpareDO-17 - Manual trip
SpareDO_18_MT	%M00178	SpareDO-18 - Manual trip
SpareDO_19_MT	%M00179	SpareDO-19 - Manual trip
SpareDO_20_MT	%M00180	SpareDO-20 - Manual trip
SpareDO_21_MT	%M00181	SpareDO-21 - Manual trip
SpareDO_22_MT	%M00182	SpareDO-22 - Manual trip
SpareDO_23_MT	%M00183	SpareDO-23 - Manual trip

Tag Name	Address	Description
SpareDO_24_MT	%M00184	SpareDO-24 - Manual trip
SpareDO_25_MT	%M00185	SpareDO-25 - Manual trip
SpareDO_26_MT	%M00186	SpareDO-26 - Manual trip
SpareDO_27_MT	%M00187	SpareDO-27 - Manual trip
SpareDO_28_MT	%M00188	SpareDO-28 - Manual trip
SpareDO_29_MT	%M00189	SpareDO-29 - Manual trip
SpareDO_30_MT	%M00190	SpareDO-30 - Manual trip
SpareDO_31_MT	%M00191	SpareDO-31 - Manual trip
SpareDO_32_MT	%M00192	SpareDO-32 - Manual trip
EY_6401_001_002_INT	%M00225	6401-EY-001/002 ESD fuel gas valve shutdown - Internal C_E use
SA_7404_001_002_INT	%M00226	7404-SA-001/002 Local alarm pump house - Internal C_E use
P2100_P2200_INT	%M00227	P2100/P2200 Oil booster pump shutdown - Internal C_E use
P3600_P3700_INT	%M00228	P3600/P3700 Water booster pump shutdown - Internal C_E use
PUMP_A_INT	%M00229	Deluge pump A Start/Stop - Internal C_E use
PUMP_B_STOP_INT	%M00230	Deluge pump B Stop - Internal C_E use
PUMP_B_START_INT	%M00231	Deluge pump B Start - Internal C_E use
PumpShutdown_INT	%M00232	Fire & Gas pump shutdown - Internal C_E use
Pump_A_PulsOut_INT	%M00233	Pulsed output for open circuit check on Pump A Out signal - Internal C_E use
Pump_B_Start_PulsOut_INT	%M00234	Pulsed output for open circuit check on Pump B Start signal - Internal C_E use
SpareDO_11_INT	%M00235	SpareDO-11 - Internal C_E use
Tag_DO_12_INT	%M00236	EDS valve open indication - Internal C_E use
Tag_DO_13_INT	%M00237	EDS valve closed indication - Internal C_E use
Tag_DO_14_INT	%M00238	EDS valve low air indication - Internal C_E use
Tag_DO_15_INT	%M00239	Panel alarm - Internal C_E use
PLC_Fault_INT	%M00240	Set when PLC fault exists - Internal C_E use
SpareDO_17_INT	%M00241	SpareDO-17 - Internal C_E use
SpareDO_18_INT	%M00242	SpareDO-18 - Internal C_E use
SpareDO_19_INT	%M00243	SpareDO-19 - Internal C_E use
SpareDO_20_INT	%M00244	SpareDO-20 - Internal C_E use
SpareDO_21_INT	%M00245	SpareDO-21 - Internal C_E use
SpareDO_22_INT	%M00246	SpareDO-22 - Internal C_E use
SpareDO_23_INT	%M00247	SpareDO-23 - Internal C_E use
SpareDO_24_INT	%M00248	SpareDO-24 - Internal C_E use
SpareDO_25_INT	%M00249	SpareDO-25 - Internal C_E use
SpareDO_26_INT	%M00250	SpareDO-26 - Internal C_E use
SpareDO_27_INT	%M00251	SpareDO-27 - Internal C_E use
SpareDO_28_INT	%M00252	SpareDO-28 - Internal C_E use
SpareDO_29_INT	%M00253	SpareDO-29 - Internal C_E use
SpareDO_30_INT	%M00254	SpareDO-30 - Internal C_E use
SpareDO_31_INT	%M00255	SpareDO-31 - Internal C_E use
SpareDO_32_INT	%M00256	SpareDO-32 - Internal C_E use
EY_6401_001_002_OC	%M00289	6401-EY-001/002 ESD fuel gas valve shutdown - Open circuit
SA_7404_001_002_OC	%M00290	7404-SA-001/002 Local alarm pump house - Open circuit
P2100_P2200_OC	%M00291	P2100/P2200 Oil booster pump shutdown - Open circuit
P3600_P3700_OC	%M00292	P3600/P3700 Water booster pump shutdown - Open circuit
PUMP_A_OC	%M00293	Deluge pump A Start/Stop - Open circuit
PUMP_B_STOP_OC	%M00294	Deluge pump B Stop - Open circuit
PUMP_B_START_OC	%M00295	Deluge pump B Start - Open circuit

Tag Name	Address	Description
PumpShutdown_OC	%M00296	Fire & Gas pump shutdown - Open circuit
Pump_A_PulsOut_OC	%M00297	Pulsed output for open circuit check on Pump A Out signal - Open circuit
Pump_B_Start_PulsOut_OC	%M00298	Pulsed output for open circuit check on Pump B Start signal - Open circuit
SpareDO_11_OC	%M00299	SpareDO-11 - Open circuit
Tag_DO_12_OC	%M00300	EDS valve open indication - Open circuit
Tag_DO_13_OC	%M00301	EDS valve closed indication - Open circuit
Tag_DO_14_OC	%M00302	EDS valve low air indication - Open circuit
Tag_DO_15_OC	%M00303	Panel alarm - Open circuit
PLC_Fault_OC	%M00304	Set when PLC fault exists - Open circuit
SpareDO_17_OC	%M00305	SpareDO-17 - Open circuit
SpareDO_18_OC	%M00306	SpareDO-18 - Open circuit
SpareDO_19_OC	%M00307	SpareDO-19 - Open circuit
SpareDO_20_OC	%M00308	SpareDO-20 - Open circuit
SpareDO_21_OC	%M00309	SpareDO-21 - Open circuit
SpareDO_22_OC	%M00310	SpareDO-22 - Open circuit
SpareDO_23_OC	%M00311	SpareDO-23 - Open circuit
SpareDO_24_OC	%M00312	SpareDO-24 - Open circuit
SpareDO_25_OC	%M00313	SpareDO-25 - Open circuit
SpareDO_26_OC	%M00314	SpareDO-26 - Open circuit
SpareDO_27_OC	%M00315	SpareDO-27 - Open circuit
SpareDO_28_OC	%M00316	SpareDO-28 - Open circuit
SpareDO_29_OC	%M00317	SpareDO-29 - Open circuit
SpareDO_30_OC	%M00318	SpareDO-30 - Open circuit
SpareDO_31_OC	%M00319	SpareDO-31 - Open circuit
SpareDO_32_OC	%M00320	SpareDO-32 - Open circuit
DG_7404_001_ALH	%M00353	7404-DG-001 Gas detector pump house - Hi Alarm
DG_7404_002_ALH	%M00354	7404-DG-002 Gas detector pump house - Hi Alarm
DG_7404_003_ALH	%M00355	7404-DG-003 Gas detector pump house - Hi Alarm
DG_7404_004_ALH	%M00356	7404-DG-004 Gas detector pump house - Hi Alarm
DG_7404_005_ALH	%M00357	7404-DG-005 Gas detector pump house - Hi Alarm
DG_7404_006_ALH	%M00358	7404-DG-006 Gas detector pump house - Hi Alarm
DG_7404_007_ALH	%M00359	7404-DG-007 Gas detector pump house - Hi Alarm
DG_7404_008_ALH	%M00360	7404-DG-008 Gas detector pump house - Hi Alarm
DG_7404_009_ALH	%M00361	7404-DG-009 Gas detector pump house - Hi Alarm
DG_7404_010_ALH	%M00362	7404-DG-010 Gas detector pump house - Hi Alarm
SpareAI_11_ALH	%M00363	SpareAI-11 - Hi Alarm
DF_7404_001_ALH	%M00364	7404-DF-001 Flame detector pump house - Fire Alarm
DF_7404_002_ALH	%M00365	7404-DF-002 Flame detector pump house - Hi Alarm
DG_7404_011R_ALH	%M00366	7404-DG-011R Open path detector - Hi Alarm
SpareAI_15_ALH	%M00367	SpareAI-15 - Hi Alarm
SpareAI_16_ALH	%M00368	SpareAI-16 - Hi Alarm
PST_7002_001_ALH	%M00369	7002-PST-001 Air acc. pressure - Hi Alarm
PST_6401_001_ALH	%M00370	6401-PST-001 Fuel gas pressure - Hi Alarm
PST_7503_001_ALH	%M00371	7503-PST-001 Deluge line pressure - Hi Alarm
FR_7503_001_002_ALH	%M00372	7503-FR-001/002 Foam release from field - Hi alarm
SpareAI_21_ALH	%M00373	SpareAI-21 - Hi Alarm
FR_7503_003_ALH	%M00374	7503-FR-003 Foam release from panel - Hi Alarm
SpareAI_23_ALH	%M00375	SpareAI-23 - Hi Alarm

Tag Name	Address	Description
SpareAI_24_ALH	%M00376	SpareAI-24 - Hi Alarm
DG_7404_001_ALHI	%M00377	7404-DG-001 Gas detector pump house - Hi Alarm C_E
DG_7404_002_ALHI	%M00378	7404-DG-002 Gas detector pump house - Hi Alarm C_E
DG_7404_003_ALHI	%M00379	7404-DG-003 Gas detector pump house - Hi Alarm C_E
DG_7404_004_ALHI	%M00380	7404-DG-004 Gas detector pump house - Hi Alarm C_E
DG_7404_005_ALHI	%M00381	7404-DG-005 Gas detector pump house - Hi Alarm C_E
DG_7404_006_ALHI	%M00382	7404-DG-006 Gas detector pump house - Hi Alarm C_E
DG_7404_007_ALHI	%M00383	7404-DG-007 Gas detector pump house - Hi Alarm C_E
DG_7404_008_ALHI	%M00384	7404-DG-008 Gas detector pump house - Hi Alarm C_E
DG_7404_009_ALHI	%M00385	7404-DG-009 Gas detector pump house - Hi Alarm C_E
DG_7404_010_ALHI	%M00386	7404-DG-010 Gas detector pump house - Hi Alarm C_E
SpareAI_11_ALHI	%M00387	SpareAI-11 - Hi Alarm C_E
DF_7404_001_ALHI	%M00388	7404-DF-001 Flame detector pump house - Hi Alarm C_E
DF_7404_002_ALHI	%M00389	7404-DF-002 Flame detector pump house - Hi Alarm C_E
DG_7404_011R_ALHI	%M00390	7404-DG-011R Open path detector - Hi Alarm C_E
SpareAI_15_ALHI	%M00391	SpareAI-15 - Hi Alarm C_E
SpareAI_16_ALHI	%M00392	SpareAI-16 - Hi Alarm C_E
PST_7002_001_ALHI	%M00393	7002-PST-001 Air acc. pressure - Hi Alarm C_E
PST_6401_001_ALHI	%M00394	6401-PST-001 Fuel gas pressure - Hi Alarm C_E
PST_7503_001_ALHI	%M00395	7503-PST-001 Deluge line pressure - Hi Alarm C_E
FR_7503_001_002_ALHI	%M00396	7503-FR-001/002 Foam release from field - Hi alarm C_E
SpareAI_21_ALHI	%M00397	SpareAI-21 - Hi Alarm C_E
FR_7503_003_ALHI	%M00398	7503-FR-003 Foam release from panel - Hi Alarm C_E
SpareAI_23_ALHI	%M00399	SpareAI-23 - Hi Alarm C_E
SpareAI_24_ALHI	%M00400	SpareAI-24 - Hi Alarm C_E
DG_7404_001_ALL	%M00401	7404-DG-001 Gas detector pump house - Lo Alarm
DG_7404_002_ALL	%M00402	7404-DG-002 Gas detector pump house - Lo Alarm
DG_7404_003_ALL	%M00403	7404-DG-003 Gas detector pump house - Lo Alarm
DG_7404_004_ALL	%M00404	7404-DG-004 Gas detector pump house - Lo Alarm
DG_7404_005_ALL	%M00405	7404-DG-005 Gas detector pump house - Lo Alarm
DG_7404_006_ALL	%M00406	7404-DG-006 Gas detector pump house - Lo Alarm
DG_7404_007_ALL	%M00407	7404-DG-007 Gas detector pump house - Lo Alarm
DG_7404_008_ALL	%M00408	7404-DG-008 Gas detector pump house - Lo Alarm
DG_7404_009_ALL	%M00409	7404-DG-009 Gas detector pump house - Lo Alarm
DG_7404_010_ALL	%M00410	7404-DG-010 Gas detector pump house - Lo Alarm
SpareAI_11_ALL	%M00411	SpareAI-11 - Lo Alarm
DF_7404_001_ALL	%M00412	7404-DF-001 Flame detector pump house - Lo Alarm
DF_7404_002_ALL	%M00413	7404-DF-002 Flame detector pump house - Lo Alarm
DG_7404_011R_ALL	%M00414	7404-DG-011R Open path detector - Lo Alarm
SpareAI_15_ALL	%M00415	SpareAI-15 - Lo Alarm
SpareAI_16_ALL	%M00416	SpareAI-16 - Lo Alarm
PST_7002_001_ALL	%M00417	7002-PST-001 Air acc. pressure - Lo Alarm
PST_6401_001_ALL	%M00418	6401-PST-001 Fuel gas pressure - Lo Alarm
PST_7503_001_ALL	%M00419	7503-PST-001 Deluge line pressure - Lo Alarm
FR_7503_001_002_ALL	%M00420	7503-FR-001/002 Foam release from field - Lo Alarm
SpareAI_21_ALL	%M00421	SpareAI-21 - Lo Alarm
FR_7503_003_ALL	%M00422	7503-FR-003 Foam release from panel - Lo Alarm
SpareAI_23_ALL	%M00423	SpareAI-23 - Lo Alarm

Tag Name	Address	Description
SpareAI_24_ALL	%M00424	SpareAI-24 - Lo Alarm
DG_7404_001_ALLI	%M00425	7404-DG-001 Gas detector pump house - Lo Alarm C_E
DG_7404_002_ALLI	%M00426	7404-DG-002 Gas detector pump house - Lo Alarm C_E
DG_7404_003_ALLI	%M00427	7404-DG-003 Gas detector pump house - Lo Alarm C_E
DG_7404_004_ALLI	%M00428	7404-DG-004 Gas detector pump house - Lo Alarm C_E
DG_7404_005_ALLI	%M00429	7404-DG-005 Gas detector pump house - Lo Alarm C_E
DG_7404_006_ALLI	%M00430	7404-DG-006 Gas detector pump house - Lo Alarm C_E
DG_7404_007_ALLI	%M00431	7404-DG-007 Gas detector pump house - Lo Alarm C_E
DG_7404_008_ALLI	%M00432	7404-DG-008 Gas detector pump house - Lo Alarm C_E
DG_7404_009_ALLI	%M00433	7404-DG-009 Gas detector pump house - Lo Alarm C_E
DG_7404_010_ALLI	%M00434	7404-DG-010 Gas detector pump house - Lo Alarm C_E
SpareAI_11_ALLI	%M00435	SpareAI-11 - Lo Alarm C_E
DF_7404_001_ALLI	%M00436	7404-DF-001 Flame detector pump house - Lo Alarm C_E
DF_7404_002_ALLI	%M00437	7404-DF-002 Flame detector pump house - Lo Alarm C_E
DG_7404_011R_ALLI	%M00438	7404-DG-011R Open path detector - Lo Alarm C_E
SpareAI_15_ALLI	%M00439	SpareAI-15 - Lo Alarm C_E
SpareAI_16_ALLI	%M00440	SpareAI-16 - Lo Alarm C_E
PST_7002_001_ALLI	%M00441	7002-PST-001 Air acc. pressure - Lo Alarm C_E
PST_6401_001_ALLI	%M00442	6401-PST-001 Fuel gas pressure - Lo Alarm C_E
PST_7503_001_ALLI	%M00443	7503-PST-001 Deluge line pressure - Lo Alarm C_E
FR_7503_001_002_ALLI	%M00444	7503-FR-001/002 Foam release from field - Lo Alarm C_E
SpareAI_21_ALLI	%M00445	SpareAI-21 - Lo Alarm C_E
FR_7503_003_ALLI	%M00446	7503-FR-003 Foam release from panel - Lo Alarm C_E
SpareAI_23_ALLI	%M00447	SpareAI-23 - Lo Alarm C_E
SpareAI_24_ALLI	%M00448	SpareAI-24 - Lo Alarm C_E
DG_7404_001_INH	%M00449	7404-DG-001 Gas detector pump house - Inhibit
DG_7404_002_INH	%M00450	7404-DG-002 Gas detector pump house - Inhibit
DG_7404_003_INH	%M00451	7404-DG-003 Gas detector pump house - Inhibit
DG_7404_004_INH	%M00452	7404-DG-004 Gas detector pump house - Inhibit
DG_7404_005_INH	%M00453	7404-DG-005 Gas detector pump house - Inhibit
DG_7404_006_INH	%M00454	7404-DG-006 Gas detector pump house - Inhibit
DG_7404_007_INH	%M00455	7404-DG-007 Gas detector pump house - Inhibit
DG_7404_008_INH	%M00456	7404-DG-008 Gas detector pump house - Inhibit
DG_7404_009_INH	%M00457	7404-DG-009 Gas detector pump house - Inhibit
DG_7404_010_INH	%M00458	7404-DG-010 Gas detector pump house - Inhibit
SpareAI_11_INH	%M00459	SpareAI-11 - Inhibit
DF_7404_001_INH	%M00460	7404-DF-001 Flame detector pump house - Inhibit
DF_7404_002_INH	%M00461	7404-DF-002 Flame detector pump house - Inhibit
DG_7404_011R_INH	%M00462	7404-DG-011R Open path detector - Inhibit
SpareAI_15_INH	%M00463	SpareAI-15 - Inhibit
SpareAI_16_INH	%M00464	SpareAI-16 - Inhibit
PST_7002_001_INH	%M00465	7002-PST-001 Air acc. pressure - Inhibit
PST_6401_001_INH	%M00466	6401-PST-001 Fuel gas pressure - Inhibit
PST_7503_001_INH	%M00467	7503-PST-001 Deluge line pressure - Inhibit
FR_7503_001_002_INH	%M00468	7503-FR-001/002 Foam release from field - Inhibit
SpareAI_21_INH	%M00469	SpareAI-21 - Inhibit
FR_7503_003_INH	%M00470	7503-FR-003 Foam release from panel - Inhibit
SpareAI_23_INH	%M00471	SpareAI-23 - Inhibit

Tag Name	Address	Description
SpareAI_24_INH	%M00472	SpareAI-24 - Inhibit
DG_7404_001_OC	%M00473	7404-DG-001 Gas detector pump house - Open circuit
DG_7404_002_OC	%M00474	7404-DG-002 Gas detector pump house - Open circuit
DG_7404_003_OC	%M00475	7404-DG-003 Gas detector pump house - Open circuit
DG_7404_004_OC	%M00476	7404-DG-004 Gas detector pump house - Open circuit
DG_7404_005_OC	%M00477	7404-DG-005 Gas detector pump house - Open circuit
DG_7404_006_OC	%M00478	7404-DG-006 Gas detector pump house - Open circuit
DG_7404_007_OC	%M00479	7404-DG-007 Gas detector pump house - Open circuit
DG_7404_008_OC	%M00480	7404-DG-008 Gas detector pump house - Open circuit
DG_7404_009_OC	%M00481	7404-DG-009 Gas detector pump house - Open circuit
DG_7404_010_OC	%M00482	7404-DG-010 Gas detector pump house - Open circuit
SpareAI_11_OC	%M00483	SpareAI-11 - Open circuit
DF_7404_001_OC	%M00484	7404-DF-001 Flame detector pump house - Open circuit
DF_7404_002_OC	%M00485	7404-DF-002 Flame detector pump house - Open circuit
DG_7404_011R_OC	%M00486	7404-DG-011R Open path detector - Open circuit
SpareAI_15_OC	%M00487	SpareAI-15 - Open circuit
SpareAI_16_OC	%M00488	SpareAI-16 - Open circuit
PST_7002_001_OC	%M00489	7002-PST-001 Air acc. pressure - Open circuit
PST_6401_001_OC	%M00490	6401-PST-001 Fuel gas pressure - Open circuit
PST_7503_001_OC	%M00491	7503-PST-001 Deluge line pressure - Open circuit
FR_7503_001_002_OC	%M00492	7503-FR-001/002 Foam release from field - Open circuit
SpareAI_21_OC	%M00493	SpareAI-21 - Open circuit
FR_7503_003_OC	%M00494	7503-FR-003 Foam release from panel - Open circuit
SpareAI_23_OC	%M00495	SpareAI-23 - Open circuit
SpareAI_24_OC	%M00496	SpareAI-24 - Open circuit
DG_7404_001_SC	%M00497	7404-DG-001 Gas detector pump house - Short circuit
DG_7404_002_SC	%M00498	7404-DG-002 Gas detector pump house - Short circuit
DG_7404_003_SC	%M00499	7404-DG-003 Gas detector pump house - Short circuit
DG_7404_004_SC	%M00500	7404-DG-004 Gas detector pump house - Short circuit
DG_7404_005_SC	%M00501	7404-DG-005 Gas detector pump house - Short circuit
DG_7404_006_SC	%M00502	7404-DG-006 Gas detector pump house - Short circuit
DG_7404_007_SC	%M00503	7404-DG-007 Gas detector pump house - Short circuit
DG_7404_008_SC	%M00504	7404-DG-008 Gas detector pump house - Short circuit
DG_7404_009_SC	%M00505	7404-DG-009 Gas detector pump house - Short circuit
DG_7404_010_SC	%M00506	7404-DG-010 Gas detector pump house - Short circuit
SpareAI_11_SC	%M00507	SpareAI-11 - Short circuit
DF_7404_001_SC	%M00508	7404-DF-001 Flame detector pump house - Short circuit
DF_7404_002_SC	%M00509	7404-DF-002 Flame detector pump house - Short circuit
DG_7404_011R_SC	%M00510	7404-DG-011R Open path detector - Short circuit
SpareAI_15_SC	%M00511	SpareAI-15 - Short circuit
SpareAI_16_SC	%M00512	SpareAI-16 - Short circuit
PST_7002_001_SC	%M00513	7002-PST-001 Air acc. pressure - Short circuit
PST_6401_001_SC	%M00514	6401-PST-001 Fuel gas pressure - Short circuit
PST_7503_001_SC	%M00515	7503-PST-001 Deluge line pressure - Short circuit
FR_7503_001_002_SC	%M00516	7503-FR-001/002 Foam release from field - Short circuit
SpareAI_21_SC	%M00517	SpareAI-21 - Short circuit
FR_7503_003_SC	%M00518	7503-FR-003 Foam release from panel - Short circuit
SpareAI_23_SC	%M00519	SpareAI-23 - Short circuit

Tag Name	Address	Description
SpareAI_24_SC	%M00520	SpareAI-24 - Short circuit
DG_7404_001_FLT1	%M00521	7404-DG-001 Gas detector pump house - Fault 1
DG_7404_002_FLT1	%M00522	7404-DG-002 Gas detector pump house - Fault 1
DG_7404_003_FLT1	%M00523	7404-DG-003 Gas detector pump house - Fault 1
DG_7404_004_FLT1	%M00524	7404-DG-004 Gas detector pump house - Fault 1
DG_7404_005_FLT1	%M00525	7404-DG-005 Gas detector pump house - Fault 1
DG_7404_006_FLT1	%M00526	7404-DG-006 Gas detector pump house - Fault 1
DG_7404_007_FLT1	%M00527	7404-DG-007 Gas detector pump house - Fault 1
DG_7404_008_FLT1	%M00528	7404-DG-008 Gas detector pump house - Fault 1
DG_7404_009_FLT1	%M00529	7404-DG-009 Gas detector pump house - Fault 1
DG_7404_010_FLT1	%M00530	7404-DG-010 Gas detector pump house - Fault 1
SpareAI_11_FLT1	%M00531	SpareAI-11 - Fault 1
DF_7404_001_FLT1	%M00532	7404-DF-001 Flame detector pump house - Fault 1
DF_7404_002_FLT1	%M00533	7404-DF-002 Flame detector pump house - Fault 1
DG_7404_011R_FLT1	%M00534	7404-DG-011R Open path detector - Fault 1 (Optic fault)
SpareAI_15_FLT1	%M00535	SpareAI-15 - Fault 1
SpareAI_16_FLT1	%M00536	SpareAI-16 - Fault 1
PST_7002_001_FLT1	%M00537	7002-PST-001 Air acc. pressure - Fault 1
PST_6401_001_FLT1	%M00538	6401-PST-001 Fuel gas pressure - Fault 1
PST_7503_001_FLT1	%M00539	7503-PST-001 Deluge line pressure - Fault 1
FR_7503_001_002_FLT1	%M00540	7503-FR-001/002 Foam release from field - Fault 1
SpareAI_21_FLT1	%M00541	SpareAI-21 - Fault 1
FR_7503_003_FLT1	%M00542	7503-FR-003 Foam release from panel - Fault 1
SpareAI_23_FLT1	%M00543	SpareAI-23 - Fault 1
SpareAI_24_FLT1	%M00544	SpareAI-24 - Fault 1
DG_7404_001_FLT2	%M00545	7404-DG-001 Gas detector pump house - Fault 2
DG_7404_002_FLT2	%M00546	7404-DG-002 Gas detector pump house - Fault 2
DG_7404_003_FLT2	%M00547	7404-DG-003 Gas detector pump house - Fault 2
DG_7404_004_FLT2	%M00548	7404-DG-004 Gas detector pump house - Fault 2
DG_7404_005_FLT2	%M00549	7404-DG-005 Gas detector pump house - Fault 2
DG_7404_006_FLT2	%M00550	7404-DG-006 Gas detector pump house - Fault 2
DG_7404_007_FLT2	%M00551	7404-DG-007 Gas detector pump house - Fault 2
DG_7404_008_FLT2	%M00552	7404-DG-008 Gas detector pump house - Fault 2
DG_7404_009_FLT2	%M00553	7404-DG-009 Gas detector pump house - Fault 2
DG_7404_010_FLT2	%M00554	7404-DG-010 Gas detector pump house - Fault 2
SpareAI_11_FLT2	%M00555	SpareAI-11 - Fault 2
DF_7404_001_FLT2	%M00556	7404-DF-001 Flame detector pump house - Fault 2
DF_7404_002_FLT2	%M00557	7404-DF-002 Flame detector pump house - Fault 2
DG_7404_011R_FLT2	%M00558	7404-DG-011R Open path detector - Fault 2
SpareAI_15_FLT2	%M00559	SpareAI-15 - Fault 2
SpareAI_16_FLT2	%M00560	SpareAI-16 - Fault 2
PST_7002_001_FLT2	%M00561	7002-PST-001 Air acc. pressure - Fault 2
PST_6401_001_FLT2	%M00562	6401-PST-001 Fuel gas pressure - Fault 2
PST_7503_001_FLT2	%M00563	7503-PST-001 Deluge line pressure - Fault 2
FR_7503_001_002_FLT2	%M00564	7503-FR-001/002 Foam release from field - Fault 2
SpareAI_21_FLT2	%M00565	SpareAI-21 - Fault 2
FR_7503_003_FLT2	%M00566	7503-FR-003 Foam release from panel - Fault 2
SpareAI_23_FLT2	%M00567	SpareAI-23 - Fault 2

Tag Name	Address	Description
SpareAI_24_FLT2	%M00568	SpareAI-24 - Fault 2
DG_7404_001_FRUP	%M00569	7404-DG-001 Gas detector pump house - First up
DG_7404_002_FRUP	%M00570	7404-DG-002 Gas detector pump house - First up
DG_7404_003_FRUP	%M00571	7404-DG-003 Gas detector pump house - First up
DG_7404_004_FRUP	%M00572	7404-DG-004 Gas detector pump house - First up
DG_7404_005_FRUP	%M00573	7404-DG-005 Gas detector pump house - First up
DG_7404_006_FRUP	%M00574	7404-DG-006 Gas detector pump house - First up
DG_7404_007_FRUP	%M00575	7404-DG-007 Gas detector pump house - First up
DG_7404_008_FRUP	%M00576	7404-DG-008 Gas detector pump house - First up
DG_7404_009_FRUP	%M00577	7404-DG-009 Gas detector pump house - First up
DG_7404_010_FRUP	%M00578	7404-DG-010 Gas detector pump house - First up
SpareAI_11_FRUP	%M00579	SpareAI-11 - First up
DF_7404_001_FRUP	%M00580	7404-DF-001 Flame detector pump house - First up
DF_7404_002_FRUP	%M00581	7404-DF-002 Flame detector pump house - First up
DG_7404_011R_FRUP	%M00582	7404-DG-011R Open path detector - First up
SpareAI_15_FRUP	%M00583	SpareAI-15 - First up
SpareAI_16_FRUP	%M00584	SpareAI-16 - First up
PST_7002_001_FRUP	%M00585	7002-PST-001 Air acc. pressure - First up
PST_6401_001_FRUP	%M00586	6401-PST-001 Fuel gas pressure - First up
PST_7503_001_FRUP	%M00587	7503-PST-001 Deluge line pressure - First up
FR_7503_001_002_FRUP	%M00588	7503-FR-001/002 Foam release from field - First up
SpareAI_21_FRUP	%M00589	SpareAI-21 - First up
FR_7503_003_FRUP	%M00590	7503-FR-003 Foam release from panel - First up
SpareAI_23_FRUP	%M00591	SpareAI-23 - First up
SpareAI_24_FRUP	%M00592	SpareAI-24 - First up
AI_LoAlarm	%M00600	Lo Alarm Out
AI_HiAlarm	%M00601	Hi Alarm Out
AI_LoFault	%M00602	Lo Fault Out
AI_HiFault	%M00604	Hi Fault Out
Voting_DG_7404_ALLI	%M00605	Voting of Gas Low Alarms in pump house
Voting_DG_7404_ALHI	%M00606	Voting of Gas High Alarms in pump house
Voting_All_Inhibits	%M00607	Voting of all Inhibits
Voting_All_Faults	%M00608	Voting of all Faults
AI_Fault1	%M00609	Fault 1 for Gas detecor
AI_Fault2	%M00610	Fault 2 for Gas detecor
DelugeFailedToRelease	%M00620	Set if Deluge fail to release after trigging start of Pump B 3 times with 30 sec. delay
Pump_Active_1_Hour	%M00621	Set if Pump A have been activated in 60 minutes.
System_Reset	%M00700	Reset all latches
System_Reset_Delay	%M00701	High when System_Reset is high and 5 sec. after this.
System_Reset_Startup	%M00702	Used for reset system at startup
System_Reset_HMI	%M00703	System reset activation signal from HMI (panel)
System_Reset_HMI_Help	%M00704	Help puls from HMI (panel) reset signal
System_AnyActions	%M00705	High first scan after C&E has caused first trip
System_Mute	%M00706	System Mute from tuch panel
System_Sound01	%M00800	Reset all latches
System_Sound02	%M00801	Reset all latches
System_Sound03	%M00802	Reset all latches
System_Sound04	%M00803	Reset all latches

Tag Name	Address	Description
System_Sound05	%M00804	Reset all latches
System_Sound06	%M00805	Reset all latches
System_Sound07	%M00806	Reset all latches
System_Sound08	%M00807	Reset all latches
System_Sound09	%M00808	Reset all latches
System_Sound10	%M00809	Reset all latches
System_Sound11	%M00810	Reset all latches
System_Sound12	%M00811	Reset all latches
System_Sound13	%M00812	Reset all latches
System_Sound14	%M00813	Reset all latches
System_Sound15	%M00814	Reset all latches
System_Sound16	%M00815	Reset all latches
System_Sound17	%M00816	Reset all latches
System_Sound18	%M00817	Reset all latches
System_Sound19	%M00818	Reset all latches
System_Sound20	%M00819	Reset all latches
System_Sound21	%M00820	Reset all latches
System_Sound22	%M00821	Reset all latches
System_Sound23	%M00822	Reset all latches
System_Sound24	%M00823	Reset all latches
System_Sound25	%M00824	Reset all latches
System_Sound26	%M00825	Reset all latches
System_Sound27	%M00826	Reset all latches
System_Sound28	%M00827	Reset all latches
System_Sound29	%M00828	Reset all latches
System_Sound30	%M00829	Reset all latches
System_Sound31	%M00830	Reset all latches
System_Sound32	%M00831	Reset all latches
System_Sound33	%M00832	Reset all latches
System_Sound34	%M00833	Reset all latches
System_Sound35	%M00834	Reset all latches
System_Sound36	%M00835	Reset all latches
System_Sound37	%M00836	Reset all latches
System_Sound38	%M00837	Reset all latches
System_Sound39	%M00838	Reset all latches
System_Sound40	%M00839	Reset all latches
EY_6401_001_002	%Q00001	6401-EY-001/002 ESD fuel gas valve shutdown
SA_7404_001_002	%Q00002	7404-SA-001/002 Local alarm pump house
P2100_P2200	%Q00003	P2100/P2200 Oil booster pump shutdown
P3600_P3700	%Q00004	P3600/P3700 Water booster pump shutdown
PUMP_A	%Q00005	Deluge pump A Start/Stop
PUMP_B_STOP	%Q00006	Deluge pump B Stop
PUMP_B_START	%Q00007	Deluge pump B Start
PumpShutdown	%Q00008	Fire & Gas pump shutdown
Pump_A_PulsOut	%Q00009	Pulsed output for open circuit check on Pump A Out signal
Pump_B_Start_PulsOut	%Q00010	Pulsed output for open circuit check on Pump B Start signal
SpareDO_11	%Q00011	SpareDO-11
Tag_DO_12	%Q00012	EDS valve open indication

Tag Name	Address	Description
Tag_DO_13	%Q00013	EDS valve closed indication
Tag_DO_14	%Q00014	EDS valve low air indication
Tag_DO_15	%Q00015	Panel alarm
PLC_Fault	%Q00016	SpareDO-16
SpareDO_17	%Q00017	SpareDO-17
SpareDO_18	%Q00018	SpareDO-18
SpareDO_19	%Q00019	SpareDO-19
SpareDO_20	%Q00020	SpareDO-20
SpareDO_21	%Q00021	SpareDO-21
SpareDO_22	%Q00022	SpareDO-22
SpareDO_23	%Q00023	SpareDO-23
SpareDO_24	%Q00024	SpareDO-24
SpareDO_25	%Q00025	SpareDO-25
SpareDO_26	%Q00026	SpareDO-26
SpareDO_27	%Q00027	SpareDO-27
SpareDO_28	%Q00028	SpareDO-28
SpareDO_29	%Q00029	SpareDO-29
SpareDO_30	%Q00030	SpareDO-30
SpareDO_31	%Q00031	SpareDO-31
SpareDO_32	%Q00032	SpareDO-32
DG_7404_001_VAL	%R00001	7404-DG-001 Gas detector pump house - Converted value
DG_7404_002_VAL	%R00003	7404-DG-002 Gas detector pump house - Converted value
DG_7404_003_VAL	%R00005	7404-DG-003 Gas detector pump house - Converted value
DG_7404_004_VAL	%R00007	7404-DG-004 Gas detector pump house - Converted value
DG_7404_005_VAL	%R00009	7404-DG-005 Gas detector pump house - Converted value
DG_7404_006_VAL	%R00011	7404-DG-006 Gas detector pump house - Converted value
DG_7404_007_VAL	%R00013	7404-DG-007 Gas detector pump house - Converted value
DG_7404_008_VAL	%R00015	7404-DG-008 Gas detector pump house - Converted value
DG_7404_009_VAL	%R00017	7404-DG-009 Gas detector pump house - Converted value
DG_7404_010_VAL	%R00019	7404-DG-010 Gas detector pump house - Converted value
SpareAI_11_VAL	%R00021	SpareAI-11 - Converted value
DF_7404_001_VAL	%R00023	7404-DF-001 Flame detector pump house - Converted value
DF_7404_002_VAL	%R00025	7404-DF-002 Flame detector pump house - Converted value
DG_7404_011R_VAL	%R00027	7404-DG-011R Open path detector - Converted value
SpareAI_15_VAL	%R00029	SpareAI-15 - Converted value
SpareAI_16_VAL	%R00031	SpareAI-16 - Converted value
PST_7002_001_VAL	%R00033	7002-PST-001 Air acc. pressure - Converted value
PST_6401_001_VAL	%R00035	6401-PST-001 Fuel gas pressure - Converted value
PST_7503_001_VAL	%R00037	7503-PST-001 Deluge line pressure - Converted value
FR_7503_001_002_VAL	%R00039	7503-FR-001/002 Foam release from field - Converted value
SpareAI_21_VAL	%R00041	SpareAI-21 - Converted value
FR_7503_003_VAL	%R00043	7503-FR-003 Foam release from panel - Converted value
SpareAI_23_VAL	%R00045	SpareAI-23 - Converted value
SpareAI_24_VAL	%R00047	SpareAI-24 - Converted value
AI_ValueIn	%R00100	The converted value moved to _VAL on analog input.
AI_LoAlarmSetpoint	%R00106	Lo Alarm Setpoint
AI_HiAlarmSetpoint	%R00108	Hi Alarm Setpoint
AI_LoFaultSetpoint	%R00110	Lo Fault Out Setpoint

Tag Name	Address	Description
AI_HiFaultSetpoint	%R00112	Hi Fault Out Setpoint
AI_TempVal	%R00120	Value used for converting analog input to correct range.
AI_ValueIn_Real	%R00122	The converted value moved to _VAL on analog input.
AI_TempVal_Real	%R00124	Value used for converting analog input to correct range.
AI_HiAlarmSetpoint_Real	%R00126	Hi Alarm Setpoint
AI_LoAlarmSetpoint_Real	%R00128	Lo Alarm Setpoint
AI_LoFaultSetpoint_Real	%R00130	Lo Fault Out Setpoint
AI_HiFaultSetpoint_Real	%R00132	Hi Fault Out Setpoint
AI_Fault1_Setpoint_Real	%R00134	Fault 1 Out Setpoint
AI_Fault2_Setpoint_Real	%R00136	Fault 2 Out Setpoint
System_ClearPLC_Fault	%R00200	Used in logic to clear I/O Fault Table
System_ClearIO_Fault	%R00201	Used in logic to clear I/O Fault Table
System_C_E_Delay	%R00202	Use in logic to delay Cause&Effect logic at startup of PLC.
System_HMI_Reset_Timer	%R00205	Resets HMI system reset signal after 5 sec.
System_Reset_HelpRegister	%R00208	Help register for System Reset at startup
Pump_Timer1	%R00209	Use in logic to delay Cause&Effect logic at startup of PLC.
[0]	%R00209	
[1]	%R00210	
[2]	%R00211	
Pump_Timer2	%R00212	Used in logic for delay of Pump B start signal.
Pump_Timer3	%R00215	Used in logic for delay of Pump B start signal.
Pump_Timer4	%R00218	Used in logic for delay of Pump B start signal.
Pump_Timer5	%R00221	Used in logic for delay of Pump B start signal.
Pump_Timer6	%R00224	Used in logic for delay of Pump B start signal.
Pump_Timer7	%R00227	Used in logic for delay of Pump B start signal.
Pump_Timer8	%R00230	Used in logic for delay of Pump B start signal.
Pump_Timer9	%R00233	Used in logic for open circuit on Pump A signal.
Pump_Timer10	%R00236	Used in logic for open circuit on Pump A signal.
Pump_Timer11	%R00239	Used in logic for open circuit on Pump A signal.
Pump_Timer12	%R00242	Used in logic for open circuit on Pump A signal.
Pump_Timer13	%R00245	Used in logic for stop Pump A/B after active in 60 minutes.
AAAAA	%R02000	
#FST_SCN	%S00001	Set to 1 when the current sweep is the first sweep
#LST_SCN	%S00002	Reset from 1 to 0 when the current sweep is the last sweep
#T_10MS	%S00003	0.01 second timer contact
#T_100MS	%S00004	0.1 second timer contact
#T_SEC	%S00005	1.0 second timer contact
#T_MIN	%S00006	1.0 minute timer contact
#ALW_ON	%S00007	Always ON
#ALW_OFF	%S00008	Always OFF
#SY_FULL	%S00009	Set when the PLC fault table fills up
#IO_FULL	%S00010	Set when the I/O fault table fills up
#OVR_PRE	%S00011	Set when an override exists in %I, %Q, %M or %G memory
#PRG_CHK	%S00013	Set when background program check is active
#PLC_BAT	%S00014	Set to indicate a bad battery in a Release 4 or later CPU
#SNPXACT	%S00017	SNP-X host is actively attached to the CPU
#SNPX_RD	%S00018	SNP-X host has read data from the CPU
#SNPX_WT	%S00019	SNP-X host has written data to the CPU

Tag Name	Address	Description
#RLTN_OK	%S00020	Set ON when a relational function using REAL data is successful
#CONTCON	%S00032	Reserved for use by the programming software for continuation contact
#PB_SUM	%SA0001	Set when PLC checksum does not match the application checksum
#OV_SWP	%SA0002	Set when PLC sweep time is longer than user specified sweep time
#APL_FLT	%SA0003	Set when an application fault occurs
#CFG_MM	%SA0009	Set when a configuration mismatch is detected during system powerup
#HRD_CPU	%SA0010	Set when the diagnostics detects a problem with the CPU hardware
#LOW_BAT	%SA0011	Set when a low battery fault occurs
#LOS_RCK	%SA0012	Set when an expansion rack stops communicating with the CPU
#LOS_IOM	%SA0014	Set when an I/O module stops communicating with the PLC CPU
#LOS_SIO	%SA0015	Set when an option module stops communicating with the PLC CPU
#ADD_RCK	%SA0017	Set when an expansion rack is added to the system
#ADD_IOM	%SA0019	Set when an I/O module is added to a rack
#ADD_SIO	%SA0020	Set when an option module is added to a rack
#HRD_SIO	%SA0027	Set when a hardware failure is detected in an option module
#SFT_SIO	%SA0031	Set when an unrecoverable fault is detected in an option module
#SBUS_ER	%SA0032	Set when a bus error occurs on the VME bus backplane.
#BAD_RAM	%SB0010	Set when the CPU detects corrupted RAM at powerup
#BAD_PWD	%SB0011	Set when a password violation occurs
#SFT_CPU	%SB0013	Set when the CPU detects an unrecoverable error in the software
#STOR_ER	%SB0014	Set when an error occurs during a programmer store operation
#ANY_FLT	%SC0009	Set when any fault occurs
#SY_FLT	%SC0010	Set when fault occurs where an entry is put in the PLC fault table
#IO_FLT	%SC0011	Set when fault occurs where an entry is put in the I/O fault table
#SY_PRES	%SC0012	Set as long as there is at least one entry in the PLC fault table
#IO_PRES	%SC0013	Set as long as there is at least one entry in the I/O fault table
#HRD_FLT	%SC0014	Set when a hardware fault occurs
#SFT_FLT	%SC0015	Set when a software fault occurs
T00002	%T00002	High 120 sec. after PLC set in run.
T00003	%T00003	
T00004	%T00004	
T00005	%T00005	
T00006	%T00006	
T00007	%T00007	
T00008	%T00008	
T00009	%T00009	
T00010	%T00010	
T00011	%T00011	For Fault 1 comparing on Gas detector
T00012	%T00012	For Fault 1 comparing on Gas detector
T00013	%T00013	For checking Open circuit on Start Pump
T00014	%T00014	For checking Open circuit on Start Pump